

Publications by Model: CMS-Flux

Total Publications: 183 (sorted newest to oldest)

1. Improved estimates of net ecosystem exchanges in mega-countries using GOSAT and OCO-2 observations

Authors: Lingyu Zhang; Fei Jiang; Wei He; Mousong Wu; Jun Wang; Weimin Ju; Hengmao Wang; Yongguang Zhang; Stephen Sitch; Jing M. Chen

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01910-w

URL: <https://doi.org/10.1038/s43247-024-01910-w>

Publication Date: 2024-11-25

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References: 72

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: N/A

2. Peplospheric Influences on Local Greenhouse Gas and Aerosol Variability at the Lamezia Terme WMO/GAW Regional Station in Calabria, Southern Italy: A Multiparameter Investigation

Authors: Francesco D'Amico (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy, Department of Biology, Ecology and Earth Sciences, University of Calabria, Via Bucci Cubo 15B, I-87036 Rende, Cosenza, Italy); Claudia Roberta Calidonna (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Ivano Ammoscato (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Daniel Gullì (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Luana Malacaria (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Salvatore Sinopoli (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Giorgia De Benedetto (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Teresa Lo Feudo (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy)

Journal: Sustainability

Publisher: MDPI AG

DOI: 10.3390/su162310175

URL: <https://doi.org/10.3390/su162310175>

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References: 142

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Abstract: One of the keys towards sustainable policies and advanced air quality monitoring is the detailed assessment of all factors that affect the surface concentrations of greenhouse gases (GHGs) and aerosols. While the development of new atmospheric tracers can pinpoint emission sources, the atmosphere itself plays a relevant role even at local scales: Its dynamics can increase, or reduce, surface concentrations of pollutants harmful to human health and the environment. PBL (planetary boundary layer), or peplospheric, variability is known to affect such concentrations. In this study, an unprecedented characterization of PBL cycles and patterns is performed at the WMO/GAW regional coastal site of Lamezia Terme (code: LMT) in Calabria, Southern Italy, in conjunction with the analysis of key GHGs and aerosols. The analysis, accounting for five months of 2024 data, indicates that

peplospheric variability and wind regimes influence the concentrations of key GHGs and aerosols. In particular, PBLH (PBL height) patterns have been tested to further influence the surface concentrations of carbon monoxide (CO), black carbon (BC), and particulate matter (PM). This research introduces four distinct wind regimes at LMT: breeze, not complete breeze, eastern synoptic, and western synoptic, each with its peculiar influences on the local transport of gases and aerosols. This research demonstrates that peplosphere monitoring needs to be considered when ensuring optimal air quality in urban and rural areas.

3. Assessment of Formaldehyde's Impact on Indoor Environments and Human Health via the Integration of Satellite Tropospheric Total Columns and Outdoor Ground Sensors

Authors: Elena Barrese (Department of Medicine, Epidemiology, Occupational and Environmental Hygiene, National Institute for Insurance Against Accidents at Work, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Marco Valentini (Department of Medicine, Epidemiology, Occupational and Environmental Hygiene, National Institute for Insurance Against Accidents at Work, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Marialuisa Scarpelli (Department of Medicine, Epidemiology, Occupational and Environmental Hygiene, National Institute for Insurance Against Accidents at Work, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Pasquale Samele (Department of Medicine, Epidemiology, Occupational and Environmental Hygiene, National Institute for Insurance Against Accidents at Work, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Luana Malacaria (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy); Francesco D'Amico (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy, Department of Biology, Ecology and Earth Sciences, University of Calabria, Via Bucci Cubo 15B, I-87036 Rende, Cosenza, Italy); Teresa Lo Feudo (Institute of Atmospheric Sciences and Climate, National Research Council of Italy, Area Industriale Comp. 15, I-88046 Lamezia Terme, Catanzaro, Italy)

Journal: Sustainability

Publisher: MDPI AG

DOI: 10.3390/su16229669

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Publication Date: 2024-11-06

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Keywords: N/A

Funding: INAIL (Italian National Institute for Insurance against Accidents at Work), CNR-ISAC (National Research Council of Italy—Institute of Atmospheric Sciences and Climate)

License: CC BY 4.0

Open Access: N/A

Abstract: Formaldehyde (HCHO) is harmful to human health and an adequate assessment of its concentrations, both in outdoor and indoor environments, is necessary in the context of sustainable policies designed to mitigate health risks. In this research, ground indoor and outdoor HCHO measurements are integrated with the analysis of tropospheric total columns obtained by satellite surveys to assess the concentrations of HCHO in a number of environments, exploiting the proximity of a World Meteorological Organization--Global Atmosphere Watch (WMO/GAW) observation site in Calabria, Southern Italy to a National Institute for Insurance against Accidents at Work (INAIL) department in the municipality of Lamezia Terme. The meteorological parameters used by the WMO

station are also used to provide additional data and test new correlations. Using statistical significance tests, this study demonstrates the presence of a correlation between indoor and outdoor HCHO concentrations, thus showing that an exchange between indoor and outdoor formaldehyde does occur. Rooms located in the local INAIL building where indoor measurements took place also demonstrate degrees of susceptibility to HCHO exposure, which are correlated with the orientation of prevailing wind corridors in the area. The new findings constitute an unprecedented characterization of HCHO hazards in Calabria and provide regulators with new tools with which to mitigate formaldehyde-related risks.

4. Characterizing and communicating uncertainty: lessons from NASA's Carbon Monitoring System

Authors: Robert E Kennedy; Shawn P Serbin; Michael C Dietze; Hans-Erik Andersen; Chad Babcock; David F Baker; Molly E Brown; Kenneth J Davis; Laura Duncanson; Sha Feng; Andrew T Hudak; Junjie Liu; Paul L Patterson; Brett Raczka; Mark A Cochrane; Edil A Sepúlveda Carlo; Rodrigo Vargas

Journal: Environmental Research Letters

Publisher: IOP Publishing

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References: 55

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Funding: NASA Carbon Monitoring System, Earth Sciences Division, Division of Environmental Biology

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Open Access: N/A

Abstract: Navigating uncertainty is a critical challenge in all fields of science, especially when translating knowledge into real-world policies or management decisions. However, the wide variance in concepts and definitions of uncertainty across scientific fields hinders effective communication. As a microcosm of diverse fields within Earth Science, NASA's Carbon Monitoring System (CMS) provides a useful crucible in which to identify cross-cutting concepts of uncertainty. The CMS convened the Uncertainty Working Group (UWG), a group of specialists across disciplines, to evaluate and synthesize efforts to characterize uncertainty in CMS projects. This paper represents efforts by the UWG to build a heuristic framework designed to evaluate data products and communicate uncertainty to both scientific and non-scientific end users. We consider four pillars of uncertainty: origins, severity, stochasticity versus incomplete knowledge, and spatial and temporal autocorrelation. Using a common vocabulary and a generalized workflow, the framework introduces a graphical heuristic accompanied by a narrative, exemplified through contrasting case studies. Envisioned as a versatile tool, this framework provides clarity in reporting uncertainty, guiding users and tempering expectations. Beyond CMS, it stands as a simple yet powerful means to communicate uncertainty across diverse scientific communities.

5. A Modeling Framework of Atmospheric CO₂ in the Mediterranean Marseille Coastal City Area, France

Authors: Brian Nathan (Institut de Recherche pour le Développement (IRD)); Irène Xueref-Remy (Institut de Recherche pour le Développement (IRD)); Thomas Lauvaux (Groupe de Spectrométrie Moléculaire et Atmosphérique, Université de Reims Champagne-Ardenne, 51100 Reims, France); Christophe Yohia (OSU Institut PYTHEAS, Service Informatique Pytheas, 13009 Marseille, France); Damien Piga (AtmoSud, 13006 Marseille, France); Jacques Piazzola (Mediterranean Institute of Oceanography (MIO), Université du Sud Toulon, 83130 Toulon, France); Tomohiro Oda (University of Maryland, Universities Space Research Association); Mélissa Milne (Mediterranean Institute of Oceanography (MIO), Aix-Marseille Université, Campus de Luminy, 13009 Marseille, France); Maria Herrmann (Pennsylvania State University); Cathy Wimart-Rousseau (National Oceanography Centre); Alexandre Armengaud (AtmoSud, 13006 Marseille, France)

Journal: Atmosphere

Publisher: MDPI AG

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References: 104

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Language: en

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Keywords: N/A

Funding: A*MIDEX project

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Open Access: N/A

Abstract: As atmospheric CO₂ emissions and the trend of urbanization both increase, the ability to accurately assess the CO₂ budget from urban environments becomes more important for effective CO₂ mitigation efforts. This task can be difficult for complex areas such as the urban-coastal Mediterranean region near Marseille, France, which contains the second most populous city in France as well as a broad coastline and nearby mountainous terrain. In this study, we establish a CO₂ modeling framework for this region for the first time using WRF-Chem and demonstrate its efficacy through comparisons against cavity-ringdown spectrometer measurements recorded at three sites: one 75 km north of the city in a forested area, one in the city center, and one at the urban/coastal border. A seasonal CO₂ analysis compares Summertime 2016 and Wintertime 2017, to which Springtime 2017 is also added due to its noticeably larger vegetation uptake values compared to Summertime. We find that there is a large biogenic signal, even in and around Marseille itself, though this may be a consequence of having limited fine-scale information on vegetation parameterization in the region. We further find that simulations without the urban heat island module had total CO₂ values 0.46 ppm closer to the measured enhancement value at the coastal Endoume site during the Summertime 2016 period than with the module turned on. This may indicate that the boundary layer on the coast is less sensitive to urban influences than it is to sea-breeze interactions, which is consistent with previous studies of the region. A back-trajectory analysis with the Lagrangian Particle Dispersion Model found 99.83% of

emissions above $100 \text{ mol km}^{-2} \text{ month}^{-1}$ captured in Summer 2016 by the three measurement towers, providing evidence of the receptors' ability to constrain the domain. Finally, a case study showcases the model's ability to capture the rapid change in CO_2 when transitioning between land-breeze and sea-breeze conditions as well as the recirculation of air from the industrial Fos region towards the Marseille metroplex. In total, the presented modeling framework should open the door to future CO_2 investigations in the region, which can inform policymakers carrying out CO_2 mitigation strategies.

6. Review of Satellite Remote Sensing of Carbon Dioxide Inversion and Assimilation

Authors: Kai Hu (Nanjing University); Xinyan Feng (Nanjing University); Qi Zhang (Nanjing University); Pengfei Shao (Nanjing University); Ziran Liu (Nanjing University); Yao Xu (University of Reading); Shiqian Wang (Economic and Technical Research Institute of State Grid Henan Electric Power Company, Zhengzhou 450052, China); Yuanyuan Wang (Economic and Technical Research Institute of State Grid Henan Electric Power Company, Zhengzhou 450052, China); Han Wang (Economic and Technical Research Institute of State Grid Henan Electric Power Company, Zhengzhou 450052, China); Li Di (State Grid Henan Electric Power Company, Zhengzhou 450003, China); Min Xia (Nanjing University)

Journal: Remote Sensing

Publisher: MDPI AG

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URL: <https://doi.org/10.3390/rs16183394>

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Cited by: 6 publications

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Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: With the rapid development of satellite remote sensing technology, carbon-cycle research, as a key focus of global climate change, has also been widely developed in terms of carbon source/sink-research methods. The internationally recognized "top-down" approach, which is based on satellite observations, is an important means to verify greenhouse gas-emission inventories. This article reviews the principles, categories, and development of satellite detection payloads for greenhouse gases and introduces inversion algorithms and datasets for satellite remote sensing of XCO₂. It emphasizes inversion methods based on machine learning and assimilation algorithms. Additionally, it presents the technology and achievements of carbon-assimilation systems used to estimate carbon fluxes. Finally, the article summarizes and prospects the future development of carbon-assimilation inversion to improve the accuracy of estimating and monitoring Earth's carbon-cycle processes.

7. On-Orbit Wavelength Calibration Error Analysis of the Spaceborne Hyperspectral Greenhouse Gas Monitoring Instrument Using the Solar Fraunhofer Lines

Authors: Yulong Guo (Chinese Academy of Sciences); Cailan Gong (Chinese Academy of Sciences); Yong Hu (Chinese Academy of Sciences); Fuqiang Zheng (Chinese Academy of Sciences); Yunmeng Liu (Chinese Academy of Sciences)

Journal: Remote Sensing

Publisher: MDPI AG

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D Program of China, Shanghai Pilot Program for Basic Research—Chinese Academy of Sciences, Shanghai Branch

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Open Access: N/A

Abstract: Accurate on-orbit wavelength calibration of the spaceborne hyperspectral payload is the key to the quantitative analysis and application of observational data. Due to the high spectral resolution of general spaceborne hyperspectral greenhouse gas (GHG) detection instruments, the common Fraunhofer lines in the solar atmosphere can be used as a reference for on-orbit wavelength calibration. Based on the performances of a GHG detection instrument under development, this study simulated the instrument's solar-viewing measurement spectra and analyzed the main sources of errors in the on-orbit wavelength calibration method of the instrument using the solar Fraunhofer lines, including the Doppler shift correction error, the instrumental measurement error, and the peak-peak algorithm error. The calibration accuracy was independently calculated for 65 Fraunhofer lines within the spectral range of the instrument. The results show that the wavelength calibration accuracy is mainly affected by the asymmetry of the Fraunhofer lines and the random error associated with instrument measurement, and it can cause calibration errors of more than 1/10 of the spectral resolution at maximum. A total of 49 Fraunhofer lines that meet the requirements for calibration accuracy were screened based on the design parameters of the instrument. Due to the uncertainty of simulation, the results in this study have inherent limitations, but provide valuable insights for quantitatively analyzing the errors of the on-orbit wavelength calibration method using the Fraunhofer lines, evaluating the influence of instrumental parameters on the calibration accuracy, and enhancing the accuracy of on-orbit wavelength calibration for similar GHG detection payloads.

8. Carbon emissions from the 2023 Canadian wildfires

Authors: Brendan Byrne; Junjie Liu; Kevin W. Bowman; Madeleine Pascolini-Campbell; Abhishek Chatterjee; Sudhanshu Pandey; Kazuyuki Miyazaki; Guido R. van der Werf; Debra Wunch; Paul O. Wennberg; Coleen M. Roehl; Saptarshi Sinha

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Publisher: Springer Science and Business Media LLC

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References: 75

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Open Access: N/A

Abstract: The 2023 Canadian forest fires have been extreme in scale and intensity with more than seven times the average annual area burned compared to the previous four decades¹. Here, we quantify the carbon emissions from these fires from May to September 2023 on the basis of inverse modelling of satellite carbon monoxide observations. We find that the magnitude of the carbon emissions is 647 TgC (570-727 TgC), comparable to the annual fossil fuel emissions of large nations, with only India, China and the USA releasing more carbon per year². We find that widespread hot-dry weather was a principal driver of fire spread, with 2023 being the warmest and driest year since at least 1980³. Although temperatures were extreme relative to the historical record, climate projections indicate that these temperatures are likely to be typical during the 2050s, even under a moderate climate mitigation scenario (shared socioeconomic pathway, SSP 2-4.5)⁴. Such conditions are likely to drive increased fire activity and suppress carbon uptake by Canadian forests, adding to concerns about the long-term durability of these forests as a carbon sink⁵⁻⁸.

9. A framework for natural resource management with geospatial machine learning: a case study of the 2021 Almora forest fires

Authors: Arpit Tiwari; Preethi Nanjundan; Ravi Ranjan Kumar; Vijay Kumar Soni

Journal: Fire Ecology

Publisher: Springer Science and Business Media LLC

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URL: <https://doi.org/10.1186/s42408-024-00293-9>

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Language: en

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Open Access: N/A

Abstract: Background Wildfires have a substantial impact on air quality and ecosystems by releasing greenhouse gases (GHGs), trace gases, and aerosols into the atmosphere. These wildfires produce both light-absorbing and merely scattering aerosols that can act as cloud condensation nuclei, altering cloud reflectivity, cloud lifetime, and precipitation frequency. Uttarakhand province in India experiences frequent wildfires that affect its protected ecosystems. Thus, a natural resource management system is needed in this region to assess the impact of wildfire hazards on land and atmosphere. We conducted an analysis of a severe fire event that occurred between January and April 2021 in the Kumaun region of Uttarakhand, by utilizing open-source geospatial data. Near-real-time satellite observations of pre- and post-fire conditions within the study area were used to detect changes in land and atmosphere. Supervised machine learning algorithm was also implemented to estimate burned above ground biomass (AGB) to monitor biomass stock. Results The study found that 21.75% of the total burned area burned with moderate to high severity, resulting in a decreased Soil Adjusted Vegetation Index value (> 0.3), a reduced Normalized Differential Moisture Index value (> 0.4), and a lowered Normalized Differential Vegetation Index (> 0.5). The AGB estimate demonstrated a significant simple determination ($r^2 = 0.001702$) and probability ($P < 2.2 \times 10^{-16}$), along with a positive correlation ($r \leq 0.24$) with vegetation and soil indices. The algorithm predicted that 17.56 tonnes of biomass per hectare burned in the Kumaun forests. This fire incident resulted in increased emissions of carbon dioxide (CO_2 ; $\sim 0.8 \times 10^{-4} \text{ kg carbon h}^{-1}$), methane (CH_4 ; $\sim 200 \times 10^{-9} \text{ mol fraction in dry air}$), carbon monoxide (CO ; $2000 \times 10^{15} \text{ molecules cm}^{-2} \text{ total column}$), and formaldehyde (HCHO ; $3500 \times 10^{13} \text{ molecules cm}^{-2} \text{ total column}$), along with increased aerosol optical thickness (varying from 0.2 to 0.5). Conclusions We believe that our proposed operational framework for managing natural resources and assessing the impact of natural hazards can be used to efficiently monitor near-real-time forest-fire-caused changes in land and atmosphere. This method makes use of openly accessible geospatial data that can be employed for several objectives, including monitoring carbon stocks, greenhouse gas emissions, criterion air pollution, and radiative forcing of the climate, among many others. Our proposed framework will assist policymakers and the scientific community in mitigating

climate change problems and in developing adaptation policies.

10. Enhanced Carbon Flux Response to Atmospheric Aridity and Water Storage Deficit During the 2015-2016 El Niño Compromised Carbon Balance Recovery in Tropical South America

Authors: Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Paul I. Palmer (University of Edinburgh, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Joanna Joiner (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Paul Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Liang Feng (University of Edinburgh); Sassan Saatchi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael Keller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, USDA Forest Service International Institute of Tropical Forestry San Juan PR USA); Marcos Longo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Lawrence Berkeley National Laboratory); David Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Paul O. Wennberg (California Institute of Technology)

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Keywords: water storage deficit, aridity, recovery, tropical South America

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Open Access: N/A

Abstract: During the 2015-2016 El Niño, the Amazon basin released almost one gigaton of carbon (GtC) into the atmosphere due to extreme temperatures and drought. The link between the drought impact and recovery of the total carbon pools and its biogeochemical drivers is still unknown. With satellite-constrained net carbon exchange and its component fluxes including gross primary production and fire emissions, we show that the total carbon loss caused by the 2015-2016 El Niño had not recovered by the end of 2018. Forest ecosystems over the Northeastern (NE) Amazon suffered a cumulative total carbon loss of ~0.6 GtC through December 2018, driven primarily by a suppression of photosynthesis whereas southeastern savannah carbon loss was driven in part by fire. We attribute the

slow recovery to the unexpected large carbon loss caused by the severe atmospheric aridity coupled with a water storage deficit during drought. We show the attenuation of carbon uptake is three times higher than expected from the pre-drought sensitivity to atmospheric aridity and ground water supply. Our study fills an important knowledge gap in our understanding of the unexpectedly enhanced response of carbon fluxes to atmospheric aridity and water storage deficit and its impact on regional post-drought recovery as a function of the vegetation types and climate perturbations. Our results suggest that the disproportionate impact of water supply and demand could compromise resiliency of the Amazonian carbon balance to future increases in extreme events.

11. Toward Low-Latency Estimation of Atmospheric CO₂ Growth Rates Using Satellite Observations: Evaluating Sampling Errors of Satellite and In Situ Observing Approaches

Authors: Sudhanshu Pandey (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John B. Miller (NOAA Global Monitoring Laboratory Boulder CO USA); Sourish Basu (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Earth System Science Interdisciplinary Center College Park MD USA); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Brad Weir (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Morgan State University Baltimore MD USA); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Frédéric Chevallier (Laboratoire des Sciences du Climat et de L'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Zhiqiang Liu (CMA Key Open Laboratory of Transforming Climate Resources to Economy Chongqing Institute of Meteorological Sciences Chongqing China); Feng Deng (University of Toronto); Christopher W. O'Dell (Colorado State University); Abhishek Chatterjee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: AGU Advances

Publisher: American Geophysical Union (AGU)

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References: 58

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Abstract: The atmospheric CO₂ growth rate is a fundamental measure of climate forcing. NOAA's growth rate estimates, derived from in situ observations at the marine boundary layer (MBL), serve as the benchmark in policy and science. However, NOAA's MBL-based method encounters challenges in accurately estimating the whole-atmosphere CO₂ growth rate at sub-annual scales. Here we introduce the Growth Rate from Satellite Observations (GRESO) method as a complementary approach to estimate the whole-atmosphere CO₂ growth rate utilizing satellite data. Satellite CO₂ observations offer extensive atmospheric coverage that extends the capability of the current NOAA benchmark. We assess the sampling errors of the GRESO and NOAA methods using 10 atmospheric transport model simulations. The simulations generate synthetic OCO-2 satellite and NOAA MBL data for calculating

CO₂ growth rates, which are compared against the global sum of carbon fluxes used as model inputs. We find good performance for the NOAA method ($R = 0.93$, RMSE = 0.12 ppm year⁻¹ or 0.25 PgC year⁻¹). GRESO demonstrates lower sampling errors ($R = 1.00$; RMSE = 0.04 ppm year⁻¹ or 0.09 PgC year⁻¹). Additionally, GRESO shows better performance at monthly scales than the NOAA method ($R = 0.76$ vs. 0.47, respectively). Due to CO₂'s atmospheric longevity, the NOAA method accurately captures growth rates over 5-year intervals. GRESO's robustness across partial coverage configurations (ocean or land data) shows that satellites can be promising tools for low-latency CO₂ growth rate information, provided the systematic biases are minimized using in situ observations. Along with accurate and calibrated NOAA in situ data, satellite-derived growth rates can provide information about the global carbon cycle at sub-annual scales.

12. Carbon and Greenhouse Gas Budgets of Europe: Trends, Interannual and Spatial Variability, and Their Drivers

Authors: Ronny Lauerwald (Université Paris-Saclay INRAE/AgroParisTech UMR EcoSys Palaiseau France); Ana Bastos (Max-Planck-Institute for Biogeochemistry Jena Germany); Matthew J. McGrath (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Ana Maria Roxana Petrescu (Vrije Universiteit Amsterdam); François Ritter (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Robbie M. Andrew (CICERO Center for International Climate Research Oslo Norway); Antoine Berchet (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Grégoire Broquet (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Dominik Brunner (Swiss Federal Laboratories for Materials Science and Technology EMPA Dübendorf Switzerland); Frédéric Chevallier (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Alessandro Cescatti (European Commission); Sara Filipek (Wageningen University); Audrey Fortems-Cheiney (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France, Science Partners Paris France); Giovanni Forzieri (Department of Civil and Environmental Engineering (DICEA) Via di Santa Marta 3 University of Florence Firenze Italy); Pierre Friedlingstein (Laboratoire de Meteorologie Dynamique Departement de Geosciences Institut Pierre-Simon Laplace CNRS-ENS-UPMC-X Ecole Normale Supérieure Paris France, University of Exeter); Richard Fuchs (Karlsruhe Institute of Technology); Christoph Gerbig (Max-Planck-Institute for Biogeochemistry Jena Germany); Sander Houweling (Vrije Universiteit Amsterdam); Piyu Ke (Tsinghua University, University of Exeter); Bas J. W. Lerink (Wageningen University); Wanjing Li (Tsinghua University); Wei Li (Tsinghua University); Xiaojun Li (INRAE UMR 1391 ISPA Université de Bordeaux Villenave d'Ornon France); Ingrid Luijkx (Wageningen University); Guillaume Monteil (Lund University); Saqr Munassar (Max-Planck-Institute for Biogeochemistry Jena Germany); Gert-Jan Nabuurs (Wageningen University); Prabir K. Patra (Research Institute for Global Change JAMSTEC Yokohama Japan, Research Institute for Humanity and Nature Kyoto Japan); Philippe Peylin (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Julia Pongratz (Department of Geography Ludwig-Maximilians-Universität München Munich Germany, Max Planck Institute for Meteorology); Pierre Regnier (Department Geoscience, Environment & Society-BGEOSYS Université Libre de Bruxelles Bruxelles Belgium); Marielle Saunois (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Mart-Jan Schelhaas (Wageningen University); Marko Scholze (Lund University); Stephen Sitch (University of Exeter); Rona L. Thompson (Norwegian Institute for Air Research (NILU)); Hanqin Tian (Department of Earth and Environmental Sciences Center for Earth System Science and Global Sustainability Schiller Institute for Integrated Science and Society Boston College Chestnut Hill MA USA); Aki Tsuruta (Finnish Meteorological Institute); Chris Wilson (University of Leeds); Jean-Pierre Wigneron (INRAE UMR 1391 ISPA Université de Bordeaux Villenave d'Ornon France); Karina Winkler (Karlsruhe Institute of Technology); Yitong Yao (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Sönke Zaehle (Max-Planck-Institute for Biogeochemistry Jena Germany); Philippe Ciais (CEA/CNRS/UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France)

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Open Access: N/A

Abstract: In the framework of the RECCAP2 initiative, we present the greenhouse gas (GHG) and carbon (C) budget of Europe. For the decade of the 2010s, we present a bottom-up (BU) estimate of GHG net-emissions of 3.9 Pg CO₂-eq. yr⁻¹ (using a global warming potential on a 100 years horizon), which are largely dominated by fossil fuel emissions. In this decade, terrestrial ecosystems acted as a net GHG sink of 0.9 Pg CO₂-eq. yr⁻¹, dominated by a CO₂ sink that was partially counterbalanced by net emissions of CH₄ and N₂O. For CH₄ and N₂O, we find good agreement between BU and top-down (TD) estimates from atmospheric inversions. However, our BU land CO₂ sink is significantly higher than the TD estimates. We further show that decadal averages of GHG net-emissions have declined by 1.2 Pg CO₂-eq. yr⁻¹ since the 1990s, mainly due to a reduction in fossil fuel emissions. In addition, based on both data driven BU and TD estimates, we also find that the land CO₂ sink has weakened over the past two decades. A large part of the European CO₂ and C sinks is located in Northern Europe. At the same time, we find a decreasing trend in sink strength in Scandinavia, which can be attributed to an increase in forest management intensity. These are partly offset by increasing CO₂ sinks in parts of Eastern Europe and Northern Spain, attributed in part to land use change. Extensive regions of high CH₄ and N₂O emissions are mainly attributed to agricultural activities and are found in Belgium, the Netherlands and the southern UK. We further analyzed interannual variability in the GHG budgets. The drought year of 2003 shows the highest net-emissions of CO₂ and of all GHGs combined.

13. Potential of ^{14}C -based vs. ΔCO -based ΔffCO_2 observations to estimate urban fossil fuel CO_2 (ffCO_2) emissions

Authors: Fabian Maier; Christian Rödenbeck; Ingeborg Levin; Christoph Gerbig; Maksym Gachkivskyi; Samuel Hammer

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Abstract: . Atmospheric transport inversions are a powerful tool for independently estimating surface CO_2 fluxes from atmospheric CO_2 concentration measurements. However, additional tracers are needed to separate the fossil fuel CO_2 (ffCO_2) emissions from non-fossil CO_2 fluxes. In this study, we focus on radiocarbon (^{14}C), the most direct tracer of ffCO_2 , and the continuously measured surrogate tracer carbon monoxide (CO), which is co-emitted with ffCO_2 during incomplete combustion. In the companion paper by Maier et al. (2024), we determined discrete ^{14}C -based and continuous ΔCO -based estimates of the ffCO_2 excess concentration (ΔffCO_2) compared with a clean-air reference for the urban Heidelberg observation site in southwestern Germany. The ΔCO -based ΔffCO_2 concentration was calculated by dividing the continuously measured ΔCO excess concentration by an average ^{14}C -based $\Delta\text{CO}/\Delta\text{ffCO}_2$ ratio. Here, we use the CarboScope inversion framework adapted for the urban domain around Heidelberg to assess the potential of both types of ΔffCO_2 observations to investigate ffCO_2 emissions and their seasonal cycle. We find that, although they are more precise, ^{14}C -based ΔffCO_2 observations from almost 100 afternoon flask samples collected in the 2 years of 2019 and 2020 are not well suited for estimating robust ffCO_2 emissions in the main footprint of this urban area, which has a very heterogeneous distribution of sources including several point sources. The benefit of the continuous ΔCO -based ΔffCO_2 estimates is that they can be averaged to reduce the impact of individual hours with an inadequate model performance. We show that the weekly averaged ΔCO -based ΔffCO_2 observations allow for a robust reconstruction of the seasonal cycle of the area source ffCO_2 emissions from temporally flat a priori emissions. In particular, the distinct COVID-19 signal - with a steep drop in emissions in spring 2020 - is clearly present in these data-driven a posteriori results. Moreover, our top-down results show a shift in the seasonality of the area source ffCO_2 emissions around Heidelberg in 2019 compared with the bottom-up estimates from the Netherlands Organization for Applied Scientific Research (TNO). This highlights the huge potential of ΔCO -based ΔffCO_2 to validate bottom-up ffCO_2 emissions at urban stations if the $\Delta\text{CO}/\Delta\text{ffCO}_2$ ratios can be determined without biases.

14. A global surface CO₂ flux dataset (2015-2022) inferred from OCO-2 retrievals using the GONGGA inversion system

Authors: Zhe Jin; Xiangjun Tian; Yilong Wang; Hongqin Zhang; Min Zhao; Tao Wang; Jinzhi Ding; Shilong Piao

Journal: Earth System Science Data

Publisher: Copernicus GmbH

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Abstract: . Accurate assessment of the size and distribution of carbon dioxide (CO₂) sources and sinks is important for efforts to understand the carbon cycle and support policy decisions regarding climate mitigation actions. Satellite retrievals of the column-averaged dry-air mole fractions of CO₂ (XCO₂) have been widely used to infer spatial and temporal variations in carbon fluxes through atmospheric inversion techniques. In this study, we present a global spatially resolved terrestrial and ocean carbon flux dataset for 2015-2022. The dataset was generated by the Global ObservationN-based system for monitoring Greenhouse Gases (GONGGA) atmospheric inversion system through the assimilation of Orbiting Carbon Observatory-2 (OCO-2) XCO₂ retrievals. We describe the carbon budget, interannual variability, and seasonal cycle for the global scale and a set of TransCom regions. The 8-year mean net biosphere exchange and ocean carbon fluxes were -2.22 ± 0.75 and -2.32 ± 0.18 Pg C yr⁻¹, absorbing approximately 23 % and 24 % of contemporary fossil fuel CO₂ emissions, respectively. The annual mean global atmospheric CO₂ growth rate was 5.17 ± 0.68 Pg C yr⁻¹, which is consistent with the National Oceanic and Atmospheric Administration (NOAA) measurement (5.24 ± 0.59 Pg C yr⁻¹). Europe has the largest terrestrial sink among the 11 TransCom land regions, followed by Boreal Asia and Temperate Asia. The dataset was evaluated by comparing posterior CO₂ simulations with Total Carbon Column Observing Network (TCCON) retrievals as well as Observation Package (ObsPack) surface flask observations and aircraft observations. Compared with CO₂ simulations using the unoptimized fluxes, the bias and root mean square error (RMSE) in posterior CO₂ simulations were largely reduced across the full range of locations, confirming that the GONGGA system improves the estimates of spatial and temporal variations in carbon fluxes by assimilating OCO-2 XCO₂ data. This dataset will improve the broader understanding of global carbon cycle dynamics and their response to climate change. The dataset can be accessed at <https://doi.org/10.5281/zenodo.8368846> (Jin et al., 2023a).

15. More Frequent Spaceborne Sampling of XCO₂ Improves Detectability of Carbon Cycle Seasonal Transitions in Arctic-Boreal Ecosystems

Authors: Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Gretchen Keppel-Aleks (University of Michigan); Stanley Sander (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Vijay Natraj (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ming Luo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jean-Francois Blavier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Len Dorsky (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ray Nassar (Environment and Climate Change Canada)

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Keywords: carbon cycle, Arctic, imaging spectroscopy, AURORA, Arctic observing mission (and 1 more)

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Abstract: Surface, aircraft, and satellite measurements indicate pervasive early cold season (August-September) CO₂ emissions across Arctic regions, consistent with increased ecosystem metabolism in plants and soils. A key remaining question is whether cold season sources will become large enough to permanently shift the Arctic into a net carbon source. Polar orbiting GHG satellites provide robust estimation of regional carbon budgets but lack sufficient spatial coverage and repeat frequency to track sink-to-source transitions in the early cold season. Mission concepts such as the Arctic Observing Mission (AOM) advocate for flying imaging spectrometers in highly elliptical orbits (HEO) over the Arctic to address sampling limitations. We perform retrieval and flux inversion simulation experiments using the AURORA mission concept, leveraging a Panchromatic imaging Fourier Transform Spectrometer (PanFTS) in HEO. Our simulations demonstrate the potential benefits of increased CO₂ sampling for detecting emissions during the early cold season.

16. Integration of surface-based and space-based atmospheric CO₂ measurements for improving carbon flux estimates using a new developed 3-GAS inversion model

Authors: Shu'an Liu; Xiaofeng Pan; Xiangyun Xiong; Tianle Sun; Lin Xue; Huifang Zhang; Junjun Fang; Jingchun Fang; Guchun Zhang; Hui Xu; Baozhang Chen

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Abstract: N/A

17. Inferred drought-induced plant allocation shifts and their impact on drought legacy at a tropical forest site

Authors: Matthew A. Worden (Department of Earth System Science Stanford University Stanford California USA); Caroline A. Famiglietti (Department of Earth System Science Stanford University Stanford California USA, HydroSat Washington District of Columbia USA); Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Shuang Ma (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Damien Bonal (AgroParisTech, INRAE, UMR Silva Université de Lorraine Nancy France); Clément Stahl (INRAE, UMR EcoFoG, AgroParisTech, CNRS, Cirad Université des Antilles, Université de Guyane Kourou France); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford California USA)

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Abstract: While droughts predominantly induce immediate reductions in plant carbon uptake, they can also exert long-lasting effects on carbon fluxes through associated changes in leaf area, soil carbon, etc. Among other mechanisms, shifts in carbon allocation due to water stress can contribute to the legacy effects of drought on carbon fluxes. However, the magnitude and impact of these allocation shifts on carbon fluxes and pools remain poorly understood. Using data from a wet tropical flux tower site in French Guiana, we demonstrate that drought-induced carbon allocation shifts can be reliably inferred by assimilating Net Biosphere Exchange (NBE) and other observations within the CARbon DATA MOdel fraMework. This model-data fusion system allows inference of optimized carbon and water cycle parameters and states from multiple observational data streams. We then examined how these inferred shifts affected the duration and magnitude of drought's impact on NBE during and after the extreme event. Compared to a static allocation scheme analogous to those typically implemented in land surface models, dynamic allocation reduced average carbon uptake during drought recovery by a factor of 2.8. Additionally, the dynamic model extended the average recovery time by 5 months. The inferred allocation shifts influenced the post-drought period by altering foliage and fine root pools, which in turn modulated gross primary productivity and heterotrophic respiration for up to a decade. These changes can create a bust-boom cycle where carbon uptake is enhanced some years after a drought,

compared to what would have occurred under drought-free conditions. Overall, allocation shifts accounted for 65% [45%-75%] of drought legacy effects in modeled NBE. In summary, drought-induced carbon allocation shifts can play a substantial role in the enduring influence of drought on cumulative land-atmosphere CO₂ exchanges and should be accounted for in ecosystem models.

18. Satellite-detected large CO₂ release in southwestern North America during the 2020-2021 drought and associated wildfires

Authors: Hui Chen; Wei He; Jinxiu Liu; Ngoc Tu Nguyen; Frédéric Chevallier; Hua Yang; Yiming Lv; Chengcheng Huang; Christian Rödenbeck; Scot M Miller; Fei Jiang; Junjie Liu; Matthew S Johnson; Sajeev Philip; Zhiqiang Liu; Ning Zeng; Sourish Basu; David F Baker

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Abstract: Southwestern North America (SWNA) continuously experienced megadroughts and large wildfires in 2020 and 2021. Here, we quantified their impact on the terrestrial carbon budget using net biome production (NBP) estimates from an ensemble of atmospheric inversions assimilating in-situ CO₂ and Carbon Observatory-2 (OCO-2) satellite XCO₂ retrievals (OCO-2 v10 MIP Extension), two satellite-based gross primary production (GPP) datasets, and two fire CO₂ emission datasets. We found that the 2020-2021 drought and associated wildfires in SWNA led to a large CO₂ loss, an ensemble mean of 95.07 TgC estimated by the satellite inversions using both nadir and glint XCO₂ retrievals (LNLG) within the OCO-2 v10 MIP, greater than 80% of SWNA's annual total carbon sink. Moreover, the carbon loss in 2020 was mainly contributed by fire emissions while in 2021 mainly contributed by drought impacts on terrestrial carbon uptake. In addition, the satellite inversions indicated the huge carbon loss was mainly contributed by fire emissions from forests and grasslands along with carbon uptake reductions due to drought impacts on grasslands and shrublands. This study provides a process understanding of how some droughts and following wildfires affect the terrestrial carbon budget on a regional scale.

19. The African Regional Greenhouse Gases Budget (2010-2019)

Authors: Yolandi Ernst (University of the Witwatersrand); Sally Archibald (University of the Witwatersrand); Heiko Balzter (University of Leicester); Frederic Chevallier (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Philippe Ciais (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Carlos Gonzalez Fischer (Department of Global Development College of Agriculture and Life Sciences Cornell University Ithaca NY USA); Benjamin Gaubert (National Center for Atmospheric Research); Thomas Higginbottom (University of Edinburgh); Steven Higgins (Plant Ecology University of Bayreuth Bayreuth Germany); Shakirudeen Lawal (North Carolina State University); Fabrice Lacroix (University of Bern); Ronny Lauerwald (INRAE AgroParisTech UMR ECOSYS Université Paris-Saclay Palaiseau France); Mauro Lourenco (National Geographic Okavango Wilderness Project Wild Bird Trust Johannesburg South Africa, University of the Witwatersrand); Carola Martens (Senckenberg Biodiversity and Climate Research Centre (SBIK-F) Frankfurt am Main Germany, Institute of Physical Geography Goethe University Frankfurt am Main Frankfurt am Main Germany); Anteneh G. Mengistu (Finnish Meteorological Institute); Lutz Merbold (Integrative Agroecology Group Strategic Research Division Agroecology and Environment Agroscope Zurich Switzerland); Edward Mitchard (University of Edinburgh); Mthokozisi Moyo (University of the Witwatersrand); Hannah Nguyen (King's College London); Michael O'Sullivan (University of Exeter); Pedro Rodríguez-Veiga (Sylvera Ltd London UK, University of Leicester); Thais Rosan (University of Exeter); Judith Rosentreter (Faculty of Science and Engineering Southern Cross University Lismore NSW Australia); Casey Ryan (University of Edinburgh); Simon Scheiter (Senckenberg Biodiversity and Climate Research Centre (SBIK-F) Frankfurt am Main Germany); Stephen Sitch (University of Exeter); Nicola Stevens (University of Oxford, University of the Witwatersrand); Torbern Tagesson (Lund University, University of Copenhagen); Hanqin Tian (Center for Earth System Science and Global Sustainability Schiller Institute for Integrated Science and Society Department of Earth and Environmental Sciences Boston College Chestnut Hill MA USA); Mengjia Wang (School of Geoscience and Technology Zhengzhou University Zhengzhou China, INRAE UMR1391 ISPA Université de Bordeaux Villenave d'Ornon France); Joel S. Woon (University of Liverpool); Bo Zheng (Tsinghua University); Yong Zhou (Utah State University); Robert J. Scholes (University of the Witwatersrand)

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Keywords: RECCAP2, sink capacity decreasing, terrestrial fluxes, bottom-up GHG component estimates, atmospheric inversions (and 1 more)

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Abstract: As part of the REgional Carbon Cycle Assessment and Processes Phase 2 (RECCAP2) project, we developed a comprehensive African Greenhouse gases (GHG) budget covering 2000 to 2019 (RECCAP1 and RECCAP2 time periods), and assessed uncertainties and trends over time. We compared bottom-up process-based models, data-driven remotely sensed products, and national GHG inventories with top-down atmospheric inversions, accounting also for lateral fluxes. We incorporated emission estimates derived from novel methodologies for termites, herbivores, and fire, which are particularly important in Africa. We further constrained global woody biomass change products with high-quality regional observations. During the RECCAP2 period, Africa's carbon sink capacity is decreasing, with net ecosystem exchange switching from a small sink of -0.61 ± 0.58 PgC yr⁻¹ in RECCAP1 to a small source in RECCAP2 at 0.16 ($-0.52/1.36$) PgC yr⁻¹. Net CO₂ emissions estimated from bottom-up approaches were 1.6 ($-0.9/5.8$) PgCO₂ yr⁻¹, net CH₄ were 77 ($56.4/93.9$) TgCH₄ yr⁻¹ and net N₂O were 2.9 ($1.4/4.9$) TgN₂O yr⁻¹. Top-down atmospheric inversions showed similar trends. Land Use Change emissions increased, representing one of the largest contributions at 1.7 ($0.8/2.7$) PgCO₂eq yr⁻¹ to the African GHG budget and almost similar to emissions from fossil fuels at 1.74 ($1.53/1.96$) PgCO₂eq yr⁻¹, which also increased from RECCAP1. Additionally, wildfire emissions decreased, while fuelwood burning increased. For most component fluxes, uncertainty is large, highlighting the need for increased efforts to address Africa-specific data gaps. However, for RECCAP2, we improved our overall understanding of many of the important components of the African GHG budget that will assist to inform climate policy and action.

20. Regional Inversion Shows Promise in Capturing Extreme-Event-Driven CO₂ Flux Anomalies but Is Limited by Atmospheric CO₂ Observational Coverage

Authors: B. Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Y. Yin (California Institute of Technology, Now at Department of Environmental Studies New York University New York NY USA); J. Yun (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); G. D. Ferreira (Colorado State University); S. M. Ogle (Colorado State University); L. Baskaran (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); L. He (Department of Global Ecology Carnegie Institution for Science Stanford CA USA); X. Li (Seoul National University); J. Xiao (University of New Hampshire); K. J. Davis (Pennsylvania State University)

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Keywords: carbon cycle, data assimilation, floods, OCO-2, extreme events (and 1 more)

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Open Access: N/A

Abstract: Extreme climate events are becoming more frequent, with poorly understood implications for carbon sequestration by terrestrial ecosystems. A better understanding will critically depend on accurate and precise quantification of ecosystems responses to these events. Taking the 2019 US Midwest floods as a case study, we investigate current capabilities for tracking regional flux anomalies with "top-down" inversion analyses that assimilate atmospheric CO₂ observations. For this analysis, we develop a regionally nested version of the NASA Carbon Monitoring System-Flux system for North America (CMS-Flux-NA) that allows high resolution atmospheric transport (0.5 degrees x 0.625 degrees). Relative to a 2018 baseline, we find the 2019 US Midwest growing season net carbon uptake is reduced by 11-57 TgC (3%-16%, range across assimilated CO₂ data sets). These estimates are found to be consistent with independent "bottom-up" estimates of carbon uptake based on vegetation remote sensing (15-78 TgC). We then investigate current limitations in tracking regional carbon budgets using "top-down" methods. In a set of observing system simulation experiments, we show that the ability of atmospheric CO₂ inversions to capture regional carbon flux anomalies is still limited by observational coverage gaps for both in situ and satellite observations. Future space-based missions that allow for daily observational coverage across North America would largely mitigate these

observational gaps, allowing for improved top-down estimates of ecosystem responses to extreme climate events.

21. Functional analysis of variance (ANOVA) for carbon flux estimates from remote sensing data

Authors: Jonathan Hobbs; Matthias Katzfuss; Hai Nguyen; Vineet Yadav; Junjie Liu

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Abstract: . The constellation of Earth-observing satellites has now produced atmospheric greenhouse gas concentration estimates covering a period of several years. Their global coverage is providing additional information on the global carbon cycle. These products can be combined with complex inversion systems to infer the magnitude of carbon sources and sinks around the globe. Multiple factors, including the atmospheric transport model and satellite product aggregation method, can impact such flux estimates. Analysis of variance (ANOVA) is a well-established statistical framework for estimating common signals while partitioning variability across factors in the analysis of experiments. Functional ANOVA extends this approach with a statistical model that incorporates spatiotemporal correlation for each ANOVA component. The approach is illustrated on inversion experiments with different satellite retrieval aggregation methods and identifies consistent significant patterns in flux increments that span large spatial scales. Functional ANOVA identifies these patterns while accounting for the uncertainty at small spatial scales that is attributed to differences in the aggregation method. Functional ANOVA is also applied to a recent flux model intercomparison project (MIP), and the relative magnitudes of inversion system effects and data source (satellite versus in situ) are similar but exhibit slightly different importance for fluxes over different continents. In all examples, the unexplained residual variability is locally sizable in magnitude but with limited spatial and temporal correlation. These common behaviors across flux inversion experiments demonstrate the diagnostic capability for functional ANOVA to simultaneously distinguish the spatiotemporal coherence of carbon cycle processes and algorithmic factors.

22. Carbon Cycle-Climate Feedbacks in the Post-Paris World

Authors: David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University)

Journal: Annual Review of Earth and Planetary Sciences

Publisher: Annual Reviews

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Abstract: The Paris Agreement calls for emissions reductions to limit climate change, but how will the carbon cycle change if it is successful? The land and oceans currently absorb roughly half of anthropogenic emissions, but this fraction will decline in the future. The amount of carbon that can be released before climate is mitigated depends on the amount of carbon the ocean and terrestrial ecosystems can absorb. Policy is based on model projections, but observations and theory suggest that climate effects emerging in today's climate will increase and carbon cycle tipping points may be crossed. Warming temperatures, drought, and a slowing growth rate of CO₂ itself will reduce land and ocean sinks and create new sources, making carbon sequestration in forests, soils, and other land and aquatic vegetation more difficult. Observations, data-assimilative models, and prediction systems are needed for managing ongoing long-term changes to land and ocean systems after achieving net-zero emissions. ■ International agreements call for stabilizing climate at 1.5 degrees above preindustrial, while the world is already seeing damaging extremes below that. ■ If climate is stabilized near the 1.5 degrees target, the driving force for most sinks will slow, while feedbacks from the warmer climate will continue to cause sources. ■ Once emissions are reduced to net zero, carbon cycle-climate feedbacks will require observations to support ongoing active management to maintain storage.

23. Bedrock regulated climatic controls on the interannual variation of land sink in South-West China karst through soil water availability

Authors: Fusheng Jiao; Xiaojuan Xu; Mingyang Zhang; Haibo Gong; Hao Sheng; Kelin Wang; Huiyu Liu

Journal: CATENA

Publisher: Elsevier BV

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Keywords: N/A

Funding: National Natural Science Foundation of China, Priority Academic Program Development of Jiangsu Higher Education Institutions

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Open Access: N/A

Abstract: N/A

24. Temporal Error Correlations in a Terrestrial Carbon Cycle Model Derived by Comparison to Carbon Dioxide Eddy Covariance Flux Tower Measurements

Authors: Daniel Wesloh (Pennsylvania State University); Klaus Keller (Dartmouth College); Sha Feng (Pacific Northwest National Laboratory, Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement Gif sur Yvette France); Kenneth J. Davis (Pennsylvania State University)

Journal: Journal of Geophysical Research: Biogeosciences

Publisher: American Geophysical Union (AGU)

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Keywords: model-data difference analysis, CO₂ flux error correlation, CO₂ flux error autocorrelation

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Open Access: N/A

Abstract: Atmospheric CO₂ flux inversions require as input an estimate of spatial and temporal correlations of errors in their estimate of the prior mean. Some previous studies have used the differences in CO₂ daily average flux estimates produced by terrestrial carbon cycle models and eddy covariance measurements to constrain the flux error correlations. Since inversions are starting to resolve the daily cycle, we set out to examine the correlations at sub-daily time scales, as well as the correlations across years. To this end, we examine the autocorrelations in the difference between net ecosystem-atmosphere exchange measurements from 75 AmeriFlux towers and temporally downscaled high-spatial-resolution flux estimates from the Carnegie-Ames-Stanford Approach (CASA) terrestrial carbon cycle model. We find that the daily cycle is prominent in these hourly autocorrelations and that these autocorrelations persist across years. We propose a family of functions to model these temporal correlations in atmospheric inversions, and use cross validation to determine which of the correlation functions best fits autocorrelation data from towers not in the training set. Correlation functions with a component that attempts to model the daily cycle in the differences match correlations from other towers better than those without. Those models that reproduce the same correlation structures at 1-year intervals while modulating the amplitudes of the correlations between those intervals improve the fit still further.

25. Satellite-Observed Increase in Aboveground Carbon over Southwest China during 2013-2021

Authors: Lei Fan (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Guanyu Dong (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Frédéric Frappart (INRAE, Bordeaux Sciences Agro, UMR 1391 ISPA, 33140 Villenave-d'Ornon, France., Université de Bordeaux, 33400 Talence, France.); Jean-Pierre Wigneron (INRAE, Bordeaux Sciences Agro, UMR 1391 ISPA, 33140 Villenave-d'Ornon, France., Université de Bordeaux, 33400 Talence, France.); Yueshan Yue (Chinese Academy of Sciences); Xiangming Xiao (University of Oklahoma); Yao Zhang (Peking University); Shengli Tao (Peking University); Lin Cao (Nanjing Forestry University); Yuechen Li (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Mingguo Ma (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Hongqian Fang (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Ling Yu (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Zhanpin Xing (Chinese Academy of Sciences); Xiaojun Li (INRAE, Bordeaux Sciences Agro, UMR 1391 ISPA, 33140 Villenave-d'Ornon, France., Université de Bordeaux, 33400 Talence, France.); Weiyu Shi (Chongqing Engineering Research Center for Remote Sensing Big Data Application, School of Geographical Sciences, Southwest University, Chongqing 400715, China., Chongqing Jinfo Mountain Karst Ecosystem National Observation and Research Station, School of Geographical Sciences, Southwest University, Chongqing 400715, China.); Xiuzhi Chen (Sun Yat-Sen University); Rasmus Fensholt (University of Copenhagen)

Journal: Journal of Remote Sensing

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Abstract: Over the past 4 decades, Southwest China has the fast vegetation growth and aboveground biomass carbon (AGC) accumulation, largely attributed to the active implementation of ecological projects. However, Southwest China has been threatened by frequent extreme drought events recently, potentially countering the expected large AGC increase caused by the ecological projects. Here, we used the L-band vegetation optical depth to quantify the AGC dynamics over Southwest China during the period 2013-2021. Our results showed a net AGC sink of 0.064 [0.057, 0.077] Pg C year⁻¹ (the range represents the maximum and minimum AGC values), suggesting that Southwest China acted as an AGC sink over the study period. Note that the AGC loss of 0.113 [0.101, 0.136] Pg C year⁻¹ was found during 2013-2014, which could mainly be attributed to the negative influence of extreme droughts on AGC changes in Southwest China, particularly in the Yunnan province. For each land use type (i.e., dense forests, persistent forests, nonforests, afforestation, and forestry), the largest AGC stock increase of 0.032 [0.028, 0.036] Pg C year⁻¹ was found in nonforests, owing to their widespread land cover rate over Southwest China. For AGC density (i.e., AGC per unit area), the afforestation areas showed the largest AGC density increase of 0.808 [0.724, 0.985] Mg C ha⁻¹ year⁻¹, reflecting the positive effect of afforestation on AGC increase. Moreover, the karst areas exhibited a higher increasing rate of AGC density than nonkarst areas, suggesting that the karst ecosystems have a high carbon sink capacity over Southwest China.

26. Weaker regional carbon uptake albeit with stronger seasonal amplitude in northern mid-latitudes estimated by higher resolution GEOS-Chem model

Authors: Zhiqiang Liu; Ning Zeng; Yun Liu; Jun Wang; Pengfei Han; Qixiang Cai

Journal: Science of The Total Environment

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Abstract: N/A

27. Neutral Tropical African CO₂ Exchange Estimated From Aircraft and Satellite Observations

Authors: Benjamin Gaubert (National Center for Atmospheric Research); Britton B. Stephens (National Center for Atmospheric Research); David F. Baker (Colorado State University); Sourish Basu (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Michael Bertolacci (School of Mathematics and Applied Statistics University of Wollongong Wollongong Australia); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rebecca Buchholz (National Center for Atmospheric Research); Abhishek Chatterjee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Frédéric Chevallier (Laboratoire des Sciences du Climat et de L'Environnement Institut Pierre-Simon Laplace CEA-CNRS-UVSQ CEDEX France); Róisín Commane (Columbia University); Noel Cressie (School of Mathematics and Applied Statistics University of Wollongong Wollongong Australia, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Feng Deng (University of Toronto); Nicole Jacobs (Colorado State University); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); Shamil S. Maksyutov (National Institute for Environmental Studies); Kathryn McKain (NOAA Global Monitoring Laboratory Boulder CO USA, University of Colorado Boulder); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhiqiang Liu (Chinese Academy of Sciences); Eric Morgan (Scripps Institution of Oceanography / University of California San Diego); Chris O'Dell (Colorado State University); Sajeew Philip (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Eric Ray (NOAA Chemical Sciences Laboratory Boulder CO USA); David Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Andrew Schuh (Colorado State University); Thomas E. Taylor (Colorado State University); Brad Weir (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Dave van Wees (BeZero Carbon Ltd Gorsuch Place Senna Building, E2 8JF London UK, Vrije Universiteit Amsterdam); Steven C. Wofsy (Harvard University); Andrew Zammit-Mangion (School of Mathematics and Applied Statistics University of Wollongong Wollongong Australia); Ning Zeng (University of Maryland)

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Abstract: Tropical lands play an important role in the global carbon cycle yet their contribution remains uncertain owing to sparse observations. Satellite observations of atmospheric carbon dioxide (CO₂) have greatly increased spatial coverage over tropical regions, providing the potential for improved estimates of terrestrial fluxes. Despite this advancement, the spread among satellite-based and in-situ atmospheric CO₂ flux inversions over northern tropical Africa (NTA), spanning 0-24 degreesN, remains large. Satellite-based estimates of an annual source of 0.8-1.45 PgC yr⁻¹ challenge our understanding of tropical and global carbon cycling. Here, we compare posterior mole fractions from the suite of inversions participating in the Orbiting Carbon Observatory 2 (OCO-2) Version 10 Model Intercomparison Project (v10 MIP) with independent in-situ airborne observations made over the tropical Atlantic Ocean by the National Aeronautics and Space Administration (NASA) Atmospheric Tomography (ATom) mission during four seasons. We develop emergent constraints on tropical African CO₂ fluxes using flux-concentration relationships defined by the model suite. We find an annual flux of 0.14 +/- 0.39 PgC yr⁻¹ (mean and standard deviation) for NTA, 2016-2018. The satellite-based flux bias suggests a potential positive concentration bias in OCO-2 B10 and earlier version retrievals over land in NTA during the dry season. Nevertheless, the OCO-2 observations provide improved flux estimates relative to the in situ observing network at other times of year, indicating stronger uptake in NTA during the wet season than the in-situ inversion estimates.

28. Water Stress Dominates 21st-Century Tropical Land Carbon Uptake

Authors: Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Victoria Meyer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA); Marcos Longo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Lawrence Berkeley National Laboratory)

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Keywords: data assimilation, carbon cycle, water stress

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Abstract: Water stress regulates land-atmosphere carbon dioxide (CO₂) exchanges in the tropics; however, its role remains poorly characterized due to the confounding roles of radiation, temperature and canopy dynamics. In particular, uncertainty stems from the relative roles of plant-available water (supply) and atmospheric water vapor deficit (demand) as mechanistic drivers of photosynthetic carbon (C) uptake variability. Using satellite measurements of gravity, CO₂ and fluorescence to constrain a mechanistic carbon-water cycle model from 2001 to 2018, we found that the interannual variability (IAV) of water stress on photosynthetic C uptake was 52% greater than the combined effects of other factors. Surprisingly, the dominance of water stress on C uptake IAV was greater in the wet tropics (94%) than in the dry tropics (26%). Plant-available water supply and atmospheric demand both contributed to the IAV of water stress on photosynthetic C uptake across the tropics, but the IAV of demand effects was 21% greater than the IAV of supply effects (33% greater in the wet tropics and 6% greater in the dry tropics). We found that the IAV of water stress on C uptake was 24% greater than the IAV of the combination of other factors in the net land-atmosphere C sink in the whole tropics, 26% greater in the wet tropics, and 7% greater in the dry tropics. Given the recent trends in tropical

precipitation and atmospheric humidity, our findings indicate that water stress----from both supply and demand----will likely dominate the climate response of land C sink across tropical ecosystems in the coming decades.

29. A Comprehensive Assessment of Anthropogenic and Natural Sources and Sinks of Australasia's Carbon Budget

Authors: Yohanna Villalobos (CSIRO Environment Canberra ACT Australia); Josep G. Canadell (CSIRO Environment Canberra ACT Australia); Elizabeth D. Keller (Victoria University of Wellington); Peter R. Briggs (CSIRO Environment Canberra ACT Australia); Beata Bukosa (NIWA National Institute of Water and Atmospheric Research Wellington New Zealand); Donna L. Giltrap (Manaaki Whenua--Landcare Research Palmerston North New Zealand); Ian Harman (CSIRO Environment Canberra ACT Australia); Timothy W. Hilton (GNS Science Lower Hutt New Zealand); Miko U. F. Kirschbaum (Manaaki Whenua--Landcare Research Palmerston North New Zealand); Ronny Lauerwald (Université Paris-Saclay INRAe AgroParisTech UMR EcoSys Palaiseau France); Liyin L. Liang (Manaaki Whenua--Landcare Research Palmerston North New Zealand); Taylor Maavara (University of Leeds); Sara E. Mikaloff-Fletcher (NIWA National Institute of Water and Atmospheric Research Wellington New Zealand); Peter J. Rayner (University of Melbourne); Laure Resplandy (Princeton University); Judith Rosentreter (Faculty of Science and Engineering Southern Cross University Lismore NSW Australia, Yale Institute for Biospheric Studies Yale University New Haven CT USA); Eva-Marie Metz (Heidelberg University); Oscar Serrano (Centre d'Estudis Avançats de Blanes Consejo Superior de Investigaciones Científicas Blanes Spain, School of Science and Centre for Marine Ecosystems Research Edith Cowan University Joondalup WA Australia); Benjamin Smith (Hawkesbury Institute for the Environment Western Sydney University Richmond NSW Australia)

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Abstract: Regional carbon budget assessments attribute and track changes in carbon sources and sinks and support the development and monitoring the efficacy of climate policies. We present a comprehensive assessment of the natural and anthropogenic carbon (C-CO₂) fluxes for Australasia as a whole, as well as for Australia and New Zealand individually, for the period from 2010 to 2019, using two approaches: bottom-up methods that integrate flux estimates from land-surface models, data-driven models, and inventory estimates; and top-down atmospheric inversions based on satellite and in situ measurements. Our bottom-up decadal assessment suggests that Australasia's net carbon balance was close to carbon neutral (-0.4 +/- 77.0 TgC yr⁻¹). However, substantial uncertainties remain in this estimate, primarily driven by the large spread between our regional terrestrial biosphere simulations and predictions from global ecosystem models. Within Australasia, Australia was a net

source of 38.2 ± 75.8 TgC yr⁻¹, and New Zealand was a net CO₂ sink of -38.6 ± 13.4 TgC yr⁻¹. The top-down approach using atmospheric CO₂ inversions indicates that fluxes derived from the latest satellite retrievals are consistent within the range of uncertainties with Australia's bottom-up budget. For New Zealand, the best agreement was found with a national scale flux inversion estimate based on in situ measurements, which provide better constrained of fluxes than satellite flux inversions. This study marks an important step toward a more comprehensive understanding of the net CO₂ balance in both countries, facilitating the improvement of carbon accounting approaches and strategies to reduce emissions.

30. An Assessment of CO₂ Uptake in the Arctic Ocean From 1985 to 2018

Authors: Sayaka Yasunaka (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan, Tohoku University); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Jens Terhaar (University of Bern, Woods Hole Oceanographic Institution); Are Olsen (University of Bergen, Bjerknes Centre for Climate Research); Ryohei Yamaguchi (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); Peter Landschützer (Flanders Marine Institute (VLIZ) Ostend Belgium, Max Planck Institute for Meteorology); Eiji Watanabe (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); Dustin Carroll (San José State University); Hanani Adiwira (Tohoku University); Jens Daniel Müller (ETH Zurich); Judith Hauck (Alfred Wegener Institute)

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Cited by: 17 publications

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Keywords: CO₂ flux, sea ice, RECCAP2, pCO₂

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Open Access: N/A

Abstract: As a contribution to the Regional Carbon Cycle Assessment and Processes phase 2 (RECCAP2) project, we present synthesized estimates of Arctic Ocean sea-air CO₂ fluxes and their uncertainties from surface ocean pCO₂-observation products, ocean biogeochemical hindcast and data assimilation models, and atmospheric inversions. For the period of 1985-2018, the Arctic Ocean was a net sink of CO₂ of 116 +/- 4 TgC yr⁻¹ in the pCO₂ products, 92 +/- 30 TgC yr⁻¹ in the models, and 91 +/- 21 TgC yr⁻¹ in the atmospheric inversions. The CO₂ uptake peaks in late summer and early autumn, and is low in winter when sea ice inhibits sea-air fluxes. The long-term mean CO₂ uptake in the Arctic Ocean is primarily caused by steady-state fluxes of natural carbon (70% +/- 15%), and enhanced by the atmospheric CO₂ increase (19% +/- 5%) and climate change (11% +/- 18%). The annual mean CO₂ uptake increased from 1985 to 2018 at a rate of 31 +/- 13 TgC yr⁻¹ dec⁻¹ in the pCO₂ products, 10 +/- 4 TgC yr⁻¹ dec⁻¹ in the models, and 32 +/- 16 TgC yr⁻¹ dec⁻¹ in the atmospheric inversions. Moreover, 77% +/- 38% of the trend in the net CO₂ uptake over time is caused by climate change, primarily due to rapid sea ice loss in recent years. Furthermore, true uncertainties may be larger than the given ensemble standard deviations due to common structural biases across all individual estimates.

31. The Southern Ocean Carbon Cycle 1985-2018: Mean, Seasonal Cycle, Trends, and Storage

Authors: Judith Hauck (Alfred Wegener Institute); Luke Gregor (ETH Zurich); Cara Nissen (University of Colorado Boulder, Alfred Wegener Institute); Lavinia Patara (GEOMAR Helmholtz Centre for Ocean Research); Mark Hague (ETH Zurich); Precious Mongwe (Southern Ocean Carbon-Climate Observatory CSIR Pretoria South Africa); Seth Bushinsky (University of Hawai'i Mānoa Honolulu HI USA); Scott C. Doney (University of Virginia); Nicolas Gruber (ETH Zurich); Corinne Le Quéré (University of East Anglia); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Matthew Mazloff (Scripps Institution of Oceanography / University of California San Diego); Pedro M. S. Monteiro (School for Climate Studies Stellenbosch University Stellenbosch South Africa, Southern Ocean Carbon-Climate Observatory CSIR Pretoria South Africa); Jens Terhaar (University of Bern, Woods Hole Oceanographic Institution)

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References: 210

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Language: en

Subjects: N/A

Keywords: carbon sink, carbon cycle, polar oceans, seasonality, CO₂ uptake (and 1 more)

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Abstract: We assess the Southern Ocean CO₂ uptake (1985-2018) using data sets gathered in the Regional Carbon Cycle Assessment and Processes Project Phase 2. The Southern Ocean acted as a sink for CO₂ with close agreement between simulation results from global ocean biogeochemistry models (GOBMs, 0.75 +/- 0.28 PgC yr⁻¹) and pCO₂-observation-based products (0.73 +/- 0.07 PgC yr⁻¹). This sink is only half that reported by RECCAP1 for the same region and timeframe. The present-day net uptake is to first order a response to rising atmospheric CO₂, driving large amounts of anthropogenic CO₂ (C_{ant}) into the ocean, thereby overcompensating the loss of natural CO₂ to the atmosphere. An apparent knowledge gap is the increase of the sink since 2000, with pCO₂-products suggesting a growth that is more than twice as strong and uncertain as that of GOBMs (0.26 +/- 0.06 and 0.11 +/- 0.03 Pg C yr⁻¹ decade⁻¹, respectively). This is despite nearly identical pCO₂ trends in GOBMs and pCO₂-products when both products are compared only at the locations where pCO₂ was measured. Seasonal analyses revealed agreement in driving processes in winter with uncertainty in the magnitude of outgassing, whereas discrepancies are more fundamental in summer, when GOBMs exhibit difficulties in simulating the effects of the non-thermal processes of biology and mixing/circulation. Ocean interior accumulation of C_{ant} points to an underestimate of C_{ant} uptake and storage in GOBMs. Future work needs to link surface fluxes and interior ocean transport, build long overdue systematic observation networks and push toward better process understanding of drivers of

the carbon cycle.

32. A simplified non-linear chemistry transport model for analyzing NO₂ column observations: STILT-NO_x

Authors: Dien Wu; Joshua L. Laughner; Junjie Liu; Paul I. Palmer; John C. Lin; Paul O. Wennberg

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Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . Satellites monitoring air pollutants (e.g., nitrogen oxides; NO_x = NO + NO₂) or greenhouse gases (GHGs) are widely utilized to understand the spatiotemporal variability in and evolution of emission characteristics, chemical transformations, and atmospheric transport over anthropogenic hotspots. Recently, the joint use of space-based long-lived GHGs (e.g., carbon dioxide; CO₂) and short-lived pollutants has made it possible to improve our understanding of emission characteristics. Some previous studies, however, lack consideration of the non-linear NO_x chemistry or complex atmospheric transport. Considering the increase in satellite data volume and the demand for emission monitoring at higher spatiotemporal scales, it is crucial to construct a local-scale emission optimization system that can handle both long-lived GHGs and short-lived pollutants in a coupled and effective manner. This need motivates us to develop a Lagrangian chemical transport model that accounts for NO_x chemistry and fine-scale atmospheric transport (STILT-NO_x) and to investigate how physical and chemical processes, anthropogenic emissions, and background may affect the interpretation of tropospheric NO₂ columns (tNO₂). Interpreting emission signals from tNO₂ commonly involves either an efficient statistical model or a sophisticated chemical transport model. To balance computational expenses and chemical complexity, we describe a simplified representation of the NO_x chemistry that bypasses an explicit solution of individual chemical reactions while preserving the essential non-linearity that links NO_x emissions to its concentrations. This NO_x chemical parameterization is then incorporated into an existing Lagrangian modeling framework that is widely applied in the GHG community. We further quantify uncertainties associated with the wind field and chemical parameterization and evaluate modeled columns against retrieved columns from the Tropospheric Monitoring Instrument (TROPOMI v2.1). Specifically, simulations with alternative model configurations of emissions, meteorology, chemistry, and inter-parcel mixing are carried out over three United States (US) power plants and two urban areas across seasons. Using the U.S. Environmental Protection Agency (EPA)-reported emissions for power plants with non-linear NO_x chemistry improves the model-data alignment in tNO₂ (a high bias of $\leq 10\%$ on an annual basis), compared to simulations using either the Emissions Database for Global Atmospheric Research (EDGAR) model or without

chemistry (bias approaching 100 %). The largest model-data mismatches are associated with substantial biases in wind directions or conditions of slower atmospheric mixing and photochemistry. More importantly, our model development illustrates (1) how NO_x chemistry affects the relationship between NO_x and CO₂ in terms of the spatial and seasonal variability and (2) how assimilating tNO₂ can quantify systematic biases in modeled wind directions and emission distribution in prior inventories of NO_x and CO₂, which laid a foundation for a local-scale multi-tracer emission optimization system.

33. Assumptions about prior fossil fuel inventories impact our ability to estimate posterior net CO₂ fluxes that are needed for verifying national inventories

Authors: Tomohiro Oda; Liang Feng; Paul I Palmer; David F Baker; Lesley E Ott

Journal: Environmental Research Letters

Publisher: IOP Publishing

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Keywords: N/A

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Abstract: Monitoring national and global greenhouse gas (GHG) emissions is a critical component of the Paris Agreement, necessary to verify collective activities to reduce GHG emissions. Top-down approaches to infer GHG emission estimates from atmospheric data are widely recognized as a useful tool to independently verify emission inventories reported by individual countries under the United Nation Framework Convention on Climate Change. Conventional top-down atmospheric inversion methods often prescribe fossil fuel CO₂ emissions (FFCO₂) and fit the resulting model values to atmospheric CO₂ observations by adjusting natural terrestrial and ocean flux estimates. This approach implicitly assumes that we have perfect knowledge of FFCO₂ and that any gap in our understanding of atmospheric CO₂ data can be explained by natural fluxes; consequently, it also limits our ability to quantify non-FFCO₂ emissions. Using two independent FFCO₂ emission inventories, we show that differences in sub-annual emission distributions are aliased to the corresponding posterior natural flux estimates. Over China, for example, where the two inventories show significantly different seasonal variations in FFCO₂, the resulting differences in national-scale flux estimates are small but are significant on the subnational scale. We compare natural CO₂ flux estimates inferred from in-situ and satellite observations. We find that sparsely distributed in-situ observations are best suited for quantifying natural fluxes and large-scale carbon budgets and less suitable for quantifying FFCO₂ errors. Satellite data provide us with the best opportunity to quantify FFCO₂ emission errors; a similar result is achievable using dense, regional in-situ measurement networks. Enhancing the top-down flux estimation capability for inventory verification requires a coordinated activity to (a) improve GHG inventories; (b) extend methods that take full advantage of measurements of trace gases that are co-emitted during combustion; and (c) improve atmospheric transport models.

34. A Gridded Inventory of Annual 2012-2018 U.S. Anthropogenic Methane Emissions

Authors: Joannes D. Maasakkers (SRON Netherlands Institute for Space Research, Leiden 3584 CA, Netherlands); Erin E. McDuffie (Climate Change Division, Environmental Protection Agency, Washington, District of Columbia 20004, United States); Melissa P. Sulprizio (Harvard University); Candice Chen (Harvard University, SRON Netherlands Institute for Space Research, Leiden 3584 CA, Netherlands); Maggie Schultz (SRON Netherlands Institute for Space Research, Leiden 3584 CA, Netherlands); Lily Brunelle (SRON Netherlands Institute for Space Research, Leiden 3584 CA, Netherlands); Ryan Thrush (SRON Netherlands Institute for Space Research, Leiden 3584 CA, Netherlands); John Steller (Climate Change Division, Environmental Protection Agency, Washington, District of Columbia 20004, United States); Christopher Sherry (Climate Change Division, Environmental Protection Agency, Washington, District of Columbia 20004, United States); Daniel J. Jacob (Harvard University); Seongeun Jeong (Lawrence Berkeley National Laboratory); Bill Irving (Climate Change Division, Environmental Protection Agency, Washington, District of Columbia 20004, United States); Melissa Weitz (Climate Change Division, Environmental Protection Agency, Washington, District of Columbia 20004, United States)

Journal: Environmental Science & Technology

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URL: <https://doi.org/10.1021/acs.est.3c05138>

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Keywords: N/A

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Open Access: N/A

Abstract: N/A

35. The consolidated European synthesis of CO₂ emissions and removals for the European Union and United Kingdom: 1990-2020

Authors: Matthew J. McGrath; Ana Maria Roxana Petrescu; Philippe Peylin; Robbie M. Andrew; Bradley Matthews; Frank Dentener; Juraj Balkovič; Vladislav Bastrikov; Meike Becker; Gregoire Broquet; Philippe Ciais; Audrey Fortems-Cheiney; Raphael Ganzenmüller; Giacomo Grassi; Ian Harris; Matthew Jones; Jürgen Knauer; Matthias Kuhnert; Guillaume Monteil; Saqr Munassar; Paul I. Palmer; Glen P. Peters; Chunjing Qiu; Mart-Jan Schelhaas; Oksana Tarasova; Matteo Vizzarri; Karina Winkler; Gianpaolo Balsamo; Antoine Berchet; Peter Briggs; Patrick Brockmann; Frédéric Chevallier; Giulia Conchedda; Monica Crippa; Stijn N. C. Dellaert; Hugo A. C. Denier van der Gon; Sara Filipek; Pierre Friedlingstein; Richard Fuchs; Michael Gauss; Christoph Gerbig; Diego Guizzardi; Dirk Günther; Richard A. Houghton; Greet Janssens-Maenhout; Ronny Lauerwald; Bas Lerink; Ingrid T. Luijkx; Géraud Moulas; Marilena Muntean; Gert-Jan Nabuurs; Aurélie Paquirissamy; Lucia Perugini; Wouter Peters; Roberto Pilli; Julia Pongratz; Pierre Regnier; Marko Scholze; Yusuf Serengil; Pete Smith; Efisio Solazzo; Rona L. Thompson; Francesco N. Tubiello; Timo Vesala; Sophia Walther

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-15-4295-2023

URL: <https://doi.org/10.5194/essd-15-4295-2023>

Publication Date: 2023-10-05

Volume/Issue: Volume 15, Issue 10

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020 Framework Programme, H2020 European Research Council, Horizon 2020

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Open Access: N/A

Abstract: . Quantification of land surface-atmosphere fluxes of carbon dioxide(CO₂) and their trends and uncertainties is essential for monitoring progress of the EU27+UK bloc as it strives to meet ambitious targets determined by both international agreements and internal regulation. This study provides a consolidated synthesis of fossil sources (CO₂ fossil) and natural (including formally managed ecosystems) sources and sinks over land (CO₂ land) using bottom-up (BU) and top-down (TD) approaches for the European Union and United Kingdom (EU27+UK), updating earlier syntheses (Petrescu et al., 2020, 2021). Given the wide scope of the work and the variety of approaches involved, this study aims to answer essential questions identified in the previous syntheses and understand the differences between datasets, particularly for poorly characterized fluxes from managed and unmanaged ecosystems. The work integrates updated emission inventory data, process-based model results, data-driven categorical model results, and inverse modeling estimates, extending the previous period 1990-2018 to the year 2020 to the extent possible. BU and TD products are compared with the European national greenhouse gas inventory (NGHGI) reported by parties including the year 2019 under the United Nations Framework Convention on Climate Change (UNFCCC). The uncertainties of the EU27+UK NGHGI were evaluated using the standard deviation reported by the EU member states

following the guidelines of the Intergovernmental Panel on Climate Change (IPCC) and harmonized by gap-filling procedures. Variation in estimates produced with other methods, such as atmospheric inversion models (TD) or spatially disaggregated inventory datasets (BU), originate from within-model uncertainty related to parameterization as well as structural differences between models. By comparing the NGHGI with other approaches, key sources of differences between estimates arise primarily in activities. System boundaries and emission categories create differences in CO₂ fossil datasets, while different land use definitions for reporting emissions from land use, land use change, and forestry (LULUCF) activities result in differences for CO₂ land. The latter has important consequences for atmospheric inversions, leading to inversions reporting stronger sinks in vegetation and soils than are reported by the NGHGI. For CO₂ fossil emissions, after harmonizing estimates based on common activities and selecting the most recent year available for all datasets, the UNFCCC NGHGI for the EU27+UK accounts for 926 $\pm$ 13 Tg C yr⁻¹, while eight other BU sources report a mean value of 948 [937, 961] Tg C yr⁻¹ (25th, 75th percentiles). The sole top-down inversion of fossil emissions currently available accounts for 875 Tg C in this same year, a value outside the uncertainty of both the NGHGI and bottom-up ensemble estimates and for which uncertainty estimates are not currently available. For the net CO₂ land fluxes, during the most recent 5-year period including the NGHGI estimates, the NGHGI accounted for -91 $\pm$ 32 Tg C yr⁻¹, while six other BU approaches reported a mean sink of -62 [-117, -49] Tg C yr⁻¹, and a 15-member ensemble of dynamic global vegetation models (DGVMs) reported -69 [-152, -5] Tg C yr⁻¹. The 5-year mean of three TD regional ensembles combined with one non-ensemble inversion of -73 Tg C yr⁻¹ has a slightly smaller spread (0th-100th percentiles of [-135, +45] Tg C yr⁻¹), and it was calculated after removing net land-atmosphere CO₂ fluxes caused by lateral transport of carbon (crop trade, wood trade, river transport, and net uptake from inland water bodies), resulting in increased agreement with the NGHGI and bottom-up approaches. Results at the category level (Forest Land, Cropland, Grassland) generally show good agreement between the NGHGI and category-specific models, but results for DGVMs are mixed. Overall, for both CO₂ fossil and net CO₂ land fluxes, we find that current independent approaches are consistent with the NGHGI at the scale of the EU27+UK. We conclude that CO₂ emissions from fossil sources have decreased over the past 30 years in the EU27+UK, while land fluxes are relatively stable: positive or negative trends larger (smaller) than 0.07 (-0.61) Tg C yr⁻² can be ruled out for the NGHGI. In addition, a gap on the order of 1000 Tg C yr⁻¹ between CO₂ fossil emissions and net CO₂ uptake by the land exists regardless of the type of approach (NGHGI, TD, BU), falling well outside all available estimates of uncertainties. However, uncertainties in top-down approaches to estimate CO₂ fossil emissions remain uncharacterized and are likely substantial, in addition to known uncertainties in top-down estimates of the land fluxes. The data used to plot the figures are available at <https://doi.org/10.5281/zenodo.8148461> (McGrath et al., 2023).

36. Resolving the Carbon-Climate Feedback Potential of Wetland CO₂ and CH₄ Fluxes in Alaska

Authors: Shuang Ma (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jennifer D. Watts (Woodwell Climate Research Center Falmouth MA USA); Gregory R. Quetin (University of California Santa Barbara); Zona Donatella (Department of Biology San Diego State University San Diego CA USA); Eugénie S. Euskirchen (University of Alaska Fairbanks); Alexander J. Norton (Australian National University); Yi Yin (California Institute of Technology); Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Renato K. Braghiere (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: CO₂, climatic sensitivities, global warming potential, northern high latitude, carbon-climate feedback (and 1 more)

Funding: Earth Sciences Division, Gordon and Betty Moore Foundation, Division of Environmental Biology (and 1 more)

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Boreal-Arctic regions are key stores of organic carbon (C) and play a major role in the greenhouse gas balance of high-latitude ecosystems. The carbon-climate (C-climate) feedback potential of northern high-latitude ecosystems remains poorly understood due to uncertainty in temperature and precipitation controls on carbon dioxide (CO₂) uptake and the decomposition of soil C into CO₂ and methane (CH₄) fluxes. While CH₄ fluxes account for a smaller component of the C balance, the climatic impact of CH₄ outweighs CO₂ (28-34 times larger global warming potential on a 100-year scale), highlighting the need to jointly resolve the climatic sensitivities of both CO₂ and CH₄. Here, we jointly constrain a terrestrial biosphere model with in situ CO₂ and CH₄ flux observations at seven eddy covariance sites using a data-model integration approach to resolve the integrated

environmental controls on land-atmosphere CO₂ and CH₄ exchanges in Alaska. Based on the combined CO₂ and CH₄ flux responses to climate variables, we find that 1970-present climate trends will induce positive C-climate feedback at all tundra sites, and negative C-climate feedback at the boreal and shrub fen sites. The positive C-climate feedback at the tundra sites is predominantly driven by increased CH₄ emissions while the negative C-climate feedback at the boreal site is predominantly driven by increased CO₂ uptake (80% from decreased heterotrophic respiration, and 20% from increased photosynthesis). Our study demonstrates the need for joint observational constraints on CO₂ and CH₄ biogeochemical processes--and their associated climatic sensitivities--for resolving the sign and magnitude of high-latitude ecosystem C-climate feedback in the coming decades.

37. Ongoing CO2 monitoring verify CO2 emissions and sinks in China during 2018-2021

Authors: Junting Zhong; Xiaoye Zhang; Lifeng Guo; Deying Wang; Changhong Miao; Xiliang Zhang

Journal: Science Bulletin

Publisher: Elsevier BV

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Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Chinese Academy of Meteorological Sciences

License: N/A

Open Access: N/A

Abstract: N/A

38. Verifying Methane Inventories and Trends With Atmospheric Methane Data

Authors: John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Sudhanshu Pandey (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Yuzhong Zhang (Harvard University, Key Laboratory of Coastal Environment and Resources of Zhejiang Province School of Engineering Westlake University Hangzhou China); Daniel H. Cusworth (University of Arizona); Zhen Qu (Harvard University); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Shuang Ma (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Joannes D. Maasakkers (SRON Netherlands Institute for Space Research Leiden The Netherlands); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Riley Duren (University of Arizona); David Crisp (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Deborah Gordon (Rocky Mountain Institute Boulder CO USA); Daniel J. Jacob (Harvard University)

Journal: AGU Advances

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2023av000871>

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Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: methane, verifying emissions, global methane pledge, satellite estimates

Funding: N/A

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Open Access: N/A

Abstract: The 2015 Paris Climate Agreement and Global Methane Pledge formalized agreement for countries to report and reduce methane emissions to mitigate near-term climate change. Emission inventories generated through surface activity measurements are reported annually or bi-annually, and evaluated periodically through a "Global Stocktake." Emissions inverted from atmospheric data support evaluation of reported inventories, but their systematic use is stifled by spatially variable biases from prior errors combined with limited sensitivity of observations to emissions (also called smoothing error), as-well-as poorly characterized information content. Here, we demonstrate a Bayesian, optimal estimation (OE) algorithm for evaluating a state-of-the-art inventory (EDGAR v6.0) using satellite-based emissions from 2009 to 2018. The OE algorithm quantifies the information content (uncertainty reduction, sectoral attribution, spatial resolution) of the satellite-based emissions and disentangles the effect of smoothing error when comparing to an inventory. We find robust differences between satellite and EDGAR for total livestock, rice, and coal emissions: 14 ± 9 , 12 ± 8 , -11 ± 6 Tg CH₄/yr respectively. EDGAR and satellite agree that livestock emissions are increasing (0.25-1.3 Tg CH₄/yr/yr), primarily in the Indo-Pakistan region, sub-tropical Africa, and the Southern Brazilian; East Asia rice emissions are also increasing, highlighting the importance of agriculture on the atmospheric methane growth rate. In contrast, low information content for the waste and fossil emission trends

confounds comparison between EDGAR and satellite; increased sampling and spatial resolution of satellite observations are therefore needed to evaluate reported changes to emissions in these sectors.

39. Response to Anonymous Referee #1

Authors: Matthew McGrath

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/essd-2022-412-ac3

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Language: N/A

Subjects: N/A

Keywords: Scope (computer science)

Funding: N/A

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Open Access: N/A

Abstract: N/A

40. Response to John Miller

Authors: Matthew McGrath

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/essd-2022-412-ac4

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Language: N/A

Subjects: N/A

Keywords: Scope (computer science)|Miller

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

41. Improved Constraints on the Recent Terrestrial Carbon Sink Over China by Assimilating OCO-2 XCO₂ Retrievals

Authors: Wei He (Nanjing University, Ministry of Natural Resources); Fei Jiang (Nanjing University, Ministry of Natural Resources); Weimin Ju (Nanjing University, Ministry of Natural Resources); Frédéric Chevallier (Laboratoire des Sciences du Climat et de L'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); David F. Baker (Colorado State University); Jun Wang (Nanjing University, Ministry of Natural Resources); Mousong Wu (Nanjing University, Ministry of Natural Resources); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); Sajeev Philip (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Hengmao Wang (Nanjing University, Ministry of Natural Resources); Michael Bertolacci (School of Mathematics and Applied Statistics University of Wollongong Wollongong NSW Australia); Zhiqiang Liu (Chinese Academy of Sciences); Ning Zeng (Chinese Academy of Sciences, Earth System Science Interdisciplinary Center College Park MD USA); Jing M. Chen (School of Geographical Sciences Fujian Normal University Fuzhou China, University of Toronto)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jd037773

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Language: en

Subjects: N/A

Keywords: atmospheric inversion, China, terrestrial carbon sink, Orbit Carbon Observatory 2 (OCO-2), atmospheric CO₂ column concentration

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: The magnitude and distribution of China's terrestrial carbon sink remain uncertain due to insufficient observational constraints; satellite column-average dry-air mole fraction carbon dioxide (XCO₂) retrievals may fill some of this gap. Here, we estimate China's carbon sink using atmospheric inversions of the Orbiting Carbon Observatory 2 (OCO-2) XCO₂ retrievals within different platforms, including the Global Carbon Assimilation System (GCAS) v2, the Copernicus Atmosphere Monitoring Service, and the OCO-2 Model Inter-comparison Project (MIP). We find that they consistently place the largest net biome production (NBP) in the south on an annual basis compared to the northeast and other main agricultural areas during peak growing season, coinciding well with the distribution of forests and crops, respectively. Moreover, the mean seasonal cycle amplitude of NBP in OCO-2 inversions is obviously larger than that of biosphere model simulations and slightly greater than surface CO₂ inversions. More importantly, the mean seasonal cycle of the OCO-2 inversions is well constrained in the temperate, tropical, and subtropical monsoon climate zones, with better inter-model consistency at a sub-regional scale compared to in situ inversions and biosphere model simulations. In addition, the

OCO-2 inversions estimate the mean annual NBP in China for 2015-2019 to be between 0.34 (GCASv2) and 0.47 ± 0.16 PgC/yr (median $\pm$ std; OCO-2 v10 MIP), and indicate the impacts of climate extremes (e.g., the 2019 drought) on the interannual variations of NBP. Our results suggest that assimilating OCO-2 XCO₂ retrievals is crucial for improving our understanding of China's terrestrial carbon sink regime.

42. The carbon sink in China as seen from GOSAT with a regional inversion system based on the Community Multi-scale Air Quality (CMAQ) and ensemble Kalman smoother (EnKS)

Authors: Xingxia Kou; Zhen Peng; Meigen Zhang; Fei Hu; Xiao Han; Ziming Li; Lili Lei

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-23-6719-2023

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Subjects: N/A

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Open Access: N/A

Abstract: . Top-down inversions of China's terrestrial carbon sink are known to be uncertain because of errors related to the relatively coarse resolution of global transport models and the sparseness of in situ observations. Taking advantage of regional chemistry transport models for mesoscale simulation and spaceborne sensors for spatial coverage, the Greenhouse Gases Observing Satellite (GOSAT) retrievals of column-mean dry mole fraction of carbon dioxide (XCO₂) were introduced in the Models-3 (a flexible software framework) Community Multi-scale Air Quality (CMAQ) and ensemble Kalman smoother (EnKS)-based regional inversion system to constrain China's biosphere sink at a spatiotemporal resolution of 64 km and 1 h. In general, the annual, monthly, and daily variation in biosphere flux was reliably delivered, attributable to the novel flux forecast model, reasonable CMAQ background simulation, well-designed observational operator, and Joint Data Assimilation Scheme (JDAS) of CO₂ concentrations and natural fluxes. The size of the assimilated biosphere sink in China was -0.47 Pg C yr⁻¹, which was comparable with most global estimates (i.e., -0.27 to -0.68 Pg C yr⁻¹). Furthermore, the seasonal patterns were recalibrated well, with a growing season that shifted earlier in the year over central and south China. Moreover, the provincial-scale biosphere flux was re-estimated, and the difference between the a posteriori and a priori flux ranged from -7.03 Tg C yr⁻¹ in Heilongjiang to 2.95 Tg C yr⁻¹ in Shandong. Additionally, better performance of the a posteriori flux in contrast to the a priori flux was statistically detectable when the simulation was fitted to independent observations, indicating sufficient to robustly constrained state variables and improved fluxes estimation. This study serves as a basis for future fine-scale top-down carbon assimilation.

43. Do State-Of-The-Art Atmospheric CO₂ Inverse Models Capture Drought Impacts on the European Land Carbon Uptake?

Authors: Wei He (Nanjing University, Chinese Academy of Sciences, Beijing Normal University, Ministry of Natural Resources); Fei Jiang (Nanjing University, Ministry of Natural Resources); Weimin Ju (Nanjing University, Ministry of Natural Resources); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jingfeng Xiao (University of New Hampshire); Ngoc Tu Nguyen (Hohai University); Mousong Wu (Nanjing University, Ministry of Natural Resources); Songhan Wang (Jiangsu Collaborative Innovation Center for Modern Crop Production Key Laboratory of Crop Physiology and Ecology in Southern China Nanjing Agricultural University Nanjing China); Jun Wang (Nanjing University, Ministry of Natural Resources); Christian Rödenbeck (Max Planck Institute for Biogeochemistry); Xing Li (Seoul National University); Marko Scholze (Lund University); Guillaume Monteil (Lund University); Hengmao Wang (Nanjing University, Ministry of Natural Resources); Yanlian Zhou (Nanjing University, Ministry of Natural Resources); Qiaoning He (School of Urban and Environmental Sciences Huaiyin Normal University Huaian China); Jing M. Chen (School of Geographical Sciences Fujian Normal University Fuzhou China, University of Toronto)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

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Abstract: The European land carbon uptake has been heavily impacted by several recent severe droughts, yet quantitative estimates of carbon uptake anomalies are uncertain. Atmospheric CO₂ inverse models (AIMs) provide observation-based estimates of the large-scale carbon flux dynamics, but how well they capture drought impacts on the terrestrial carbon uptake is poorly known. Here we assessed the capacity of state-of-the-art AIMs in monitoring drought impacts on the European carbon uptake over 2001-2015 using observations of environmental variability and vegetation function and made comparisons with bottom-up estimates of carbon uptake anomalies. We found that global inversions with only limited surface CO₂ observations give divergent estimates of drought impacts. Regional inversions assimilating denser CO₂ observations over Europe demonstrated some improved consistency, with all inversions capturing a reduction in carbon uptake during the 2012 drought. However, they failed to capture the reduction caused by the 2015 drought. Finally, we found that a set of inversions that assimilated satellite XCO₂ or assimilated environmental variables plus surface CO₂

observations better captured carbon uptake anomalies induced by both the 2012 and 2015 droughts. In addition, the recent Orbiting Carbon Observatory-2 XCO₂ inversions showed good potential in capturing drought impacts, with better performances for larger-scale droughts like the 2018 drought. These results suggest that surface CO₂ observations may still be too sparse to fully capture the impact of drought on the carbon cycle at subcontinental scales over Europe, and satellite XCO₂ and ancillary environmental data can be used to improve observational constraints in atmospheric inversion systems.

44. Anomalous Net Biome Exchange Over Amazonian Rainforests Induced by the 2015/16 El Niño: Soil Dryness-Shaped Spatial Pattern but Temperature-dominated Total Flux

Authors: Jun Wang (Nanjing University); Ning Zeng (University of Maryland); Meirong Wang (Nanjing University); Fei Jiang (Nanjing University); Frédéric Chevallier (Laboratoire des Sciences du Climat et de L'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Sean Crowell (University of Oklahoma); Wei He (Nanjing University); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhiqiang Liu (Chinese Academy of Sciences); Scot M. Miller (Johns Hopkins University); Sajeev Philip (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Hengmao Wang (Nanjing University); Mousong Wu (Nanjing University); Weimin Ju (Nanjing University); Shuzhuang Feng (Nanjing University); Mengwei Jia (Nanjing University)

Journal: Geophysical Research Letters

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References: 70

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Keywords: OCO-2 MIP, climate drivers, net biome exchange, 2015/16 El Niño, Amazonian rainforests

Funding: National Key Research and Development Program of China, Natural Science Foundation of Jiangsu Province, National Natural Science Foundation of China (and 3 more)

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Open Access: N/A

Abstract: The magnitude and spatial pattern of anomalous net biome exchange (NBE) induced by the 2015/16 El Niño over Amazonian rainforests remain uncertain. We here investigated them using multi-model posterior NBE products in the Orbiting Carbon Observatory-2 (OCO-2) version 10 modeling intercomparison project. Results suggest that relative to the annual NBE average in 2017/18, larger anomalous carbon release occurred over the eastern and northern Amazonian rainforests in 2015/16, with a total flux of approximately 0.4 PgC yr⁻¹ after assimilating satellite-observed column CO₂ concentrations (XCO₂) over land. We further find that this anomalous spatial pattern was predominantly determined by soil dryness, while the total positive NBE anomaly was dominated by higher temperature with its contribution of approximately 68~70%. We believe that atmospheric inversions assimilating more satellite-observed XCO₂ in future can provide us more comprehensive understanding how Amazonian rainforests cope with the abiotic stresses induced by El Niño events.

45. A Robust Estimate of Continental-Scale Terrestrial Carbon Sinks Using GOSAT XCO₂ Retrievals

Authors: Lingyu Zhang (Nanjing University); Fei Jiang (Nanjing University, Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China); Wei He (Nanjing University); Mousong Wu (Nanjing University); Jun Wang (Nanjing University); Weimin Ju (Nanjing University, Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China); Hengmao Wang (Nanjing University); Yongguang Zhang (Nanjing University, Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China); Stephen Sitch (University of Exeter); Anthony P. Walker (Oak Ridge National Laboratory); Xu Yue (Nanjing University); Shuzhuang Feng (Nanjing University); Mengwei Jia (Nanjing University); Jing M. Chen (University of Toronto)

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Abstract: Satellite XCO₂ retrievals could improve the estimates of surface carbon fluxes, but it remains unknown on what scales these estimates are robust. Here, we use the time-dependent Bayesian synthesis top-down method and prior net ecosystem exchanges (NEEs) from 12 terrestrial biosphere models (TBMs) to infer the monthly carbon fluxes of 51 land regions with constraints by GOSAT XCO₂ retrievals. We find that the uncertainty (standard deviation of 12 TBMs) reduction rates (uncertainty reduction rate (URR)) decrease significantly at decreasing spatial scales. On the continental-scale, the mean URR is about 57%, and the annual and seasonal cycle estimates of NEE are rather robust. The evaluation shows that the posterior CO₂ concentrations are significantly improved at the continental scale. Our study suggests that the GOSAT XCO₂ can only promise a robust continental-scale NEE estimate, and improving the XCO₂ accuracy is an effective way to achieve robust estimates on smaller scales under current spatial coverage.

46. National CO2 budgets (2015-2020) inferred from atmospheric CO2 observations in support of the global stocktake

Authors: Brendan Byrne; David F. Baker; Sourish Basu; Michael Bertolacci; Kevin W. Bowman; Dustin Carroll; Abhishek Chatterjee; Frédéric Chevallier; Philippe Ciais; Noel Cressie; David Crisp; Sean Crowell; Feng Deng; Zhu Deng; Nicholas M. Deutsch; Manvendra K. Dubey; Sha Feng; Omaira E. García; David W. T. Griffith; Benedikt Herkommer; Lei Hu; Andrew R. Jacobson; Rajesh Janardanan; Sujong Jeong; Matthew S. Johnson; Dylan B. A. Jones; Rigel Kivi; Junjie Liu; Zhiqiang Liu; Shamil Maksyutov; John B. Miller; Scot M. Miller; Isamu Morino; Justus Notholt; Tomohiro Oda; Christopher W. O'Dell; Young-Suk Oh; Hirofumi Ohyama; Prabir K. Patra; Hélène Peiro; Christof Petri; Sajeev Philip; David F. Pollard; Benjamin Poulter; Marine Remaud; Andrew Schuh; Mahesh K. Sha; Kei Shiomi; Kimberly Strong; Colm Sweeney; Yao Té; Hanqin Tian; Voltaire A. Velazco; Mihalís Vrekoussis; Thorsten Warneke; John R. Worden; Debra Wunch; Yuanzhi Yao; Jeongmin Yun; Andrew Zammit-Mangion; Ning Zeng

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Abstract: . Accurate accounting of emissions and removals of CO₂ is critical for the planning and verification of emission reduction targets in support of the Paris Agreement. Here, we present a pilot dataset of country-specific net carbon exchange (NCE; fossil plus terrestrial ecosystem fluxes) and terrestrial carbon stock changes aimed at informing countries' carbon budgets. These estimates are based on "top-down" NCE outputs from the v10 Orbiting Carbon Observatory (OCO-2) modeling intercomparison project (MIP), wherein an ensemble of inverse modeling groups conducted standardized experiments assimilating OCO-2 column-averaged dry-air mole fraction (XCO₂) retrievals (ACOS v10), in situ CO₂ measurements or combinations of these data. The v10 OCO-2 MIP NCE estimates are combined with "bottom-up" estimates of fossil fuel emissions and lateral carbon fluxes to estimate changes in terrestrial carbon stocks, which are impacted by anthropogenic and natural drivers. These flux and stock change estimates are reported annually (2015-2020) as both a global 1° x 1° gridded dataset and a country-level dataset and are available for download from the Committee on Earth Observation Satellites' (CEOS) website: <https://doi.org/10.48588/npf6-sw92> (Byrne et al., 2022). Across the v10 OCO-2 MIP experiments, we obtain increases in the ensemble median terrestrial carbon stocks of 3.29-4.58 Pg CO₂ yr⁻¹ (0.90-1.25 Pg C yr⁻¹). This is a result of broad increases in

terrestrial carbon stocks across the northern extratropics, while the tropics generally have stock losses but with considerable regional variability and differences between v10 OCO-2 MIP experiments. We discuss the state of the science for tracking emissions and removals using top-down methods, including current limitations and future developments towards top-down monitoring and verification systems.

47. Regional CO₂ Inversion Through Ensemble-Based Simultaneous State and Parameter Estimation: TRACE Framework and Controlled Experiments

Authors: Hans W. Chen (Chalmers University of Technology, Pennsylvania State University, Lund University); Fuqing Zhang (Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement CEA CNRS UVSQ/IPSL Université Paris-Saclay Gif-sur-Yvette France, Pennsylvania State University); Marko Scholze (Lund University); Kenneth J. Davis (Pennsylvania State University); Richard B. Alley (Pennsylvania State University)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Keywords: CO₂, data assimilation, flux estimation, ensemble methods, Kalman filter (and 1 more)

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Abstract: Atmospheric inversions provide estimates of carbon dioxide (CO₂) fluxes between the surface and atmosphere based on atmospheric CO₂ concentration observations. The number of CO₂ observations is projected to increase severalfold in the next decades from expanding in situ networks and next-generation CO₂-observing satellites, providing both an opportunity and a challenge for inversions. This study introduces the TRACE Regional Atmosphere-Carbon Ensemble (TRACE) system, which employ an ensemble-based simultaneous state and parameter estimation (ESSPE) approach to enable the assimilation of large volumes of observations for constraining CO₂ flux parameters. TRACE uses an online full-physics mesoscale atmospheric model and assimilates observations serially in a coupled atmosphere-carbon ensemble Kalman filter. The data assimilation system was tested in a series of observing system simulation experiments using in situ observations for a regional domain over North America in summer. Under ideal conditions with known prior flux parameter error covariances, TRACE reduced the error in domain-integrated monthly CO₂ fluxes by about 97% relative to the prior flux errors. In a more realistic scenario with unknown prior flux error statistics, the corresponding relative error reductions ranged from 80.6% to 88.5% depending on the specification of prior flux parameter error correlations. For regionally integrated fluxes on a spatial scale of 106 km², the sum of absolute errors was reduced by 34.5%-50.9% relative to the prior flux errors. Moreover, TRACE produced posterior uncertainty estimates that were consistent with the true errors. These initial experiments show that the ESSPE approach in TRACE provides a promising method for advancing CO₂ inversion techniques.

48. Constraint of satellite CO₂ retrieval on the global carbon cycle from a Chinese atmospheric inversion system

Authors: Zhe Jin; Tao Wang; Hongqin Zhang; Yilong Wang; Jinzhi Ding; Xiangjun Tian

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

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Funding: N/A

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Open Access: N/A

Abstract: N/A

49. Attributing Past Carbon Fluxes to CO₂ and Climate Change: Respiration Response to CO₂ Fertilization Shifts Regional Distribution of the Carbon Sink

Authors: Gregory R. Quetin (University of California Santa Barbara, Department of Earth System Science Stanford University Stanford CA USA); Caroline A. Famiglietti (Department of Earth System Science Stanford University Stanford CA USA); Nathan C. Dadap (Department of Earth System Science Stanford University Stanford CA USA); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Noah S. Diffenbaugh (Department of Earth System Science Stanford University Stanford CA USA, Woods Institute for the Environment Stanford University Stanford CA USA); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anna T. Trugman (University of California Santa Barbara); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA, Woods Institute for the Environment Stanford University Stanford CA USA)

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Subjects: N/A

Keywords: carbon cycle, climate, model data fusion, remote sensing, respiration (and 1 more)

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Abstract: Over the past century, increased atmospheric CO₂ concentrations have enhanced photosynthesis through CO₂ fertilization across the globe. However, the increased growth has also led to greater respiration rates--both from vegetation (autotrophic respiration) and through the breakdown of plant litter and soil organic matter (heterotrophic respiration). The resulting change in carbon flux--and its spatial distribution--that can be attributed to increasing CO₂ and climate change remains unknown. We used the Carbon Data Model Framework, a model-data fusion system that assimilates global observations from satellites and other sources to create an ensemble of observationally constrained carbon cycle representations, to determine the photosynthesis and respiration fluxes that can be attributed to increased atmospheric CO₂ and associated climate change from 1920 to 2015. Across the globe, the response of photosynthesis and respiration to atmospheric CO₂ dominates their response to climate alone. The regional distribution of the carbon sink attributable to climate change and CO₂ is strongly influenced by the 'loss ratio of carbon gained'--the fraction of enhanced photosynthesis that is lost to respiration. While the wet tropics' attributable photosynthesis flux is 1.4 times larger than that of the temperate region, the attributable flux of net carbon uptake is actually 1.25

larger in the temperate region, due to the wet tropics' greater heterotrophic respiration response to enhanced plant growth. At the global scale, the loss ratio of carbon gained is $83 \pm 0.6\%$. Our results highlight the importance of the respiration responses to enhanced plant growth in regulating the land carbon sink.

50. Characterizing Average Seasonal, Synoptic, and Finer Variability in Orbiting Carbon Observatory-2 XCO₂ Across North America and Adjacent Ocean Basins

Authors: Kayla A. Mitchell (University of Virginia); Scott C. Doney (University of Virginia); Gretchen Keppel-Aleks (University of Michigan)

Journal: Journal of Geophysical Research: Atmospheres

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Cited by: 7 publications

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Keywords: synoptic variability, correlated error, mesoscale variability, carbon flux estimation, OCO-2 (and 1 more)

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Open Access: N/A

Abstract: Variations in atmosphere total column-mean CO₂ (XCO₂) collected by the National Aeronautics and Space Administration's Orbiting Carbon Observatory-2 satellite can be used to constrain surface carbon fluxes if the influence of atmospheric transport and observation errors on the data is known and accounted for. Due to sparse validation data, the portions of fine-scale variability in XCO₂ driven by fluxes, transport, or retrieval errors remain uncertain, particularly over the ocean. To better understand these drivers, we characterize variability in OCO-2 Level 2 version 10 XCO₂ from the seasonal scale, synoptic-scale (order of days, thousands of kilometers), and mesoscale (within-day, hundreds of kilometers) for 10 biomes over North America and adjacent ocean basins. Seasonal and synoptic variations in XCO₂ reflect real geophysical drivers (transport and fluxes), following large-scale atmospheric circulation and the north-south distribution of biosphere carbon uptake. In contrast, geostatistical analysis of mesoscale and finer variability shows that real signals are obscured by systematic biases across the domain. Spatial correlations in along-track XCO₂ are much shorter and spatially coherent variability is much larger in magnitude than can be attributed to fluxes or transport. We characterize random and coherent along-track XCO₂ variability in addition to quantifying uncertainty in XCO₂ aggregates across typical lengths used in inverse modeling. Even over the ocean, correlated errors decrease the independence and increase uncertainty in XCO₂. We discuss the utility of computing geostatistical parameters and demonstrate their importance for XCO₂ science applications spanning from data reprocessing and algorithm development to error estimation and carbon flux inference.

51. An inversion model based on GEOS-Chem for estimating global and China's terrestrial carbon fluxes in 2019

Authors: Chong-Yuan Wu; Xiao-Ye Zhang; Li-Feng Guo; Jun-Ting Zhong; De-Ying Wang; Chang-Hong Miao; Xiang Gao; Xi-Liang Zhang

Journal: Advances in Climate Change Research

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Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: N/A

52. Global net biome CO₂ exchange predicted comparably well using parameter-environment relationships and plant functional types

Authors: Caroline A. Famiglietti (Department of Earth System Science Stanford University Stanford California USA); Matthew Worden (Department of Earth System Science Stanford University Stanford California USA); Gregory R. Quetin (Department of Geography University of California at Santa Barbara Santa Barbara California USA); T. Luke Smallman (University of Edinburgh); Uma Dayal (Department of Earth System Science Stanford University Stanford California USA); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mathew Williams (University of Edinburgh); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford California USA)

Journal: Global Change Biology

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Abstract: Accurate estimation and forecasts of net biome CO₂ exchange (NBE) are vital for understanding the role of terrestrial ecosystems in a changing climate. Prior efforts to improve NBE predictions have predominantly focused on increasing models' structural realism (and thus complexity), but parametric error and uncertainty are also key determinants of model skill. Here, we investigate how different parameterization assumptions propagate into NBE prediction errors across the globe, pitting the traditional plant functional type (PFT)-based approach against a novel top-down, machine learning-based "environmental filtering" (EF) approach. To do so, we simulate these contrasting methods for parameter assignment within a flexible model-data fusion framework of the terrestrial carbon cycle (CARDAMOM) at a global scale. In the PFT-based approach, model parameters from a small number of select locations are applied uniformly within regions sharing similar land cover characteristics. In the EF-based approach, a pixel's parameters are predicted based on underlying relationships with climate, soil, and canopy properties. To isolate the role of parametric from structural uncertainty in our analysis, we benchmark the resulting PFT-based and EF-based NBE predictions with estimates from CARDAMOM's Bayesian optimization approach (whereby "true" parameters consistent with a suite of data constraints are retrieved on a pixel-by-pixel basis). When considering the mean absolute error of NBE predictions across time, we find that the EF-based approach matches or outperforms the PFT-based approach at 55% of pixels--a narrow majority. However, NBE estimates from the EF-based approach are susceptible to compensation between errors in component flux

predictions and predicted parameters can align poorly with the assumed "true" values. Overall, though, the EF-based approach is comparable to conventional approaches and merits further investigation to better understand and resolve these limitations. This work provides insight into the relationship between terrestrial biosphere model performance and parametric uncertainty, informing efforts to improve model parameterization via PFT-free and trait-based approaches.

53. Evaluation of wetland CH₄ in the Joint UK Land Environment Simulator (JULES) land surface model using satellite observations

Authors: Robert J. Parker; Chris Wilson; Edward Comyn-Platt; Garry Hayman; Toby R. Marthews; A. Anthony Bloom; Mark F. Lunt; Nicola Gedney; Simon J. Dadson; Joe McNorton; Neil Humpage; Hartmut Boesch; Martyn P. Chipperfield; Paul I. Palmer; Dai Yamazaki

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Abstract: . Wetlands are the largest natural source of methane. The ability to model the emissions of methane from natural wetlands accurately is critical to our understanding of the global methane budget and how it may change under future climate scenarios. The simulation of wetland methane emissions involves a complicated system of meteorological drivers coupled to hydrological and biogeochemical processes. The Joint UK Land Environment Simulator (JULES) is a process-based land surface model that underpins the UK Earth System Model (UKESM) and is capable of generating estimates of wetland methane emissions. In this study, we use GOSAT satellite observations of atmospheric methane along with the TOMCAT global 3-D chemistry transport model to evaluate the performance of JULES in reproducing the seasonal cycle of methane over a wide range of tropical wetlands. By using an ensemble of JULES simulations with differing input data and process configurations, we investigate the relative importance of the meteorological driving data, the vegetation, the temperature dependency of wetland methane production and the wetland extent. We find that JULES typically performs well in replicating the observed methane seasonal cycle. We calculate correlation coefficients to the observed seasonal cycle of between 0.58 and 0.88 for most regions; however, the seasonal cycle amplitude is typically underestimated (by between 1.8 and 19.5 ppb). This level of performance is comparable to that typically provided by state-of-the-art data-driven wetland CH₄ emission inventories. The meteorological driving data are found to be the most significant factor in determining the ensemble performance, with temperature dependency and vegetation having moderate effects. We find that neither wetland extent configuration outperforms the other, but this does lead to poor performance in some regions. We focus in detail on three African wetland regions (Sudd, Southern Africa and Congo) where we find the performance of JULES to be poor and explore the reasons for this in detail. We find that neither wetland extent configuration used is sufficient in representing the wetland distribution in these regions (underestimating the wetland seasonal cycle amplitude by 11.1, 19.5 and 10.1 ppb).

respectively, with correlation coefficients of 0.23, 0.01 and 0.31). We employ the Catchment-based Macro-scale Floodplain (CaMa-Flood) model to explicitly represent river and floodplain water dynamics and find that these JULES-CaMa-Flood simulations are capable of providing a wetland extent that is more consistent with observations in this regions, highlighting this as an important area for future model development.

54. Global-Scale Evaluation of XCO₂ Products from GOSAT, OCO-2 and CarbonTracker Using Direct Comparison and Triple Collocation Method

Authors: Yuanyuan Chen (Zhejiang University); Jiefeng Cheng (Zhejiang Geophysics Spatial Planning Technology Co., Ltd., Hangzhou 310000, China); Xiaodong Song (Zhejiang University); Shuo Liu (Zhejiang University); Yuan Sun (Qiqihar Meteorological Service, Heilongjiang Meteorological Bureau, Qiqihar 161000, China); Dajiang Yu (China Meteorological Administration); Shuangxi Fang (Zhejiang University)

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Abstract: Triple collocation (TC) shows potential in estimating the errors of various geographical data in the absence of the truth. In this study, the TC techniques are first applied to evaluate the performances of multiple column-averaged dry air CO₂ mole fraction (XCO₂) estimates derived from the Greenhouse Gases Observing Satellite (GOSAT), the Orbiting Carbon Observatory 2 (OCO-2) and the CarbonTracker model (CT2019B) at a global scale. A direct evaluation with the Total Carbon Column Observing Network (TCCON) measurements is also employed for comparison. Generally, the TC-based evaluation results are consistent with the direct evaluation results on the overall performances of three XCO₂ products, in which the CT2019B performs best, followed by OCO-2 and GOSAT. Correlation coefficient estimates of the TC show higher consistency and stronger robustness than root mean square error estimates. TC-based error estimates show that most of the terrestrial areas have larger error than the marine areas overall, especially for the GOSAT and CT2019B datasets. The OCO-2 performs well in areas where CT2019B or GOSAT have large errors, such as most of China except the northwest, and Russia. This study provides a reference for characterizing the performances of multiple CO₂ products from another perspective.

55. Multi-year observations reveal a larger than expected autumn respiration signal across northeast Eurasia

Authors: Brendan Byrne; Junjie Liu; Yonghong Yi; Abhishek Chatterjee; Sourish Basu; Rui Cheng; Russell Doughty; Frédéric Chevallier; Kevin W. Bowman; Nicholas C. Parazoo; David Crisp; Xing Li; Jingfeng Xiao; Stephen Sitch; Bertrand Guenet; Feng Deng; Matthew S. Johnson; Sajeev Philip; Patrick C. McGuire; Charles E. Miller

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Abstract: . Site-level observations have shown pervasive cold season CO₂ release across Arctic and boreal ecosystems, impacting annual carbon budgets. Still, the seasonality of CO₂ emissions are poorly quantified across much of the high latitudes due to the sparse coverage of site-level observations. Space-based observations provide the opportunity to fill some observational gaps for studying these high-latitude ecosystems, particularly across poorly sampled regions of Eurasia. Here, we show that data-driven net ecosystem exchange (NEE) from atmospheric CO₂ observations implies strong summer uptake followed by strong autumn release of CO₂ over the entire cold northeastern region of Eurasia during the 2015-2019 study period. Combining data-driven NEE with satellite-based estimates of gross primary production (GPP), we show that this seasonality implies less summer heterotrophic respiration (Rh) and greater autumn Rh than would be expected given an exponential relationship between respiration and surface temperature. Furthermore, we show that this seasonality of NEE and Rh over northeastern Eurasia is not captured by the TRENDY v8 ensemble of dynamic global vegetation models (DGVMs), which estimate that 47 %-57 % (interquartile range) of annual Rh occurs during August-April, while the data-driven estimates suggest 59 %-76 % of annual Rh occurs over this period. We explain this seasonal shift in Rh by respiration from soils at depth during the zero-curtain period, when sub-surface soils remain unfrozen up to several months after the surface has frozen. Additional impacts of physical processes related to freeze-thaw dynamics may contribute to the seasonality of Rh. This study confirms a significant and spatially extensive early cold season CO₂ efflux in the permafrost-rich region of northeast Eurasia and suggests that autumn Rh from subsurface soils in the northern high latitudes is not well captured by current DGVMs.

56. Remote-Sensing Derived Trends in Gross Primary Production Explain Increases in the CO₂ Seasonal Cycle Amplitude

Authors: Liyin He (California Institute of Technology); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yi Yin (California Institute of Technology); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Subjects: N/A

Keywords: satellite remote sensing, net ecosystem exchange, peak season photosynthesis, seasonal cycle of atmospheric CO₂, climate change

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License: N/A

Open Access: N/A

Abstract: An increase in the seasonal cycle amplitude (SCA) of atmospheric CO₂ since the 1960s has been observed in the Northern Hemisphere (NH). However, the underlying dominant drivers are still debated. The peak season CO₂ uptake by vegetation is critical in shaping the CO₂ seasonality. Using satellite-upscaled gross primary production (GPP) from FLUXCOM and near-infrared reflectance of vegetation (NIRV), we demonstrate that peak GPP has increased across the NH over the last two decades. We relate this productivity increase to changes in the CO₂ SCA using an atmospheric transport model. The increased photosynthesis has strongly contributed to CO₂ SCA trends, but with substantial latitudinal and longitudinal variations. Despite a general increase in the CO₂ SCA, there are distinct regional differences. These differences are mainly controlled by regional biosphere carbon fluxes, with the remainder explained by non-biome factors, including large-scale atmospheric transport, changes in fossil fuel combustion, biomass burning and oceanic fluxes. Using the global flask and in situ CO₂ measurement sites, we find that SCA trends at high latitude are mainly driven by increasingly productive natural ecosystems, whereas mid latitude sites around the Midwest United States are mainly impacted by intensified agriculture and atmospheric transport. Averaging across the 15 long-term surface sites, forests contribute 26% (7%) to the SCA trends, while crops contribute 17% (24%) and the combined shrubland, grassland and wetland regions contribute 23% (37%) for simulations driven by FLUXCOM (NIRv) ecosystem fluxes. Our findings demonstrate that satellite inferred trends of ecosystem fluxes can capture the observed CO₂ SCA trend.

57. Evaluating Global Atmospheric Inversions of Terrestrial Net Ecosystem Exchange CO₂ Over North America on Seasonal and Sub-Continental Scales

Authors: Yu Yan Cui (Pennsylvania State University); Li Zhang (Pennsylvania State University); Andrew R. Jacobson (University of Colorado Boulder); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); Sajeev Philip (National Aeronautics and Space Administration (NASA) Ames Research Center, Centre for Atmospheric Sciences, Indian Institute of Technology Delhi, Universities Space Research Association); David Baker (Colorado State University); Frederic Chevallier (Laboratoire des Sciences du Climat et de L'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Andrew E. Schuh (Colorado State University); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Sean Crowell (University of Oklahoma); Hélène E. Peiro (University of Oklahoma); Feng Deng (University of Toronto); Sourish Basu (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Kenneth J. Davis (Pennsylvania State University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 29

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

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Abstract: Atmospheric inversion estimates of net ecosystem exchange (NEE) of CO₂ are increasingly relevant to climate policy. We evaluated sub-continental, seasonal estimates of CO₂ NEE from nine global inversion systems that participated in the Orbiting Carbon Observatory-2 model intercomparison project (OCO-2 v9 MIP), using 98 research flights conducted over the central and eastern United States from 2016 to 2018 as part of the Atmospheric Carbon and Transport - America mission. We found that the seasonal amplitude of NEE in the central and eastern United States is underestimated in these models and model-data biases are largest for those inversions with the smallest seasonal flux amplitudes. These results were independent of whether the inversions used satellite or in situ data. The largest NEE biases were observed in the Midwest croplands and eastern forests. Future experiments are needed to determine the causes of the persistent biases and if they are associated with biases in annual flux estimates.

58. Consistency of seasonal variability in regional CO₂ fluxes from GOSAT-IM, NASA-GEOS, and NOAA-CT

Authors: M Krishnapriya; Rabindra K Nayak; C V Naidu; Shaik Allaudheen; A Bhuvanachandra; V K Dadhwal; M V R Seshasai

Journal: Journal of Earth System Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12040-022-01934-w

URL: <https://doi.org/10.1007/s12040-022-01934-w>

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References: 75

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: FLUXNET, triple point analysis, harmonic analysis, root mean square error, Blending (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

59. Constraining China's land carbon sink from emerging satellite CO₂ observations: Progress and challenges

Authors: Yilong Wang (Chinese Academy of Sciences); Xiangjun Tian (Chinese Academy of Sciences, University of Chinese Academy Sciences Beijing China); Frédéric Chevallier (Laboratoire des Sciences du Climat et de l'Environnement, LSCE-IPSL (CEA-CNRS-UVSQ) Université Paris-Saclay Gif-sur-Yvette France); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); Sajeev Philip (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); David F. Baker (Colorado State University); Andrew E. Schuh (Colorado State University); Feng Deng (University of Toronto); Xingying Zhang (China Meteorological Administration); Lu Zhang (China Meteorological Administration); Dan Zhu (Peking University); Xuhui Wang (Peking University)

Journal: Global Change Biology

Publisher: Wiley

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References: 46

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Keywords: N/A

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Abstract: Land carbon sink is a vital component for the achievement of China's ambitious carbon neutrality goal, but its magnitude is poorly known. Atmospheric observations and inverse models are valuable tools to constrain the China's land carbon sink. Space-based CO₂ measurements from satellites form an emerging data stream for application of such atmospheric inversions. Here, we reviewed the satellite missions that is dedicated to the monitoring of CO₂, and the recent progresses on the inversion of China's land carbon sink using satellite CO₂ measurements. We summarized the limitations and challenges in current space platforms, retrieval algorithms, and the inverse modeling. It is shown that there are large uncertainties of contemporary satellite-based estimates of China's land carbon sink. We discussed future opportunities of continuous improvements in three aspects to better constrain China's land carbon sink with space-based CO₂ measurements.

60. A review of datasets and methods for deriving spatiotemporal distributions of atmospheric CO₂

Authors: Changpei He; Mingrui Ji; Michael L. Grieneisen; Yu Zhan

Journal: Journal of Environmental Management

Publisher: Elsevier BV

DOI: 10.1016/j.jenvman.2022.116101

URL: <https://doi.org/10.1016/j.jenvman.2022.116101>

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References: 105

Cited by: 24 publications

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Funding: N/A

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Open Access: N/A

Abstract: N/A

61. Global and regional carbon budget for 2015-2020 inferred from OCO-2 based on an ensemble Kalman filter coupled with GEOS-Chem

Authors: Yawen Kong; Bo Zheng; Qiang Zhang; Kebin He

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-22-10769-2022

URL: <https://doi.org/10.5194/acp-22-10769-2022>

Publication Date: 2022-08-25

Volume/Issue: Volume 22, Issue 16

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, China Association for Science and Technology, Tsinghua Shenzhen International Graduate School

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Open Access: N/A

Abstract: . Understanding carbon sources and sinks across the Earth's surface is fundamental in climate science and policy; thus, these topics have been extensively studied but have yet to be fully resolved and are associated with massive debate regarding the sign and magnitude of the carbon budget from global to regional scales. Developing new models and estimates based on state-of-the-art algorithms and data constraints can provide valuable knowledge and contribute to a final ensemble model in which various optimal carbon budget estimates are integrated, such as the annual global carbon budget paper. Here, we develop a new atmospheric inversion system based on the 4D local ensemble transform Kalman filter (4D-LETKF) coupled with the GEOS-Chem global transport model to infer surface-to-atmosphere net carbon fluxes from Orbiting Carbon Observatory-2 (OCO-2) V10r XCO₂ retrievals. The 4D-LETKF algorithm is adapted to an OCO-2-based global carbon inversion system for the first time in this work. On average, the mean annual terrestrial and oceanic fluxes between 2015 and 2020 are estimated as - 2.02 and - 2.34 GtC yr⁻¹, respectively, compensating for 21 % and 24 %, respectively, of global fossil carbon dioxide (CO₂) emissions (9.80 GtC yr⁻¹). Our inversion results agree with the CO₂ atmospheric growth rates reported by the National Oceanic and Atmospheric Administration (NOAA) and reduce the modeled CO₂ concentration biases relative to the prior fluxes against surface and aircraft measurements. Our inversion-based carbon fluxes are broadly consistent with those provided by other global atmospheric inversion models, although discrepancies still occur in the land-ocean flux partitioning schemes and seasonal flux amplitudes over boreal and tropical regions, possibly due to the sparse observational constraints of the OCO-2 satellite and the divergent prior fluxes used in different inversion models. Four sensitivity experiments are performed herein to vary the prior fluxes and uncertainties in our inversion system, suggesting that regions that lack OCO-2 coverage are sensitive to the priors, especially over the tropics and high latitudes. In the further development of our inversion system, we will optimize the data-assimilation configuration to fully utilize current observations and increase the spatial and seasonal representativeness of the prior fluxes over regions that lack observations.

62. Toward a long-term atmospheric CO₂ inversion for elucidating natural carbon fluxes: technical notes of NISMON-CO₂ v2021.1

Authors: Yosuke Niwa; Kentaro Ishijima; Akihiko Ito; Yosuke Iida

Journal: Progress in Earth and Planetary Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40645-022-00502-6

URL: <https://doi.org/10.1186/s40645-022-00502-6>

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Funding: Japan Society for the Promotion of Science, Environmental Restoration and Conservation Agency

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Abstract: Accurate estimates of the carbon dioxide (CO₂) fluxes at the earth's surface are imperative for comprehending the carbon cycle mechanisms and providing reliable global warming predictions. Furthermore, they can also provide valuable science-based information that will be helpful in reducing human-induced CO₂ emissions. Inverse analysis is a prominent method of quantitatively estimating spatiotemporal variations in CO₂ fluxes; however, it involves a certain level of uncertainty and requires technical refinement, specifically to improve the horizontal resolution so that local fluxes can be compared with other estimates made at the regional or national level. In this study, a novel set of inversion schemes was incorporated into a state-of-the-art inverse analysis system named NISMON-CO₂. The introduced schemes include a grid conversion, observational weighting, and anisotropic prior error covariance, the details of which are described. Moreover, pseudo-observation experiments were performed to examine the effect of the new schemes and to assess the reliability of NISMON-CO₂ for long-term analysis with practical inhomogeneous observations. The experiment results evidently demonstrate the advantages of the grid conversion scheme for high-resolution flux estimates (1 degrees x 1 degrees), with notable improvements being achieved through the observational weighting and anisotropic prior error covariance. Furthermore, the estimated seasonal and interannual variations in regional CO₂ fluxes were confirmed to be reliable, although some potential bias in terms of global land-ocean partitioning was observed. Thus, these results are useful for interpreting the flux variations that result from real-observation inverse analysis by NISMON-CO₂ ver. 2021.1.

63. Improving the joint estimation of CO₂ and surface carbon fluxes using a constrained ensemble Kalman filter in COLA (v1.0)

Authors: Zhiqiang Liu; Ning Zeng; Yun Liu; Eugenia Kalnay; Ghassem Asrar; Bo Wu; Qixiang Cai; Di Liu; Pengfei Han

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-5511-2022

URL: <https://doi.org/10.5194/gmd-15-5511-2022>

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References: 59

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Dream Project of Ministry of Science and Technology of the People's Republic of China

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Open Access: N/A

Abstract: . Atmospheric inversion of carbon dioxide (CO₂) measurements to better understand carbon sources and sinks has made great progress over the last 2 decades. However, most of the studies, including a four-dimensional variational ensemble Kalman filter and Bayesian synthesis approaches, directly obtain only fluxes, while CO₂ concentration is derived with the forward model as part of a post-analysis. Kang et al. (2012) used the local ensemble transform Kalman filter (LETKF), which updates the CO₂, surface carbon flux (SCF), and meteorology fields simultaneously. Following this track, a system with a short assimilation window and a long observation window was developed (Liu et al., 2019). However, this data assimilation system faces the challenge of maintaining carbon mass conservation. To overcome this shortcoming, here we apply a constrained ensemble Kalman filter (CEnKF) approach to ensure the conservation of global CO₂ mass. After a standard LETKF procedure, an additional assimilation is used to adjust CO₂ at each model grid point and to ensure the consistency between the analysis and the first guess of the global CO₂ mass. Compared to an observing system simulation experiment without mass conservation, the CEnKF significantly reduces the annual global SCF bias from ~ 0.2 to less than 0.06 Gt and slightly improves the seasonal and annual performance over tropical and southern extratropical regions. We show that this system can accurately track the spatial distribution of annual mean SCF. And the system reduces the seasonal flux root mean square error from a priori to analysis by 48 %-90 %, depending on the continental region. Moreover, the 2015-2016 El Niño impact is well captured with anomalies mainly in the tropics.

64. GHG Monitoring Project for the Global Stocktake 2023: implications of the COP26 Japan Pavilion seminar

Authors: Tomohiro Oda

Journal: Carbon Balance and Management

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s13021-022-00211-z

URL: <https://doi.org/10.1186/s13021-022-00211-z>

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References: 3

Cited by: N/A publications

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Open Access: N/A

Abstract: During the 2021 Glasgow Climate Change Conference (COP26), a hybrid seminar event "Greenhouse gas (GHG) Monitoring Project for the Global Stocktake 2023" was held at the COP26 Japan Pavilion on 2nd of November 2011. The participants presented and discussed science-based GHG monitoring approaches in support of the Global Stocktake. This report summarizes the five research talks given at the event.

65. Interannual variability in the Australian carbon cycle over 2015-2019, based on assimilation of Orbiting Carbon Observatory-2 (OCO-2) satellite data

Authors: Yohanna Villalobos; Peter J. Rayner; Jeremy D. Silver; Steven Thomas; Vanessa Haverd; Jürgen Knauer; Zoë M. Loh; Nicholas M. Deutscher; David W. T. Griffith; David F. Pollard

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-22-8897-2022

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References: 68

Cited by: 9 publications

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Open Access: N/A

Abstract: . In this study, we employ a regional inverse modelling approach to estimate monthly carbon fluxes over the Australian continent for 2015-2019 using the assimilation of the total column-averaged mole fractions of carbon dioxide from the Orbiting Carbon Observatory-2 (OCO-2, version 9) satellite. Subsequently, we study the carbon cycle variations and relate their fluctuations to anomalies in vegetation productivity and climate drivers. Our 5-year regional carbon flux inversion suggests that Australia was a carbon sink averaging -0.46 ± 0.08 PgC yr⁻¹ (excluding fossil fuel emissions), largely influenced by a strong carbon uptake (-1.04 PgC yr⁻¹) recorded in 2016. Australia's semi-arid ecosystems, such as sparsely vegetated regions (in central Australia) and savanna (in northern Australia), were the main contributors to the carbon uptake in 2016. These regions showed relatively high vegetation productivity, high rainfall, and low temperature in 2016. In contrast to the large carbon sink found in 2016, the large carbon outgassing recorded in 2019 coincides with an unprecedented rainfall deficit and higher-than-average temperatures across Australia. Comparison of the posterior column-averaged CO₂ concentration with Total Carbon Column Observing Network (TCCON) stations and in situ measurements offers limited insight into the fluxes assimilated with OCO-2. However, the lack of these monitoring stations across Australia, mainly over ecosystems such as savanna and areas with sparse vegetation, impedes us from providing strong conclusions. To a certain extent, we found that the flux anomalies across Australia are consistent with the ensemble means of the OCO-2 Model Intercomparison Project (OCO-2 MIP) and FLUXCOM (2015-2018), which estimate an anomalous carbon sink for Australia in 2016 of -1.09 and -0.42 PgC yr⁻¹ respectively. More accurate estimates of OCO-2 retrievals, with the addition of ocean glint data into our system, and a better understanding of the error in the atmospheric transport modelling will yield further insights into the difference in the magnitude of our carbon flux estimates.

66. A 10-year global monthly averaged terrestrial net ecosystem exchange dataset inferred from the ACOS GOSAT v9 XCO₂ retrievals (GCAS2021)

Authors: Fei Jiang; Weimin Ju; Wei He; Mousong Wu; Hengmao Wang; Jun Wang; Mengwei Jia; Shuzhuang Feng; Lingyu Zhang; Jing M. Chen

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-14-3013-2022

URL: <https://doi.org/10.5194/essd-14-3013-2022>

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Keywords: N/A

Funding: National Key Research and Development Program of China, Fundamental Research Funds for the Central Universities

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Open Access: N/A

Abstract: . A global gridded net ecosystem exchange (NEE) of CO₂ dataset is vital in global and regional carbon cycle studies. Top-down atmospheric inversion is one of the major methods to estimate the global NEE; however, the existing global NEE datasets generated through inversion from conventional CO₂ observations have large uncertainties in places where observational data are sparse. Here, by assimilating the GOSAT ACOS v9 XCO₂ product, we generate a 10-year (2010–2019) global monthly terrestrial NEE dataset using the Global Carbon Assimilation System, version 2 (GCASv2), which is named GCAS2021. It includes gridded (1° × 1°), globally, latitudinally, and regionally aggregated prior and posterior NEE and ocean (OCN) fluxes and prescribed wildfire (FIRE) and fossil fuel and cement (FFC) carbon emissions. Globally, the decadal mean NEE is -3.73 ± 0.52 PgC yr⁻¹, with an interannual amplitude of 2.73 PgC yr⁻¹. Combining the OCN flux and FIRE and FFC emissions, the net biosphere flux (NBE) and atmospheric growth rate (AGR) as well as their inter-annual variabilities (IAVs) agree well with the estimates of the Global Carbon Budget 2020. Regionally, our dataset shows that eastern North America, the Amazon, the Congo Basin, Europe, boreal forests, southern China, and Southeast Asia are carbon sinks, while the western United States, African grasslands, Brazilian plateaus, and parts of South Asia are carbon sources. In the TRANSCom land regions, the NBEs of temperate N. America, northern Africa, and boreal Asia are between the estimates of CMS-Flux NBE 2020 and CT2019B, and those in temperate Asia, Europe, and Southeast Asia are consistent with CMS-Flux NBE 2020 but significantly different from CT2019B. In the RECCAP2 regions, except for Africa and South Asia, the NBEs are comparable with the latest bottom-up estimate of Ciais et al. (2021). Compared with previous studies, the IAVs and seasonal cycles of NEE of this dataset could clearly reflect the impacts of extreme climates and large-scale climate anomalies on the carbon flux. The evaluations also show that the posterior CO₂ concentrations at remote sites and on a regional scale, as well as on vertical CO₂ profiles in the Asia-Pacific region, are all consistent with independent CO₂ measurements from surface flask and aircraft CO₂ observations, indicating that this

dataset captures surface carbon fluxes well. We believe that this dataset can contribute to regional- or national-scale carbon cycle and carbon neutrality assessment and carbon dynamics research. The dataset can be accessed at <https://doi.org/10.5281/zenodo.5829774> (Jiang, 2022).

67. Regional and seasonal partitioning of water and temperature controls on global land carbon uptake variability

Authors: Kai Wang; Ana Bastos; Philippe Ciais; Xuhui Wang; Christian Rödenbeck; Pierre Gentine; Frédéric Chevallier; Vincent W. Humphrey; Chris Huntingford; Michael O'Sullivan; Sonia I. Seneviratne; Stephen Sitch; Shilong Piao

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-022-31175-w

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References: 79

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Abstract: Global fluctuations in annual land carbon uptake (NEEIAV) depend on water and temperature variability, yet debate remains about local and seasonal controls of the global dependences. Here, we quantify regional and seasonal contributions to the correlations of globally-averaged NEEIAV against terrestrial water storage (TWS) and temperature, and respective uncertainties, using three approaches: atmospheric inversions, process-based vegetation models, and data-driven models. The three approaches agree that the tropics contribute over 63% of the global correlations, but differ on the dominant driver of the global NEEIAV, because they disagree on seasonal temperature effects in the Northern Hemisphere (NH, >25 degreesN). In the NH, inversions and process-based models show inter-seasonal compensation of temperature effects, inducing a global TWS dominance supported by observations. Data-driven models show weaker seasonal compensation, thereby estimating a global temperature dominance. We provide a roadmap to fully understand drivers of global NEEIAV and discuss their implications for future carbon-climate feedbacks.

68. Quantifying and Reducing Uncertainty in Global Carbon Cycle Predictions: Lessons and Perspectives From 15 Years of Data Assimilation Studies With the ORCHIDEE Terrestrial Biosphere Model

Authors: N. MacBean (Indiana University); C. Bacour (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvett France, NOVELTIS Labège Toulouse France); N. Raoult (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvett France); V. Bastrikov (Science Partners Paris France); E. N. Koffi (European Commission); S. Kuppel (Institut de Recherche pour le Développement (IRD)); F. Maignan (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvett France); C. Ottlé (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvett France); M. Peaucelle (Computational & Applied Vegetation Ecology Laboratory (CAVELAB) Ghent Universtiy Ghent Belgium, INRAE Université de Bordeaux UMR ISPA Villenave-d'Ornon Bordeaux France); D. Santaren (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvett France); P. Peylin (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvett France)

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Abstract: Predicting terrestrial carbon, C, budgets and carbon-climate feedbacks strongly relies on our ability to accurately model interactions between vegetation, C and water cycles, and the atmosphere. However, C fluxes simulated by global, process-based terrestrial biosphere models (TBMs) remain subject to large uncertainties, partly due to unknown or poorly calibrated parameters. This is because TBMs have not routinely been confronted against C cycle related datasets within a statistical data assimilation (DA) system. In this review, we present 15 years' development of a C cycle DA system for optimizing C cycle parameters of the ORCHIDEE TBM. We analyze the impact of assimilating multiple different C cycle related datasets on regional to global-scale gross and net CO₂ fluxes. We find that assimilating atmospheric CO₂ data is crucial for improving (increasing) ORCHIDEE predictions of the terrestrial land C sink. The improvement is predominantly due to the global-scale constraint these data provide for optimizing initial soil C stocks, which are likely in error due to inaccurate assumptions about

steady state spin-up and incomplete knowledge of land use change histories. When comparing the data-constrained ORCHIDEE land C sink estimates to the CAMS atmospheric inversion, we show that while the two approaches agree on the global C sink magnitude, they continue to differ in how the global C sink is partitioned between the northern hemisphere and tropics. We also discuss technical challenges faced in our C cycle DA studies, in particular the difficulty in characterizing the error covariance matrix due to unknown observation biases and/or model-data inconsistencies. We offer our perspectives on how to tackle these challenges that we hope can serve as a roadmap for other TBM groups wishing to develop C cycle DA systems.

69. The NASA Carbon Monitoring System Phase 2 synthesis: scope, findings, gaps and recommended next steps

Authors: George C Hurtt; Arlyn Andrews; Kevin Bowman; Molly E Brown; Abhishek Chatterjee; Vanessa Escobar; Lola Fatoyinbo; Peter Griffith; Maddie Guy; Sean P Healey; Daniel J Jacob; Robert Kennedy; Steven Lohrenz; Megan E McGroddy; Valeria Morales; Thomas Nehrkorn; Lesley Ott; Sassan Saatchi; Edil Sepulveda Carlo; Shawn P Serbin; Hanqin Tian

Journal: Environmental Research Letters

Publisher: IOP Publishing

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References: 196

Cited by: 15 publications

Language: N/A

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Abstract: Underlying policy efforts to address global climate change is the scientific need to develop the methods to accurately measure and model carbon stocks and fluxes across the wide range of spatial and temporal scales in the Earth system. Initiated in 2010, the NASA Carbon Monitoring System is one of the most ambitious relevant science initiatives to date, exploiting the satellite remote sensing resources, computational capabilities, scientific knowledge, airborne science capabilities, and end-to-end system expertise that are major strengths of the NASA Earth Science program. Here we provide a synthesis of 'Phase 2' activities (2011-2019), encompassing 79 projects, 482 publications, and 136 data products. Our synthesis addresses four key questions: What has been attempted? What major results have been obtained? What major gaps and uncertainties remain? and What are the recommended next steps? Through this review, we take stock of what has been accomplished and identify future priorities toward meeting the nation's needs for carbon monitoring reporting and verification.

70. Global nature run data with realistic high-resolution carbon weather for the year of the Paris Agreement

Authors: Anna Agustí-Panareda; Joe McNorton; Gianpaolo Balsamo; Bianca C. Baier; Nicolas Bousserez; Souhail Boussetta; Dominik Brunner; Frédéric Chevallier; Margarita Choulga; Michail Diamantakis; Richard Engelen; Johannes Flemming; Claire Granier; Marc Guevara; Hugo Denier van der Gon; Nellie Elguindi; Jean-Mathieu Haussaire; Martin Jung; Greet Janssens-Maenhout; Rigel Kivi; Sébastien Massart; Dario Papale; Mark Parrington; Miha Razinger; Colm Sweeney; Alex Vermeulen; Sophia Walther

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41597-022-01228-2

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Open Access: N/A

Abstract: The CO₂ Human Emissions project has generated realistic high-resolution 9 km global simulations for atmospheric carbon tracers referred to as nature runs to foster carbon-cycle research applications with current and planned satellite missions, as well as the surge of in situ observations. Realistic atmospheric CO₂, CH₄ and CO fields can provide a reference for assessing the impact of proposed designs of new satellites and in situ networks and to study atmospheric variability of the tracers modulated by the weather. The simulations spanning 2015 are based on the Copernicus Atmosphere Monitoring Service forecasts at the European Centre for Medium Range Weather Forecasts, with improvements in various model components and input data such as anthropogenic emissions, in preparation of a CO₂ Monitoring and Verification Support system. The relative contribution of different emissions and natural fluxes towards observed atmospheric variability is diagnosed by additional tagged tracers in the simulations. The evaluation of such high-resolution model simulations can be used to identify model deficiencies and guide further model improvements.

71. Review of Byrne et al: Multi-year observations reveal a larger than expected autumn respiration signal across northeast Eurasia

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

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Keywords: Seasonality|Ecosystem respiration

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Abstract: N/A

72. Attribution of Space-Time Variability in Global-Ocean Dissolved Inorganic Carbon

Authors: Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Jonathan M. Lauderdale (Massachusetts Institute of Technology); Jess F. Adkins (California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Holger Brix (University of California Los Angeles, Institute of Coastal Ocean Dynamics Helmholtz-Zentrum Hereon Geesthacht Germany); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michelle M. Gierach (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Chris Hill (Massachusetts Institute of Technology); Oliver Jahn (Massachusetts Institute of Technology); Peter Landschützer (Max Planck Institute for Meteorology); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Matt R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ariane Verdy (Scripps Institution of Oceanography / University of California San Diego); Daniel B. Whitt (National Aeronautics and Space Administration (NASA) Ames Research Center); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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Keywords: equatorial, sink, ocean, budget, carbon (and 1 more)

Funding: National Aeronautics and Space Administration

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Abstract: The inventory and variability of oceanic dissolved inorganic carbon (DIC) is driven by the interplay of physical, chemical, and biological processes. Quantifying the spatiotemporal variability of these drivers is crucial for a mechanistic understanding of the ocean carbon sink and its future trajectory. Here, we use the Estimating the Circulation and Climate of the Ocean-Darwin ocean biogeochemistry state estimate to generate a global-ocean, data-constrained DIC budget and investigate how spatial and seasonal-to-interannual variability in three-dimensional circulation, air-sea

CO₂ flux, and biological processes have modulated the ocean sink for 1995-2018. Our results demonstrate substantial compensation between budget terms, resulting in distinct upper-ocean carbon regimes. For example, boundary current regions have strong contributions from vertical diffusion while equatorial regions exhibit compensation between upwelling and biological processes. When integrated across the full ocean depth, the 24-year DIC mass increase of 64 Pg C (2.7 Pg C year⁻¹) primarily tracks the anthropogenic CO₂ growth rate, with biological processes providing a small contribution of 2% (1.4 Pg C). In the upper 100 m, which stores roughly 13% (8.1 Pg C) of the global increase, we find that circulation provides the largest DIC gain (6.3 Pg C year⁻¹) and biological processes are the largest loss (8.6 Pg C year⁻¹). Interannual variability is dominated by vertical advection in equatorial regions, with the 1997-1998 El Niño-Southern Oscillation causing the largest year-to-year change in upper-ocean DIC (2.1 Pg C). Our results provide a novel, data-constrained framework for an improved mechanistic understanding of natural and anthropogenic perturbations to the ocean sink.

73. Definitions and methods to estimate regional land carbon fluxes for the second phase of the REgional Carbon Cycle Assessment and Processes Project (RECCAP-2)

Authors: Philippe Ciais; Ana Bastos; Frédéric Chevallier; Ronny Lauerwald; Ben Poulter; Josep G. Canadell; Gustaf Hugelius; Robert B. Jackson; Atul Jain; Matthew Jones; Masayuki Kondo; Ingrid T. Luijkx; Prabir K. Patra; Wouter Peters; Julia Pongratz; Ana Maria Roxana Petrescu; Shilong Piao; Chunjing Qiu; Celso Von Randow; Pierre Regnier; Marielle Saunois; Robert Scholes; Anatoly Shvidenko; Hanqin Tian; Hui Yang; Xuhui Wang; Bo Zheng

Journal: Geoscientific Model Development

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References: 145

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Open Access: N/A

Abstract: . Regional land carbon budgets provide insights into the spatial distribution of the land uptake of atmospheric carbon dioxide and can be used to evaluate carbon cycle models and to define baselines for land-based additional mitigation efforts. The scientific community has been involved in providing observation-based estimates of regional carbon budgets either by downscaling atmospheric CO₂ observations into surface fluxes with atmospheric inversions, by using inventories of carbon stock changes in terrestrial ecosystems, by upscaling local field observations such as flux towers with gridded climate and remote sensing fields, or by integrating data-driven or process-oriented terrestrial carbon cycle models. The first coordinated attempt to collect regional carbon budgets for nine regions covering the entire globe in the RECCAP-1 project has delivered estimates for the decade 2000–2009, but these budgets were not comparable between regions due to different definitions and component fluxes being reported or omitted. The recent recognition of lateral fluxes of carbon by human activities and rivers that connect CO₂ uptake in one area with its release in another also requires better definitions and protocols to reach harmonized regional budgets that can be summed up to a globe scale and compared with the atmospheric CO₂ growth rate and inversion results. In this study, using the international initiative RECCAP-2 coordinated by the Global Carbon Project, which aims to be an update to regional carbon budgets over the last 2 decades based on observations for 10 regions covering the globe with a better harmonization than the precursor project, we provide recommendations for using atmospheric inversion results to match bottom-up carbon accounting and models, and we define the different component fluxes of the net land atmosphere carbon exchange that should be reported by each research group in charge of each region. Special attention is given to lateral fluxes, inland water

fluxes, and land use fluxes.

74. The status of carbon neutrality of the world's top 5 CO₂ emitters as seen by carbon satellites

Authors: Fei Jiang; Wei He; Weimin Ju; Hengmao Wang; Mousong Wu; Jun Wang; Shuzhuang Feng; Lingyu Zhang; Jing M. Chen

Journal: Fundamental Research

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Abstract: N/A

75. China's Terrestrial Carbon Sink Over 2010-2015 Constrained by Satellite Observations of Atmospheric CO₂ and Land Surface Variables

Authors: Wei He (Nanjing University, Ministry of Natural Resources); Fei Jiang (Nanjing University, Ministry of Natural Resources); Mousong Wu (Nanjing University, Lund University, Ministry of Natural Resources); Weimin Ju (Nanjing University, Ministry of Natural Resources); Marko Scholze (Lund University); Jing M. Chen (School of Geographical Sciences Fujian Normal University Fuzhou China, University of Toronto); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hengmao Wang (Nanjing University, Ministry of Natural Resources); Jun Wang (Nanjing University, Ministry of Natural Resources); Songhan Wang (Jiangsu Collaborative Innovation Center for Modern Crop Production Key Laboratory of Crop Physiology and Ecology in Southern China Nanjing Agricultural University Nanjing China); Yanlian Zhou (Nanjing University, Ministry of Natural Resources); Chunhua Zhang (School of Resources and Environmental Engineering Ludong University Yantai China); Ngoc Tu Nguyen (Hohai University); Yang Shen (Nanjing University, Ministry of Natural Resources); Zhi Chen (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Biogeosciences

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Keywords: atmospheric inversion, data assimilation, atmospheric CO₂ concentration, multisource satellite observations, China's carbon sink

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Open Access: N/A

Abstract: The magnitude and distribution of China's terrestrial carbon sink remain uncertain due to insufficient constraints at large scales, whereby satellite data offer great potential for reducing the uncertainty. Here, we present two carbon sink estimates for China constrained either by satellite CO₂ column concentrations (XCO₂) within the Global Carbon Assimilation System or by remotely sensed soil moisture and Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) in addition to in situ CO₂ observations within the Carbon Cycle Data Assimilation System. They point to a moderate size of carbon sinks of 0.34 ± 0.14 (mean ± unc.) and 0.43 ± 0.09 PgC/yr during 2010-2015, which are supported by an inventory-based estimate for forest and soil carbon sink (0.26 PgC/yr) and fall in the range of contemporary ensemble atmospheric inversions (0.25-0.48 PgC/yr). They also agree reasonably well on interannual variations, which reflect the carbon sink anomalies induced by regional droughts in southwest China. Furthermore, their spatial distributions are broadly consistent that of the

forest inventory-based estimate, indicating that the largest carbon sinks locate in central and eastern China. Their estimates for forest carbon sink coincide fairly well with the inventory-based estimate across different regions, especially when aggregated to the north and south of China. Although enhanced recently by afforestation, China's carbon sink was also significantly weakened by regional droughts, which were often not fully represented in previous in situ CO₂-based inversions due to insufficient observations. Our results suggest that satellite-based atmospheric CO₂ and land surface observations are vital in characterizing terrestrial net carbon fluxes at regional scales.

76. Resolving temperature limitation on spring productivity in an evergreen conifer forest using a model-data fusion framework

Authors: Stephanie G. Stettz; Nicholas C. Parazoo; A. Anthony Bloom; Peter D. Blanken; David R. Bowling; Sean P. Burns; Cédric Bacour; Fabienne Maignan; Brett Raczka; Alexander J. Norton; Ian Baker; Mathew Williams; Mingjie Shi; Yongguang Zhang; Bo Qiu

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Keywords: N/A

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Open Access: N/A

Abstract: . The flow of carbon through terrestrial ecosystems and the response to climate are critical but highly uncertain processes in the global carbon cycle. However, with a rapidly expanding array of in situ and satellite data, there is an opportunity to improve our mechanistic understanding of the carbon (C) cycle's response to land use and climate change. Uncertainty in temperature limitation on productivity poses a significant challenge to predicting the response of ecosystem carbon fluxes to a changing climate. Here we diagnose and quantitatively resolve environmental limitations on the growing-season onset of gross primary production (GPP) using nearly 2 decades of meteorological and C flux data (2000-2018) at a subalpine evergreen forest in Colorado, USA. We implement the CARbonData-MODEL fraMework (CARDAMOM) model-data fusion network to resolve the temperature sensitivity of spring GPP. To capture a GPP temperature limitation - a critical component of the integrated sensitivity of GPP to temperature - we introduced a cold-temperature scaling function in CARDAMOM to regulate photosynthetic productivity. We found that GPP was gradually inhibited at temperatures below 6.0 °C (± 2.6 °C) and completely inhibited below -7.1 °C (± 1.1 °C). The addition of this scaling factor improved the model's ability to replicate spring GPP at interannual and decadal timescales ($r=0.88$), relative to the nominal CARDAMOM configuration ($r=0.47$), and improved spring GPP model predictability outside of the data assimilation training period ($r=0.88$). While cold-temperature limitation has an important influence on spring GPP, it does not have a significant impact on integrated growing-season GPP, revealing that other environmental controls, such as precipitation, play a more important role in annual productivity. This study highlights growing-season onset temperatures as a key limiting factor for spring growth in winter-dormant evergreen forests, which is critical in understanding future responses to climate change.

77. Four years of global carbon cycle observed from the Orbiting Carbon Observatory 2 (OCO-2) version 9 and in situ data and comparison to OCO-2 version 7

Authors: H      Peiro; Sean Crowell; Andrew Schuh; David F. Baker; Chris O'Dell; Andrew R. Jacobson; Fr        Chevallier; Junjie Liu; Annmarie Eldering; David Crisp; Feng Deng; Brad Weir; Sourish Basu; Matthew S. Johnson; Sajeev Philip; Ian Baker

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Keywords: N/A

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Open Access: N/A

Abstract: . The Orbiting Carbon Observatory 2 (OCO-2) satellite has been providing information to estimate carbon dioxide (CO₂) fluxes at global and regional scales since 2014 through the combination of CO₂ retrievals with top-down atmospheric inversion methods. Column average CO₂ dry-air mole fraction retrievals have been constantly improved. A bias correction has been applied in the OCO-2 version 9 retrievals compared to the previous OCO-2 version 7r improving data accuracy and coverage. We study an ensemble of 10 atmospheric inversions all characterized by different transport models, data assimilation algorithms, and prior fluxes using first OCO-2 v7 in 2015-2016 and then OCO-2 version 9 land observations for the longer period 2015-2018. Inversions assimilating in situ (IS) measurements have also been used to provide a baseline against which the satellite-driven results are compared. The time series at different scales (going from global to regional scales) of the models emissions are analyzed and compared to each experiment using either OCO-2 or IS data. We then evaluate the inversion ensemble based on the dataset from the Total Carbon Column Observing Network (TCCON), aircraft, and in situ observations, all independent from assimilated data. While we find a similar constraint of global total carbon emissions between the ensemble spread using IS and both OCO-2 retrievals, differences between the two retrieval versions appear over regional scales and particularly in tropical Africa. A difference in the carbon budget between v7 and v9 is found over this region, which seems to show the impact of corrections applied in retrievals. However, the lack of data in the tropics limits our conclusions, and the estimation of carbon emissions over tropical Africa require further analysis.

78. OCO-2 Satellite-Imposed Constraints on Terrestrial Biospheric CO₂ Fluxes Over South Asia

Authors: Sajeev Philip (National Aeronautics and Space Administration (NASA) Ames Research Center, Centre for Atmospheric Sciences, Indian Institute of Technology Delhi, Universities Space Research Association); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); David F. Baker (Colorado State University, NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA); Sourish Basu (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Yogesh K. Tiwari (Indian Institute of Tropical Meteorology); Nuggehalli K. Indira (CSIR Fourth Paradigm Institute); Michel Ramonet (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL (CEA-CNRS-UVSQ) Université Paris-Saclay Gif-sur-Yvette France); Benjamin Poulter (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Atmospheres

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Abstract: The spatiotemporal variability of terrestrial biospheric carbon dioxide (CO₂) fluxes over South Asia has large uncertainty. The Orbiting Carbon Observatory 2 (OCO-2) satellite provides much-needed retrievals of column-average CO₂ on a global-scale, with the highest sensitivity to surface CO₂ fluxes and spatiotemporal resolution available to-date. This study conducted global inverse model simulations, assimilating in situ (IS) data and OCO-2 retrievals, to assess optimized CO₂ net ecosystem exchange (NEE) fluxes for South Asia. Annual Net Biome Exchange (NBE = NEE + biomass burning) fluxes over South Asia were estimated to be near neutral (0.04 +/- 0.14 PgC yr⁻¹) using both IS and OCO-2 observations. The most robust result found by assimilating OCO-2 observations was the constraint imposed on the seasonal cycle of NBE fluxes. The amplitude of the seasonal cycle of NEE was found to be larger than previously assumed. The OCO-2 inversion led to an NBE seasonal amplitude of 0.34 PgC month⁻¹, which was larger compared to IS constrained NBE (0.19 PgC month⁻¹) and MSTMIP ensemble mean NEE (0.16 PgC month⁻¹). Moreover, OCO-2 data imposed a phase shift in the NBE seasonal cycle predicted by the prior model. The larger magnitude of NEE seasonality, and phase shift, simulated when assimilating OCO-2 observations are in general agreement with previous studies assimilating regional aircraft observations in addition to global IS observations. This result suggests that OCO-2 provides valuable data that allows for the estimate of NBE on a regional scale in a similar manner as regional in situ aircraft networks.

79. Multi-Season Evaluation of CO₂ Weather in OCO-2 MIP Models

Authors: Li Zhang (Pennsylvania State University); Kenneth J. Davis (Pennsylvania State University); Andrew E. Schuh (Colorado State University); Andrew R. Jacobson (University of Colorado Boulder); Sandip Pal (Texas Tech University); Yu Yan Cui (Pennsylvania State University); David Baker (Colorado State University); Sean Crowell (University of Oklahoma); Frederic Chevallier (Laboratoire des Sciences du Climat et de L'Environnement LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Marine Remaud (Laboratoire des Sciences du Climat et de L'Environnement LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Brad Weir (Now at Morgan State University Baltimore MD USA, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Sajeev Philip (National Aeronautics and Space Administration (NASA) Ames Research Center, Centre for Atmospheric Sciences, Indian Institute of Technology Delhi, Universities Space Research Association); Matthew S. Johnson (National Aeronautics and Space Administration (NASA) Ames Research Center); Feng Deng (University of Toronto); Sourish Basu (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland)

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Keywords: N/A

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Abstract: The ability of current global models to simulate the transport of CO₂ by mid-latitude, synoptic-scale weather systems (i.e., CO₂ weather) is important for inverse estimates of regional and global carbon budgets but remains unclear without comparisons to targeted measurements. Here, we evaluate ten models that participated in the Orbiting Carbon Observatory-2 model intercomparison project (OCO-2 MIP version 9) with intensive aircraft measurements collected from the Atmospheric Carbon Transport (ACT)-America mission. We quantify model-data differences in the spatial variability of CO₂ mole fractions, mean winds, and boundary layer depths in 27 mid-latitude cyclones spanning four seasons over the central and eastern United States. We find that the OCO-2 MIP models are able to simulate observed CO₂ frontal differences with varying degrees of success in summer and spring, and most underestimate frontal differences in winter and autumn. The models may underestimate the observed boundary layer-to-free troposphere CO₂ differences in spring and autumn due to model errors in boundary layer height. Attribution of the causes of model biases in other seasons remains

elusive. Transport errors, prior fluxes, and/or inversion algorithms appear to be the primary cause of these biases since model performance is not highly sensitive to the CO₂ data used in the inversion. The metrics presented here provide new benchmarks regarding the ability of atmospheric inversion systems to reproduce the CO₂ structure of mid-latitude weather systems. Controlled experiments are needed to link these metrics more directly to the accuracy of regional or global flux estimates.

80. WOMBAT v1.0: a fully Bayesian global flux-inversion framework

Authors: Andrew Zammit-Mangion; Michael Bertolacci; Jenny Fisher; Ann Stavert; Matthew Rigby; Yi Cao; Noel Cressie

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . WOMBAT (the Wollongong Methodology for Bayesian Assimilation of Trace-gases) is a fully Bayesian hierarchical statistical framework for flux inversion of trace gases from flask, in situ, and remotely sensed data. WOMBAT extends the conventional Bayesian synthesis framework through the consideration of a correlated error term, the capacity for online bias correction, and the provision of uncertainty quantification on all unknowns that appear in the Bayesian statistical model. We show, in an observing system simulation experiment (OSSE), that these extensions are crucial when the data are indeed biased and have errors that are spatio-temporally correlated. Using the GEOS-Chem atmospheric transport model, we show that WOMBAT is able to obtain posterior means and variances on non-fossil-fuel CO₂ fluxes from Orbiting Carbon Observatory-2 (OCO-2) data that are comparable to those from the Model Intercomparison Project (MIP) reported in Crowell et al. (2019). We also find that WOMBAT's predictions of out-of-sample retrievals obtained from the Total Column Carbon Observing Network (TCCON) are, for the most part, more accurate than those made by the MIP participants.

81. Comments on Global Carbon Budget 2021 (essd-2021-386)

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Journal: Unknown Journal

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Language: N/A

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Open Access: N/A

Abstract: N/A

82. The Carbon Cycle of Southeast Australia During 2019-2020: Drought, Fires, and Subsequent Recovery

Authors: B. Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Yin (California Institute of Technology); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); K. Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. J. Norton (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Joiner (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); D. F. Pollard (National Institute of Water and Atmospheric Research Ltd. (NIWA) Lauder New Zealand); D. W. T. Griffith (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); V. A. Velazco (Deutscher Wetterdienst German Meteorological Service Meteorological Observatory Hohenpeissenberg Germany, Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); N. M. Deutscher (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); N. B. Jones (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); C. Paton-Walsh (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia)

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Keywords: N/A

Funding: National Aeronautics and Space Administration, Jet Propulsion Laboratory, Ames Research Center (and 2 more)

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Open Access: N/A

Abstract: 2019 was the hottest and driest year on record for southeast Australia leading to bushfires of unprecedented extent. Ecosystem carbon losses due to drought and fire are believed to have been substantial, but have not been well quantified. Here, we utilize space-based measurements of trace gases (TROPOspheric Monitoring Instrument XCO, Orbiting Carbon Observatory 2) and up-scaled GPP (FluxSat GPP) to quantify the carbon cycle anomalies resulting from drought and fire in southeast Australia during the 2019-2020 growing season. We find that biomass burning released 113-236 TgC

of CO₂ while drought and fire-induced anomalies in net ecosystem exchange reduced growing season carbon uptake by an additional 19-52 TgC of CO₂. These carbon losses were concentrated during the spring and early summer, when hot-dry conditions were most severe. A shift to cooler conditions with above average rainfall during February is found to result in a partial recovery and greening in unburned ecosystems, but not in fire-impacted areas. The net 2019-2020 carbon loss substantially exceeded interannual variations in net uptake over 2010-2019 estimated from top-down constraints (~5sigma anomaly), and exceeded Australia's annual fossil fuel emissions (~104 TgC year⁻¹). Top-down constraints show that the regional carbon budget is strongly regulated by climate variability, and suggest that cool-wet conditions are required for a rapid recovery of carbon stocks. This has implications for the regional carbon budget as more frequent climate-change-driven heat and drought events may increase the frequency of fire events and the recovery time of ecosystems, threatening the carbon stocks of the region.

83. Was Australia a sink or source of CO₂ in 2015? Data assimilation using OCO-2 satellite measurements

Authors: Yohanna Villalobos; Peter J. Rayner; Jeremy D. Silver; Steven Thomas; Vanessa Haverd; Jürgen Knauer; Zoë M. Loh; Nicholas M. Deutscher; David W. T. Griffith; David F. Pollard

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Abstract: . In this study, we present the assimilation of data from the Orbiting Carbon Observatory-2 (OCO-2) (land nadir and glint data, version 9) to estimate the Australian carbon surface fluxes for the year 2015. To perform this estimation, we used both a regional-scale atmospheric transport-dispersion model and a four-dimensional variational assimilation scheme. Our results suggest that Australia was a carbon sink of -0.41 ± 0.08 PgC yr⁻¹ compared to the prior estimate 0.09 ± 0.20 PgC yr⁻¹ (excluding fossil fuel emissions). Most of the carbon uptake occurred in northern Australia over the savanna ecotype and in the western region over areas with sparse vegetation. Analysis of the enhanced vegetation index (EVI) suggests that the majority of the carbon uptake over the savanna ecosystem was due to an increase of vegetation productivity (positive EVI anomalies) amplified by an anomalous increase of rainfall in summer. Further from this, a slight increase of carbon uptake in Western Australia over areas with sparse vegetation (the largest ecosystem in Australia) was noted due to increased land productivity in the area caused by positive rainfall anomalies. The stronger carbon uptake estimate in this ecosystem was partially due to the land surface model (CABLE-BIOS3) underestimating the gross primary productivity of the ecosystem. To evaluate the accuracy of our carbon flux estimates from OCO-2 retrievals, we compare our posterior concentration fields against the column-averaged carbon retrievals from the Total Carbon Column Observing Network (TCCON) and ground-based in situ monitoring sites located around our domain. The validation analysis against TCCON shows that our system is able to reduce bias mainly in the summer season. Comparison with surface in situ observations was less successful, particularly over oceanic monitoring sites that are strongly affected by oceanic fluxes and subject to less freedom by the inversion. For stations located far from the coast, the comparison with in situ data was more variable, suggesting difficulties matching the column-integrated and surface data by the inversion, most likely linked to model vertical transport. Comparison of our fluxes against the OCO-2 model intercomparison (MIP) was encouraging. The annual carbon uptake estimated by our inversion falls within the ensemble of the OCO-2 MIP global

inversions and presents a similar seasonal pattern.

84. Improving Representation of Tropical Wetland Methane Emissions With CYGNSS Inundation Maps

Authors: Cynthia Gerlein-Safdi (Lawrence Berkeley National Laboratory, University of Michigan); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Genevieve Plant (University of Michigan); Eric A. Kort (University of Michigan); Christopher S. Ruf (University of Michigan)

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Abstract: Wetlands are the single largest source of methane to the atmosphere and their emissions are expected to respond to a changing climate. Inaccuracy and uncertainty in inundation extent drives differences in modeled wetland emissions and impacts representation of wetland emissions on inter-annual and seasonal time frames. Existing wetland maps are based on optical or NIR satellite data obscured by clouds and vegetation, often leading to underestimates in wetlands extent, especially in the Tropics. Here, we present new inundation maps based on the Cyclone Global Navigation Satellite System (CYGNSS) satellite constellation, operating in L-band that is not impacted by clouds or vegetation, providing reliable observations through canopy and cloudy periods. We map the temporal and spatial dynamics of the Pantanal and Sudd wetlands, two of the largest wetlands in the world, using CYGNSS data and a computer vision algorithm. We link these inundation maps to methane fluxes via WetCHARTs, a global wetland methane emissions model ensemble. We contrast CYGNSS-modeled methane emissions with WetCHARTs standard runs that use monthly rainfall data from ECMWF re-analysis (ERA5), as well as the commonly used SWAMPS wetland maps. We find that the CYGNSS-based inundation maps modify the methane emissions in multiple ways. The seasonality of inundation and methane emissions is shifted by two months because of the lag in wetland recharge following peak rainfall. Both inundation and methane emissions also respond non-linearly to wet-season precipitation totals, leading to large interannual variability in emissions. Finally, the annual magnitude of emissions is found to be greater than previously estimated.

85. The NASA Atmospheric Tomography (ATom) Mission: Imaging the Chemistry of the Global Atmosphere

Authors: Chelsea R. Thompson (University of Colorado Boulder); Steven C. Wofsy (Harvard University); Michael J. Prather (University of California Irvine); Paul A. Newman (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Thomas F. Hanisco (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Thomas B. Ryerson (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); David W. Fahey (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Eric C. Apel (National Center for Atmospheric Research); Charles A. Brock (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); William H. Brune (Pennsylvania State University); Karl Froyd (University of Colorado Boulder); Joseph M. Katich (University of Colorado Boulder); Julie M. Nicely (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Jeff Peischl (University of Colorado Boulder); Eric Ray (University of Colorado Boulder); Patrick R. Veres (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Siyuan Wang (National Center for Atmospheric Research); Hannah M. Allen (California Institute of Technology); Elizabeth Asher (National Center for Atmospheric Research); Huisheng Bian (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Donald Blake (University of California Irvine); Ilann Bourgeois (University of Colorado Boulder); John Budney (Harvard University); T. Paul Bui (National Aeronautics and Space Administration (NASA) Ames Research Center); Amy Butler (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Pedro Campuzano-Jost (University of Colorado Boulder); Cecilia Chang (National Aeronautics and Space Administration (NASA) Ames Research Center); Mian Chin (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Róisín Commane (Harvard University); Gus Correa (Columbia University); John D. Crounse (California Institute of Technology); Bruce Daube (Harvard University); Jack E. Dibb (University of New Hampshire); Joshua P. DiGangi (National Aeronautics and Space Administration (NASA) Langley Research Center); Glenn S. Diskin (National Aeronautics and Space Administration (NASA) Langley Research Center); Maximilian Dollner (University of Vienna, Vienna, Austria;); James W. Elkins (NOAA Global Monitoring Laboratory, Boulder, Colorado;); Arlene M. Fiore (Columbia University); Clare M. Flynn (University of California Irvine); Hao Guo (University of California Irvine); Samuel R. Hall (National Center for Atmospheric Research); Reem A. Hannun (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Alan Hills (National Center for Atmospheric Research); Eric J. Hints (University of Colorado Boulder); Alma Hodzic (National Center for Atmospheric Research); Rebecca S. Hornbrook (National Center for Atmospheric Research); L. Greg Huey (Georgia Institute of Technology); Jose L. Jimenez (University of Colorado Boulder); Ralph F. Keeling (Scripps Institution of Oceanography); Michelle J. Kim (California Institute of Technology); Agnieszka Kupc (NOAA Chemical Sciences Laboratory, Boulder, Colorado, and University of Vienna, Vienna, Austria;); Forrest Lacey (National Center for Atmospheric Research); Leslie R. Lait (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Jean-Francois Lamarque (National Center for Atmospheric Research); Junhua Liu (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Kathryn McKain (University of Colorado Boulder); Simone Meinardi (University of California Irvine); David O. Miller (Pennsylvania State University); Stephen A. Montzka (NOAA Global Monitoring Laboratory, Boulder, Colorado;); Fred L. Moore (University of Colorado Boulder); Eric J. Morgan (Scripps Institution of Oceanography); Daniel M. Murphy (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Lee T. Murray (University of Rochester, Rochester, New York); Benjamin A. Nault (University of Colorado Boulder); J. Andrew Neuman (University of Colorado Boulder); Louis Nguyen (National Aeronautics and Space Administration (NASA) Langley Research Center); Yenny Gonzalez (Harvard University); Andrew Rollins (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Karen Rosenlof (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Maryann Sargent (Harvard University); Gregory Schill (University of Colorado Boulder); Joshua P. Schwarz (NOAA Chemical Sciences Laboratory, Boulder, Colorado;); Jason M. St. Clair (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space

Flight Center, University of Maryland); Stephen D. Steenrod (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Britton B. Stephens (National Center for Atmospheric Research); Susan E. Strahan (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Sarah A. Strode (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Colm Sweeney (NOAA Global Monitoring Laboratory, Boulder, Colorado); Alexander B. Thames (Pennsylvania State University); Kirk Ullmann (National Center for Atmospheric Research); Nicholas Wagner (University of Colorado Boulder); Rodney Weber (Georgia Institute of Technology); Bernadett Weinzierl (University of Vienna, Vienna, Austria); Paul O. Wennberg (California Institute of Technology); Christina J. Williamson (University of Colorado Boulder); Glenn M. Wolfe (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Linghan Zeng (Georgia Institute of Technology)

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Abstract: This article provides an overview of the NASA Atmospheric Tomography (ATom) mission and a summary of selected scientific findings to date. ATom was an airborne measurements and modeling campaign aimed at characterizing the composition and chemistry of the troposphere over the most remote regions of the Pacific, Southern, Atlantic, and Arctic Oceans, and examining the impact of anthropogenic and natural emissions on a global scale. These remote regions dominate global chemical reactivity and are exceptionally important for global air quality and climate. ATom data provide the in situ measurements needed to understand the range of chemical species and their reactions, and to test satellite remote sensing observations and global models over large regions of the remote atmosphere. Lack of data in these regions, particularly over the oceans, has limited our understanding of how atmospheric composition is changing in response to shifting anthropogenic emissions and physical climate change. ATom was designed as a global-scale tomographic sampling mission with extensive geographic and seasonal coverage, tropospheric vertical profiling, and detailed speciation of reactive compounds and pollution tracers. ATom flew the NASA DC-8 research aircraft over four seasons to collect a comprehensive suite of measurements of gases, aerosols, and radical species from the remote troposphere and lower stratosphere on four global circuits from 2016 to 2018. Flights maintained near-continuous vertical profiling of 0.15-13-km altitudes on long meridional transects of the Pacific and Atlantic Ocean basins. Analysis and modeling of ATom data have led to the significant early

findings highlighted here.

86. Emergent constraints on tropical atmospheric aridity--carbon feedbacks and the future of carbon sequestration

Authors: Armineh Barkhordarian; Kevin W Bowman; Noel Cressie; Jeffrey Jewell; Junjie Liu

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Abstract: Carbon-climate feedbacks, which amplifies or attenuates atmospheric CO₂ from fossil fuel emissions, are one of the largest sources of uncertainty in climate projections. However, these feedbacks depend both on temperature and its coupling to water and energy cycles, especially in the tropics. We show that atmospheric aridity--quantified as vapor pressure deficit (VPD)--is a good proxy for this coupling. Tropical VPD is strongly correlated to the global CO₂ growth rate (CGR) with observed present-day sensitivities of -2.5 ± 0.4 GtC mb⁻¹ yr⁻¹. The sensitivity of CGR to tropical VPD interannual variability has increased by a factor of 1.7 ± 0.3 in the 21st century. A combination of causality and statistical analysis point to mechanistic moisture drivers of the VPD-CGR sensitivities, independent of temperature. Observational records provide evidence that tropical atmospheric aridity is linked to both water deficit and spatially correlated with evaporative fraction suggesting that CGR variability is indirectly driven by land--atmosphere coupling (compound soil and atmospheric drought). This coupling is manifest as a kind of carbon-climate feedback in CMIP6 Earth System Models where long-term increases in tropical VPD reduce tropical carbon storage but with a substantial inter-model range [-1.4 to -59.4 GtC mb⁻¹]. However, by employing a hierarchical emergent constraint, the best estimate of atmospheric aridity--carbon cycle feedback (ϕ_{TL}) is -19 ± 10 GtC mb⁻¹, which is 28% lower than model estimates with an uncertainty reduction of 50%. Our results bridge the role of moisture and land-atmosphere coupling on net carbon variability to the vulnerability of carbon storage in a changing climate.

87. Satellite Constraints on the Latitudinal Distribution and Temperature Sensitivity of Wetland Methane Emissions

Authors: Shuang Ma (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yuzhong Zhang (Institute of Advanced Technology Westlake Institute for Advanced Study Hangzhou China, Key Laboratory of Coastal Environment and Resources of Zhejiang Province School of Engineering Westlake University Hangzhou China); Benjamin Poulter (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Daniel H. Cusworth (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yi Yin (California Institute of Technology); Sudhanshu Pandey (SRON Netherlands Institute for Space Research Utrecht The Netherlands); Joannes D. Maasakkers (SRON Netherlands Institute for Space Research Utrecht The Netherlands); Xiao Lu (Harvard University, Sun Yat-Sen University); Lu Shen (Harvard University); Jianxiong Sheng (Massachusetts Institute of Technology); Christian Frankenberg (California Institute of Technology); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Daniel J. Jacob (Harvard University)

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Abstract: Wetland methane (CH₄) emissions comprise about one-third of the global CH₄ source. The latitudinal distribution and climate sensitivity of wetland CH₄ fluxes are the key determinants of the global CH₄-climate feedback. However, large differences exist between bottom-up estimates, informed by ground-based flux measurements, and top-down estimates derived from spaceborne total column CH₄. Despite the extensive coverage of satellite CH₄ concentration observations, challenges remain with using top-down estimates to test bottom-up models, mainly because of the uncertainties in the satellite retrievals, the model representation errors, the variable prior emissions, and the confounding role of the posterior error covariance structures. Here, we use satellite-based top-down CH₄ flux estimates (2010-2012) to test and refine 42 bottom-up estimates of wetland emissions that use a range of hypothesized wetland extents and process controls. Our comparison between bottom-up models and

satellite-based fluxes innovatively accounts for cross-correlations and spatial uncertainties typically found in top-down inverse estimates, such that only the information from satellite observations and the atmospheric transport model is kept as a constraint. We present a satellite-constrained wetland CH₄ ensemble product derived from assembling the highest-performance bottom-up models, which estimates global wetland CH₄ emissions of 148 (117-189, 5th-95th percentile) Tg CH₄ yr⁻¹. We find that tropical wetland emissions contribute 72% (63%-85%) to the global wetland total. We also find that a lower-than-expected temperature sensitivity agrees better with atmospheric CH₄ measurements. Overall, our approach demonstrates the potential for using satellites to quantitatively refine bottom-up wetland CH₄ emission estimates, their latitudinal distributions, and their sensitivity to climate.

88. Covariation of Airborne Biogenic Tracers (CO₂, COS, and CO) Supports Stronger Than Expected Growing Season Photosynthetic Uptake in the Southeastern US

Authors: Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bianca C. Baier (NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA, Cooperative Institute for Research in Environmental Sciences University of Colorado-Boulder Boulder CO USA); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Le Kuai (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yoichi Shiga (Universities Space Research Association); Ian Baker (Colorado State University); Mary E. Whelan (Department of Environmental Sciences Rutgers, The State University of New Jersey New Brunswick NJ USA); Sha Feng (Pennsylvania State University); Maarten Krol (Wageningen University, Utrecht University); Colm Sweeney (NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA); Benjamin R. Runkle (University of Arkansas); Elahe Tajfar (University of Arkansas); Kenneth J. Davis (Pennsylvania State University)

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Abstract: The Atmospheric Carbon Transport (ACT)-America Earth Venture mission conducted five airborne campaigns across four seasons from 2016 to 2019, to study the transport and fluxes of Greenhouse gases across the eastern United States. Unprecedented spatial sampling of atmospheric tracers (CO₂, carbon monoxide [CO], carbonyl sulfide [COS]) related to biospheric processes offers opportunities to improve our qualitative and quantitative understanding of seasonal and spatial patterns of biospheric carbon uptake. Here, we examine co-variation of boundary layer enhancements of CO₂, CO, and COS across three diverse regions: the crop-dominated Midwest, evergreen-dominated South, and deciduous broadleaf-dominated Northeast. To understand the biogeochemical processes controlling these tracers, we compare the observed co-variation to simulated co-variation resulting from model- and satellite- constrained surface carbon fluxes. We found indication of a common terrestrial

biogenic sink of CO₂ and COS and secondary production of CO from biogenic sources in summer throughout the eastern US, driven by stomatal conductance. Upper Midwest crops drive CO₂ and COS depletion from early to late summer. Northeastern temperate forests drive CO₂ and COS depletion in late summer. The unprecedented ACT-America flask samples uncovered evidence that southern humid temperate forests photosynthesize and absorb CO₂ and COS, and emit CO precursors, deep into the growing season. Satellite- constrained carbon fluxes capture much of the observed seasonal and spatial variability, but underestimate the magnitude of net CO₂ and COS depletion in the South, indicating a stronger than expected net sink of CO₂ in late summer. Additional sampling of the South will more accurately constrain underlying biological processes and climate sensitivities governing southern carbon dynamics.

89. Examining CO₂ Model Observation Residuals Using ACT-America Data

Authors: Tobias Gerken (James Madison University, Pennsylvania State University); Sha Feng (Pacific Northwest National Laboratory, Pennsylvania State University); Klaus Keller (Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement CEA CNRS UVSQ/IPSL Université Paris-Saclay Gif-sur-Yvette CEDEX France); Joshua P. DiGangi (National Aeronautics and Space Administration (NASA) Langley Research Center); Yonghoon Choi (National Aeronautics and Space Administration (NASA) Langley Research Center, Science Systems and Applications Inc. Hampton VA USA); Bianca Baier (NOAA Global Monitoring Laboratory Boulder CO USA, Cooperative Institute for Research in Environmental Sciences University of Colorado-Boulder Boulder CO USA); Kenneth J. Davis (Pennsylvania State University)

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Abstract: Atmospheric inversion typically relies on the specification of prior flux and atmospheric model transport errors, which have large uncertainties. Here, we used ACT-America airborne observations to compare model observation mismatch in the eastern U.S. and during four climatological seasons for the mesoscale WRF(-Chem) and global scale CarbonTracker/TM5 (CT) models. Models used identical surface carbon fluxes, and CT was used as boundary condition for WRF. Both models showed reasonable agreement with observations, and residuals follow near symmetric peaked (i.e., non-Gaussian) distribution with near-zero bias of both models (CT: ppm; WRF: ppm). We also found large magnitude residuals at the tails of the distribution that contribute considerably to overall bias. Atmospheric boundary-layer biases (1-10 ppm) were much larger than free tropospheric biases (0.5-1 ppm) and were of same magnitude as model-model differences, whereas free tropospheric biases were mostly governed by background conditions. Results revealed systematic differences in atmospheric transport, most pronounced in the warm and cold sectors of synoptic systems, highlighting the importance of transport for residuals. While CT could reproduce the principal dynamics associated with synoptic systems, WRF showed a clearer distinction for differences across fronts. Variograms were used to quantify spatial correlation of residuals and showed characteristic residual length scales of approximately 100-300 km. Our findings suggest that inclusion of synoptic weather-dependent and non-Gaussian error structure may benefit inversion systems.

90. A Data-Driven Global Soil Heterotrophic Respiration Dataset and the Drivers of Its Inter-Annual Variability

Authors: Yitong Yao (LSCE/IPSL CEA-CNRS-UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Philippe Ciais (LSCE/IPSL CEA-CNRS-UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Nicolas Viovy (LSCE/IPSL CEA-CNRS-UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Wei Li (Tsinghua University); Fabio Cresto-Aleina (LSCE/IPSL CEA-CNRS-UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Hui Yang (LSCE/IPSL CEA-CNRS-UVSQ Laboratoire des Sciences du Climat et de l'Environnement Université Paris-Saclay Gif-sur-Yvette France); Emilie Joetzjer (CNRS CNRM Université de Toulouse Météo-France Toulouse France); Ben Bond-Lamberty (University of Maryland, Pacific Northwest National Laboratory)

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Abstract: Soil heterotrophic respiration (SHR) is important for carbon-climate feedbacks because of its sensitivity to soil carbon, climatic conditions and nutrient availability. However, available global SHR estimates have either a coarse spatial resolution or rely on simple upscaling formulations. To better quantify the global distribution of SHR and its response to climate variability, we produced a new global SHR data set using Random Forest, up-scaling 455 point data from the Global Soil Respiration Database (SRDB 4.0) with gridded fields of climatic, edaphic and productivity. We estimated a global total SHR of Pg C yr⁻¹ over 1985-2013 with a significant increasing trend of 0.03 Pg C yr⁻². Among the inputs to generate SHR products, the choice of soil moisture datasets contributes more to the difference among SHR ensemble. Water availability dominates SHR inter-annual variability (IAV) at the global scale; more precisely, temperature strongly controls the SHR IAV in tropical forests, while water availability dominates in extra-tropical forest and semi-arid regions. Our machine-learning SHR ensemble of data-driven gridded estimates and outputs from process-based models (TRENDYv6) shows agreement for a strong association between water variability and SHR IAV at the global scale, but ensemble members exhibit different ecosystem-level SHR IAV controllers. The important role of water availability in driving SHR suggests both a direct effect limiting decomposition and an indirect effect on litter available from productivity. Considering potential uncertainties remaining in our data-driven SHR datasets, we call for more scientifically designed SHR observation network and

deep-learning methods making maximum use of observation data.

91. Changes in global terrestrial live biomass over the 21st century

Authors: Liang Xu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Sassan S. Saatchi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Yan Yang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yifan Yu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Julia Pongratz (Ludwig-Maximilians-Universität Munich, Luisenstr. 37, 80333 Munich, Germany., Max Planck Institute for Meteorology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yi Yin (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Grant Domke (U.S. Department of Agriculture, Forest Service, St. Paul, MN, USA.); Ronald E. McRoberts (University of Minnesota); Christopher Woodall (U.S. Forest Service, Northern Research Station, Durham, NH, USA.); Gert-Jan Nabuurs (Wageningen University); Sergio de-Miguel (Department of Crop and Forest Sciences, University of Lleida, Lleida, Spain., Joint Research Unit CTFC - AGROTECNIO, Solsona, Lleida, Spain.); Michael Keller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, USDA Forest Service, International Institute of Tropical Forestry, San Juan, Puerto Rico.); Nancy Harris (World Resources Institute, 10 G Street NE, Washington, DC, USA.); Sean Maxwell (School of Earth and Environmental Sciences, University of Queensland, Brisbane, QLD, Australia.); David Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: Live woody vegetation is the largest reservoir of biomass carbon, with its restoration considered one of the most effective natural climate solutions. However, terrestrial carbon fluxes remain the largest uncertainty in the global carbon cycle. Here, we develop spatially explicit estimates

of carbon stock changes of live woody biomass from 2000 to 2019 using measurements from ground, air, and space. We show that live biomass has removed 4.9 to 5.5 PgC year⁻¹ from the atmosphere, offsetting 4.6 $\pm$ 0.1 PgC year⁻¹ of gross emissions from disturbances and adding substantially (0.23 to 0.88 PgC year⁻¹) to the global carbon stocks. Gross emissions and removals in the tropics were four times larger than temperate and boreal ecosystems combined. Although live biomass is responsible for more than 80% of gross terrestrial fluxes, soil, dead organic matter, and lateral transport may play important roles in terrestrial carbon sink.

92. Resolving temperature limitation on spring productivity in an evergreen conifer forest using a model-data fusion framework

Authors: Stephanie G. Stettz; Nicholas C. Parazoo; A. Anthony Bloom; Peter D. Blanken; David R. Bowling; Sean P. Burns; Cédric Bacour; Fabienne Maignan; Brett Raczka; Alexander J. Norton; Ian Baker; Mathew Williams; Mingjie Shi; Yongguang Zhang; Bo Qiu

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Abstract: . The flow of carbon through terrestrial ecosystems and the response to climate is a critical but highly uncertain process in the global carbon cycle. However, with a rapidly expanding array of in situ and satellite data, there is an opportunity to improve our mechanistic understanding of the carbon (C) cycle's response to land use and climate change. Uncertainty in temperature limitation on productivity pose a significant challenge to predicting the response of ecosystem carbon fluxes to a changing climate. Here we diagnose and quantitatively resolve environmental limitations on growing season onset of gross primary production (GPP) using nearly two decades of meteorological and C flux data (2000-2018) at a subalpine evergreen forest in Colorado USA. We implement the CARDAMOM model-data fusion network to resolve the temperature sensitivity of spring GPP. To capture a GPP temperature limitation - a critical component of integrated sensitivity of GPP to temperature - we introduced a cold temperature scaling function in CARDAMOM to regulate photosynthetic productivity. We found that GPP was gradually inhibited at temperature below 6.0 degreesC (± 2.6 degreesC) and completely inhibited below -7.1 degreesC (± 1.1 degreesC). The addition of this scaling factor improved the model's ability to replicate spring GPP at interannual and decadal time scales ($r = 0.88$), relative to the nominal CARDAMOM configuration ($r = 0.47$), and improved spring GPP model predictability outside of the data assimilation training period ($r = 0.88$). While cold temperature limitation has an important influence on spring GPP, it does not have a significant impact on integrated growing season GPP, revealing that other environmental controls, such as precipitation, play a more important role in annual productivity. This study highlights growing season onset temperature as a key limiting factor for spring growth in winter-dormant evergreen forests, which is critical in understanding future responses to climate change.

93. Joint CO₂ Mole Fraction and Flux Analysis Confirms Missing Processes in CASA Terrestrial Carbon Uptake Over North America

Authors: Sha Feng (Pacific Northwest National Laboratory, Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement CEA, CNRS, UVSQ/IPSL Université Paris-Saclay Orme des Merisiers Gif-sur-Yvette France, Pennsylvania State University); Christopher A. Williams (Graduate School of Geography Clark University Worcester MA USA); Kenneth J. Davis (Pennsylvania State University); Yu Zhou (Graduate School of Geography Clark University Worcester MA USA); Ian Baker (Colorado State University); Zachary R. Barkley (Pennsylvania State University); Daniel Wesloh (Pennsylvania State University)

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Abstract: Terrestrial biosphere models (TBMs) play a key role in the detection and attribution of carbon cycle processes at local to global scales and in projections of the coupled carbon-climate system. TBM evaluation commonly involves direct comparison to eddy-covariance flux measurements. We use atmospheric CO₂ mole fraction ([CO₂]) measured in situ from aircraft and tower, in addition to flux-measurements from summer 2016 to evaluate the Carnegie-Ames-Stanford-Approach (CASA) TBM. WRF-Chem is used to simulate [CO₂] using biogenic CO₂ fluxes from a CASA parameter-based ensemble and CarbonTracker version 2017 (CT2017) in addition to transport and CO₂ boundary condition ensembles. The resulting "super ensemble" of modeled [CO₂] demonstrates that the biosphere introduces the majority of uncertainty to the simulations. Both aircraft and tower [CO₂] data show that the CASA ensemble net ecosystem exchange (NEE) of CO₂ is biased high (NEE too positive) and identify the maximum light use efficiency E_{max} a key parameter that drives the spread of the CASA ensemble in summer 2016. These findings are verified with flux-measurements. The direct comparison of the CASA flux ensemble with flux-measurements confirms missing sink processes in CASA. Separating the daytime and nighttime flux, we discover that the underestimated net uptake results from missing sink processes that result in overestimation of respiration. NEE biases are smaller in the CT2017 posterior biogenic fluxes, which assimilate observed [CO₂]. Flux tower analyses reveal an unrealistic overestimation of nighttime respiration in CT2017 which we attribute to limited flexibility in the inversion strategy.

94. Evaluation of CarbonTracker's Inverse Estimates of North American Net Ecosystem Exchange of CO₂ From Different Observing Systems Using ACT-America Airborne Observations

Authors: Yu Yan Cui (Pennsylvania State University); Andrew R. Jacobson (University of Colorado Boulder, NOAA Earth System Research Laboratory Global Monitoring Laboratory Boulder CO USA); Sha Feng (Pacific Northwest National Laboratory, Pennsylvania State University); Daniel Wesloh (Pennsylvania State University); Zachary R. Barkley (Pennsylvania State University); Li Zhang (Pennsylvania State University); Tobias Gerken (James Madison University, Pennsylvania State University); Klaus Keller (Pennsylvania State University); David Baker (Colorado State University); Kenneth J Davis (Pennsylvania State University)

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Keywords: regional evaluation, biogenic CO₂ fluxes, Aircraft campaign

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Open Access: N/A

Abstract: Quantification of regional terrestrial carbon dioxide (CO₂) fluxes is critical to our understanding of the carbon cycle. We evaluate inverse estimates of net ecosystem exchange (NEE) of CO₂ fluxes in temperate North America, and their sensitivity to the observational data used to drive the inversions. Specifically, we consider the state-of-the-science CarbonTracker global inversion system, which assimilates (a) in situ measurements (IS), (b) the Orbiting Carbon Observatory-2 (OCO-2) v9 column CO₂ (XCO₂) retrievals over land (LNLG), (c) OCO-2 v9 XCO₂ retrievals ocean-glint (OG), and (d) a combination of all these observational constraints (LNLGOGIS). We use independent CO₂ observations from the Atmospheric Carbon and Transport (ACT)--America aircraft mission to evaluate the inversions. We diagnose errors in the flux estimates using the differences between modeled and observed biogenic CO₂ mole fractions, influence functions from a Lagrangian transport model, Bayesian inference, and root-mean-square error (RMSE) and bias metrics. The IS fluxes have the smallest RMSE among the four products, followed by LNLG. Both IS and LNLG outperform the OG and LNLGOGIS inversions with regard to RMSE. Regional errors do not differ markedly across the four sets of posterior fluxes. The CarbonTracker inversions appear to overestimate the seasonal cycle of NEE in the Midwest and Western Canada, and overestimate dormant season NEE across the Central and Eastern US. The CarbonTracker inversions may overestimate annual NEE in the Central and Eastern US. The success of the LNLG inversion with respect to independent observations bodes well for satellite-based inversions in regions with more limited in situ observing networks.

95. Atmospheric Carbon and Transport - America (ACT-America) Data Sets: Description, Management, and Delivery

Authors: Y. Wei (Oak Ridge National Laboratory); R. Shrestha (Oak Ridge National Laboratory); S. Pal (Texas Tech University); T. Gerken (James Madison University); S. Feng (Pacific Northwest National Laboratory, Pennsylvania State University); J. McNelis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, Oak Ridge National Laboratory); D. Singh (Oak Ridge National Laboratory); M. M. Thornton (Oak Ridge National Laboratory); A. G. Boyer (Now at Reviewbox Knoxville TN USA, Oak Ridge National Laboratory); M. A. Shook (National Aeronautics and Space Administration (NASA) Langley Research Center); G. Chen (National Aeronautics and Space Administration (NASA) Langley Research Center); B. C. Baier (University of Colorado Boulder); Z. R. Barkley (Pennsylvania State University); J. D. Barrick (National Aeronautics and Space Administration (NASA) Langley Research Center); J. R. Bennett (Bay Area Environmental Research Institute Moffett Field CA USA); E. V. Browell (National Aeronautics and Space Administration (NASA) Langley Research Center); J. F. Campbell (National Aeronautics and Space Administration (NASA) Langley Research Center); L. J. Campbell (Pennsylvania State University); Y. Choi (National Aeronautics and Space Administration (NASA) Langley Research Center); J. Collins (National Aeronautics and Space Administration (NASA) Langley Research Center); J. Dobler (Spectral Sensor Solutions LLC Fort Wayne IN USA); M. Eckl (Deutsches Zentrum für Luft- und Raumfahrt (DLR) Institut für Physik der Atmosphäre Bavaria Germany); A. Fiehn (Deutsches Zentrum für Luft- und Raumfahrt (DLR) Institut für Physik der Atmosphäre Bavaria Germany); A. Fried (University of Colorado Boulder); J. P. Digangi (National Aeronautics and Space Administration (NASA) Langley Research Center); R. Barton-Grimley (National Aeronautics and Space Administration (NASA) Langley Research Center); H. Halliday (National Aeronautics and Space Administration (NASA) Langley Research Center, Now at U.S. Environmental Protection Agency Washington, D.C USA); T. Klausner (Deutsches Zentrum für Luft- und Raumfahrt (DLR) Institut für Physik der Atmosphäre Bavaria Germany); S. Kooi (National Aeronautics and Space Administration (NASA) Langley Research Center); J. Kostinek (Institute of Atmospheric Physics German Aerospace Center Mecklenburg-Vorpommern Germany); T. Lauvaux (Laboratory for Sciences of Climate and Environment Paris France); B. Lin (National Aeronautics and Space Administration (NASA) Langley Research Center); M. J. McGill (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); B. Meadows (National Aeronautics and Space Administration (NASA) Langley Research Center); N. L. Miles (Pennsylvania State University); A. R. Nehrir (National Aeronautics and Space Administration (NASA) Langley Research Center); J. B. Nowak (National Aeronautics and Space Administration (NASA) Langley Research Center); M. Obland (National Aeronautics and Space Administration (NASA) Langley Research Center); C. O'Dell (Colorado State University); R. M. P. Fao (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); S. J. Richardson (Pennsylvania State University); D. Richter (University of Colorado Boulder); A. Roiger (Deutsches Zentrum für Luft- und Raumfahrt (DLR) Institut für Physik der Atmosphäre Bavaria Germany); C. Sweeney (NOAA Global Monitoring Laboratory Boulder CO USA); J. Walega (University of Colorado Boulder); P. Weibring (University of Colorado Boulder); C. A. Williams (Graduate School of Geography Clark University Worcester MA USA); M. M. Yang (Deutsches Zentrum für Luft- und Raumfahrt (DLR) Institut für Physik der Atmosphäre Bavaria Germany); Y. Zhou (Graduate School of Geography Clark University Worcester MA USA); K. J. Davis (Pennsylvania State University)

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Keywords: data description, trace gases, meteorological properties, atmospheric CO₂, atmospheric CH₄ (and 1 more)

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Abstract: The ACT-America project is a NASA Earth Venture Suborbital-2 mission designed to study the transport and fluxes of greenhouse gases. The open and freely available ACT-America data sets provide airborne in situ measurements of atmospheric carbon dioxide, methane, trace gases, aerosols, clouds, and meteorological properties, airborne remote sensing measurements of aerosol backscatter, atmospheric boundary layer height and columnar content of atmospheric carbon dioxide, tower-based measurements, and modeled atmospheric mole fractions and regional carbon fluxes of greenhouse gases over the Central and Eastern United States. We conducted 121 research flights during five campaigns in four seasons during 2016-2019 over three regions of the US (Mid-Atlantic, Midwest and South) using two NASA research aircraft (B-200 and C-130). We performed three flight patterns (fair weather, frontal crossings, and OCO-2 underflights) and collected more than 1,140 h of airborne measurements via level-leg flights in the atmospheric boundary layer, lower, and upper free troposphere and vertical profiles spanning these altitudes. We also merged various airborne in situ measurements onto a common standard sampling interval, which brings coherence to the data, creates geolocated data products, and makes it much easier for the users to perform holistic analysis of the ACT-America data products. Here, we report on detailed information of data sets collected, the workflow for data sets including storage and processing of the quality controlled and quality assured harmonized observations, and their archival and formatting for users. Finally, we provide some important information on the dissemination of data products including metadata and highlights of applications of ACT-America data sets.

96. Five years of variability in the global carbon cycle: comparing an estimate from the Orbiting Carbon Observatory-2 and process-based models

Authors: Zichong Chen; Deborah N Huntzinger; Junjie Liu; Shilong Piao; Xuhui Wang; Stephen Sitch; Pierre Friedlingstein; Peter Anthoni; Almut Arneth; Vladislav Bastrikov; Daniel S Goll; Vanessa Haverd; Atul K Jain; Emilie Joetzjer; Etsushi Kato; Sebastian Lienert; Danica L Lombardozzi; Patrick C McGuire; Joe R Melton; Julia E M S Nabel; Julia Pongratz; Benjamin Poulter; Hanqin Tian; Andrew J Wiltshire; Sönke Zaehle; Scot M Miller

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Keywords: N/A

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Open Access: N/A

Abstract: Year-to-year variability in CO₂ fluxes can yield insight into climate-carbon cycle relationships, a fundamental yet uncertain aspect of the terrestrial carbon cycle. In this study, we use global observations from NASA's Orbiting Carbon Observatory-2 (OCO-2) satellite for years 2015-2019 and a geostatistical inverse model to evaluate 5 years of interannual variability (IAV) in CO₂ fluxes and its relationships with environmental drivers. OCO-2 launched in late 2014, and we specifically evaluate IAV during the time period when OCO-2 observations are available. We then compare inferences from OCO-2 with state-of-the-art process-based models (terrestrial biosphere model, TBMs). Results from OCO-2 suggest that the tropical grasslands biome (including grasslands, savanna, and agricultural lands within the tropics) makes contributions to global IAV during the 5 year study period that are comparable to tropical forests, a result that differs from a majority of TBMs. Furthermore, existing studies disagree on the environmental variables that drive IAV during this time period, and the analysis using OCO-2 suggests that both temperature and precipitation make comparable contributions. TBMs, by contrast, tend to estimate larger IAV during this time and usually estimate larger relative contributions from the extra-tropics. With that said, TBMs show little consensus on both the magnitude and the contributions of different regions to IAV. We further find that TBMs show a wide range of responses on the relationships of CO₂ fluxes with annual anomalies in temperature and precipitation, and these relationships across most of the TBMs have a larger magnitude than inferred from OCO-2. Overall, the findings of this study highlight large uncertainties in process-based estimates of IAV during recent years and provide an avenue for evaluating these processes against inferences from OCO-2.

97. Carbon Dioxide Distribution, Origins, and Transport Along a Frontal Boundary During Summer in Mid-Latitudes

Authors: Arkayan Samaddar (Pennsylvania State University); Sha Feng (Pacific Northwest National Laboratory, Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement CEA CNRS UVSQ/IPSL Université Paris-Saclay Orme des Merisiers Gif-sur-Yvette Cedex France); Zachary R. Barkley (Pennsylvania State University); Sandip Pal (Texas Tech University); Kenneth J. Davis (Pennsylvania State University)

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Subjects: N/A

Keywords: mechanisms of CO₂ transport², frontal transport of CO₂, high resolution simulation of CO

Funding: Earth Sciences Division, Pennsylvania State University

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Abstract: Synoptic weather systems are a major driver of spatial gradients in atmospheric CO₂ mole fractions. During frontal passages, air masses from different regions meet at the frontal boundary creating significant gradients in CO₂ mole fractions. We quantitatively describe the atmospheric transport of CO₂ mole fractions during a mid-latitude cold front passage and explore the impact of various sources of CO₂. We focus here on a cold front passage over Lincoln, Nebraska on August 4th, 2016 observed by aircraft during the Atmospheric Carbon and Transport-America campaign. A band of air with elevated CO₂ was located along the frontal boundary. Observed and simulated differences in CO₂ across the front were as high as 25 ppm. Numerical simulations using Weather Research and Forecasting Model with Chemistry at cloud resolving resolutions (3 km), coupled with CO₂ surface fluxes and boundary conditions from CarbonTracker (CT-NRTv2017x), were performed to explore atmospheric transport at the front. Model results demonstrate that the frontal CO₂ difference in the upper troposphere can be explained largely by inflow from outside of North America. This difference is modified in the atmospheric boundary layer and lower troposphere by continental surface fluxes, dominated in this case by biogenic and fossil fuel fluxes. Horizontal and vertical advection are found to be responsible for the transport of CO₂ mole fractions along the frontal boundary. We show that cold front passages lead to large CO₂ transport events including a significant contribution from vertical advection, and that midcontinent frontal boundaries are formed from a complex mixture of CO₂ sources.

98. GIS-Based Approach to Spatio-Temporal Interpolation of Atmospheric CO₂ Concentrations in Limited Monitoring Dataset

Authors: Yaroslav Bezyk (Wroclaw University of Science and Technology); Izabela Sówka (Wroclaw University of Science and Technology); Maciej Górka (University of Wroclaw); Jan Blachowski (Wroclaw University of Science and Technology)

Journal: Atmosphere

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Funding: N/A

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Open Access: N/A

Abstract: Understanding the magnitude and distribution of the mixes of the near-ground carbon dioxide (CO₂) components spatially (related to the surface characteristics) and temporally (over seasonal timescales) is critical to evaluating present and future climate impacts. Thus, the application of in situ measurement approaches, combined with the spatial interpolation methods, will help to explore variations in source contribution to the total CO₂ mixing ratios in the urban atmosphere. This study presents the spatial characteristic and temporal trend of atmospheric CO₂ levels observed within the city of Wroclaw, Poland for the July 2017-August 2018 period. The seasonal variability of atmospheric CO₂ around the city was directly measured at the selected sites using flask sampling with a Picarro G2201-I Cavity Ring-Down Spectroscopy (CRDS) technique. The current work aimed at determining the accuracy of the interpolation techniques and adjusting the interpolation parameters for estimating the magnitude of CO₂ time series/seasonal variability in terms of limited observations during the vegetation and non-vegetation periods. The objective was to evaluate how different interpolation methods will affect the assessment of air pollutant levels in the urban environment and identify the optimal sampling strategy. The study discusses the schemes for optimization of the interpolation results that may be adopted in areas where no observations are available, which is based on the kriging error predictions for an appropriate spatial density of measurement locations. Finally, the interpolation results were extended regarding the average prediction bias by exploring additional experimental configurations and introducing the limitation of the future sampling strategy on the seasonal representation of the CO₂ levels in the urban area.

99. Co-evolution of carbon cycle and air quality fluxes constrained by CMS-Flux and MOMO-Chem assimilation systems

Authors: Kevin Bowman; Kazuyuki Miyazaki; Junjie Liu; Anthony Bloom

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Carbon flux|Carbon fibers|Atmospheric carbon cycle

Funding: N/A

License: N/A

Open Access: N/A

Abstract: <p>Rapid regional changes in anthropogenic emissions in response to the COVID-19 pandemic have underscored the contribution of fossil fuel (FF) emission uncertainty to regional carbon budgets. Typical methods for spatially-explicit FF emissions are dependent on national reporting, which can incur substantial latencies. However, the concomitant changes in short-lived pollutants from common emission sources point to opportunities to develop independent low-latency estimates of fossil fuel emissions and to better understand anthropogenic processes. Here we combine state-of-the-art Multiple Model Multi Constituent chemical data assimilation system (MOMO-Chem) with bottom-up FF emissions to repartition the net carbon fluxes from the NASA Carbon Monitoring System Flux (CMS-Flux) project. To that end, we implement a novel Kalman filtering algorithm that predicts emission ratio co-evolution of air quality (AQ) and carbon species. Based upon top-down estimates of AQ emissions, FF CO₂ emissions and uncertainties can be rapidly determined. We show overall good agreement between predicted FF fluxes and the latest bottom-up inventories. These data are in turn used to interpret the decadal evolution of CMS-Flux net carbon exchange. This approach is an important step in quantifying both regional fossil fuel and natural carbon fluxes contributions to the atmospheric CO₂ growth rate.</p>

100. Carbon Monitoring System Flux Net Biosphere Exchange 2020 (CMS-Flux NBE 2020)

Authors: Junjie Liu; Latha Baskaran; Kevin Bowman; David Schimel; A. Anthony Bloom; Nicholas C. Parazoo; Tomohiro Oda; Dustin Carroll; Dimitris Menemenlis; Joanna Joiner; Roisin Commene; Bruce Daube; Lucianna V. Gatti; Kathryn McKain; John Miller; Britton B. Stephens; Colm Sweeney; Steven Wofsy

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Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . Here we present a global and regionally resolved terrestrial netbiosphere exchange (NBE) dataset with corresponding uncertainties between 2010–2018: Carbon Monitoring System Flux Net Biosphere Exchange 2020 (CMS-Flux NBE 2020). It is estimated using the NASA Carbon Monitoring System Flux (CMS-Flux) top-down flux inversion system that assimilates column CO₂ observations from the Greenhouse Gases Observing Satellite (GOSAT) and NASA's Observing Carbon Observatory 2 (OCO-2). The regional monthly fluxes are readily accessible as tabular files, and the gridded fluxes are available in NetCDF format. The fluxes and their uncertainties are evaluated by extensively comparing the posterior CO₂ mole fractions with CO₂ observations from aircraft and the NOAA marine boundary layer reference sites. We describe the characteristics of the dataset as the global total, regional climatological mean, and regional annual fluxes and seasonal cycles. We find that the global total fluxes of the dataset agree with atmospheric CO₂ growth observed by the surface-observation network within uncertainty. Averaged between 2010 and 2018, the tropical regions range from close to neutral in tropical South America to a net source in Africa; these contrast with the extra-tropics, which are a net sink of 2.5±0.3 Gt C/year. The regional satellite-constrained NBE estimates provide a unique perspective for understanding the terrestrial biosphere carbon dynamics and monitoring changes in regional contributions to the changes of atmospheric CO₂ growth rate. The gridded and regional aggregated dataset can be accessed at <https://doi.org/10.25966/4v02-c391> (Liu et al., 2020).

101. Regional-Scale, Sector-Specific Evaluation of Global CO₂ Inversion Models Using Aircraft Data From the ACT-America Project

Authors: B. J. Gaudet (Pacific Northwest National Laboratory, Pennsylvania State University); K. J. Davis (Pennsylvania State University); S. Pal (Texas Tech University); A. R. Jacobson (University of Colorado Boulder); A. Schuh (Colorado State University); T. Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement Gif-sur-Yvette France); S. Feng (Pacific Northwest National Laboratory, Pennsylvania State University); E. V. Browell (National Aeronautics and Space Administration (NASA) Langley Research Center)

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Keywords: N/A

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Abstract: We use 148 airborne vertical profiles of CO₂ for frontal cases from the summer 2016 Atmospheric Carbon and Transport--America (ACT-America) campaign to evaluate the skill of 10 global CO₂ in situ inversion models from the version 7 Orbiting Carbon Observatory-2 (OCO-2) Model Intercomparison Project (MIP). Model errors (model posterior--observed CO₂ dry air mole fractions) were categorized by region (Mid-Atlantic, Midwest, and South), frontal sector (warm or cold), and transport model (predominantly Tracer Model 5 (TM5) and Goddard Earth Observing System--Chemistry [GEOS-Chem]). All inversions assimilated the same CO₂ observations. Overall, the median inversion profiles reproduce the general structures of the observations (enhanced/depleted low-level CO₂ in warm/cold sectors), but 1) they underestimate the magnitude of the warm/cold sector mole fraction difference, and 2) the spread among individual inversions can be quite large (>5 ppm). Uniquely in the Mid-Atlantic, inversion biases segregated according to atmospheric transport model, where TM5 inversions biases were -3 to -4 ppm in warm sectors, while those of GEOS-Chem were +2 to +3 ppm in cold sectors. The large spread among the mean posterior CO₂ profiles is not explained by the different atmospheric transport models. These results show that the inversion systems themselves are the dominant cause of this spread, and that the aircraft campaign data are clearly able to identify these large biases. Future controlled experiments should identify which inversions best reproduce mid-latitude CO₂ mole fractions, and how inversion system components are linked to system performance.

102. Satellite Observations of the Tropical Terrestrial Carbon Balance and Interactions With the Water Cycle During the 21st Century

Authors: John Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Sassan Saatchi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Michael Keller (EMBRAPA Agricultural Informatics Campinas Brazil, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, U.S. Department of Agriculture Forest Service International Institute of Tropical Forestry Campinas Brazil); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Nicholas Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Joshua B. Fisher (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, University of California Los Angeles); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Kristen Fahy (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); David Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Rong Fu (University of California Los Angeles); Sarah Worden (University of California Los Angeles); Yi Yin (California Institute for Technology Geological and Planetary Science Pasadena CA USA); Pierre Gentile (Columbia University); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA); Gregory R. Quetin (Department of Earth System Science Stanford University Stanford CA USA); Mathew Williams (University of Edinburgh); Helen Worden (National Center for Atmospheric Research); Mingjie Shi (University of California Los Angeles, Pacific Northwest National Labs); Armineh Barkhordarian (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, University of Hamburg)

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Abstract: A constellation of satellites is now in orbit providing information about terrestrial carbon and water storage and fluxes. These combined observations show that the tropical biosphere has changed significantly in the last 2 decades from the combined effects of climate variability and land use. Large areas of forest have been cleared in both wet and dry forests, increasing the source of carbon to the atmosphere. Concomitantly, tropical fire emissions have declined, at least until 2016, from changes in land-use practices and rainfall, increasing the net carbon sink. Measurements of carbon stocks and

fluxes from disturbance and recovery and of vegetation photosynthesis show significant regional variability of net biosphere exchange and gross primary productivity across the tropics and are tied to seasonal and interannual changes in water fluxes and storage. Comparison of satellite based estimates of evapotranspiration, photosynthesis, and the deuterium content of water vapor with patterns of total water storage and rainfall demonstrate the presence of vegetation-atmosphere interactions and feedback mechanisms across tropical forests. However, these observations of stocks, fluxes and inferred interactions between them do not point unambiguously to either positive or negative feedbacks in carbon and water exchanges. These ambiguities highlight the need for assimilation of these new measurements with Earth System models for a consistent assessment of process interactions, along with focused field campaigns that integrate ground, aircraft and satellite measurements, to quantify the controlling carbon and water processes and their feedback mechanisms.

103. Using satellite data to identify the methane emission controls of South Sudan's wetlands

Authors: Sudhanshu Pandey; Sander Houweling; Alba Lorente; Tobias Borsdorff; Maria Tsivlidou; A. Anthony Bloom; Benjamin Poulter; Zhen Zhang; Ilse Aben

Journal: Biogeosciences

Publisher: Copernicus GmbH

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License: CC BY 4.0

Open Access: N/A

Abstract: . The TROPOspheric Monitoring Instrument (TROPOMI) provides observations of atmospheric methane (CH₄) at an unprecedented combination of high spatial resolution and daily global coverage. Hu et al. (2018) reported unexpectedly large methane enhancements over South Sudan in these observations. Here we assess methane emissions from the wetlands of South Sudan using 2 years (December 2017–November 2019) of TROPOMI total column methane observations. We estimate annual wetland emissions of 7.4 ± 3.2 Tg yr⁻¹, which agrees with the multiyear GOSAT inversion of Lunt et al. (2019) but is an order of magnitude larger than estimates from wetland process models. This disagreement may be explained by the underestimation (by up to 4 times) of inundation extent by the hydrological schemes used in those models. We investigate the seasonal cycle of the emissions and find the lowest emissions during the June–August season when the process models show the largest emissions. Using satellite–altimetry-based river water height measurements, we infer that this seasonal mismatch is likely due to a seasonal mismatch in inundation extent. In models, inundation extent is controlled by regional precipitation scaled to static wetland extent maps, whereas the actual inundation extent is driven by water inflow from rivers like the White Nile and the Sobat. We find the lowest emissions in the highest precipitation and lowest temperature season (June–August, JJA) when models estimate large emissions. In general, our emission estimates show better agreement in terms of both seasonal cycle and annual mean with model estimates that use a stronger temperature dependence. This suggests that temperature might be a stronger control for the South Sudan wetland emissions than currently assumed by models. Our findings demonstrate the use of satellite instruments for quantifying emissions from inaccessible and uncertain tropical wetlands, providing clues for the improvement of process models and thereby improving our understanding of the currently uncertain contribution of wetlands to the global methane budget.

104. Lagged effects regulate the inter-annual variability of the tropical carbon balance

Authors: A. Anthony Bloom; Kevin W. Bowman; Junjie Liu; Alexandra G. Konings; John R. Worden; Nicholas C. Parazoo; Victoria Meyer; John T. Reager; Helen M. Worden; Zhe Jiang; Gregory R. Quetin; T. Luke Smallman; Jean-François Exbrayat; Yi Yin; Sassan S. Saatchi; Mathew Williams; David S. Schimel

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Abstract: . Inter-annual variations in the tropical land carbon (C) balance are a dominant component of the global atmospheric CO₂ growth rate. Currently, the lack of quantitative knowledge on processes controlling net tropical ecosystem C balance on inter-annual timescales inhibits accurate understanding and projections of land-atmosphere C exchanges. In particular, uncertainty on the relative contribution of ecosystem C fluxes attributable to concurrent forcing anomalies (concurrent effects) and those attributable to the continuing influence of past phenomena (lagged effects) stifles efforts to explicitly understand the integrated sensitivity of a tropical ecosystem to climatic variability. Here we present a conceptual framework - applicable in principle to any land biosphere model - to explicitly quantify net biospheric exchange (NBE) as the sum of anomaly-induced concurrent changes and climatology-induced lagged changes to terrestrial ecosystem C states ($NBE = NBECON + NBELAG$). We apply this framework to an observation-constrained analysis of the 2001-2015 tropical C balance: we use a data-model integration approach (CARbon DATA-MODEL framework - CARDAMOM) to merge satellite-retrieved land-surface C observations (leaf area, biomass, solar-induced fluorescence), soil C inventory data and satellite-based atmospheric inversion estimates of CO₂ and CO fluxes to produce a data-constrained analysis of the 2001-2015 tropical C cycle. We find that the inter-annual variability of both concurrent and lagged effects substantially contributes to the 2001-2015 NBE inter-annual variability throughout 2001-2015 across the tropics ($NBECON$ IAV = 80 % of total NBE IAV, $r = 0.76$; $NBELAG$ IAV = 64 % of NBE IAV, $r = 0.61$), and the prominence of $NBELAG$ IAV persists across both wet and dry tropical ecosystems. The magnitude of lagged effect variations on NBE across the tropics is largely attributable to lagged effects on net primary productivity (NPP; $NPPLAG$ IAV 113 % of $NBELAG$ IAV, $r = -0.93$, p value < 0.05), which emerge due to the dependence of NPP on inter-annual variations in foliar C and plant-available H₂O states. We conclude that concurrent and lagged effects need to be explicitly and jointly resolved to retrieve an accurate understanding of the processes

regulating the present-day and future trajectory of the terrestrial land C sink.

105. Observational Constraints on the Response of High-Latitude Northern Forests to Warming

Authors: Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Paul O. Wennberg (California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yi Yin (California Institute of Technology); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: AGU Advances

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Abstract: Since the 1960s, carbon cycling in the high-latitude northern forest (HLNF) has experienced dramatic changes: Most of the forest is greening and net carbon uptake from the atmosphere has increased. During the same time period, the CO₂ seasonal cycle amplitude (SCA) has increased by ~50% or more. Disentangling complex processes that drive these changes has been challenging. In this study, we substitute spatial sensitivity to temperature for time to quantify the impact of temperature increase on gross primary production (GPP), total ecosystem respiration (TER), the fraction of Photosynthetic Active Radiation (fPAR), and the resulted contribution of these changes in amplifying the CO₂ SCA over the HLNF since 1960s. We use the spatial heterogeneity of GPP inferred from solar-induced chlorophyll Fluorescence in combination with net ecosystem exchange (NEE) inferred from column CO₂ observations made between 2015 and 2017 from NASA's Orbiting Carbon Observatory-2. We find that three quarters of the spatial variations in GPP can be explained by the spatial variation in the growing season mean temperature (GSMT). The long term hindcast captures both the magnitude and spatial variability of the trends in observed fPAR. We estimate that between 1960 and 2010, the increase in GSMT enhanced both GPP and the SCA of NEE by ~20%. The calculated enhancement of NEE due to increase in GSMT contributes 56-72% of the trend in the CO₂ SCA at high latitudes, much larger than simulations by most biogeochemical models.

106. Exploring constraints on a wetland methane emission ensemble (WetCHARTs) using GOSAT observations

Authors: Robert J. Parker; Chris Wilson; A. Anthony Bloom; Edward Comyn-Platt; Garry Hayman; Joe McNorton; Hartmut Boesch; Martyn P. Chipperfield

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Keywords: N/A

Funding: Natural Environment Research Council, National Aeronautics and Space Administration, Horizon 2020

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Open Access: N/A

Abstract: . Wetland emissions contribute the largest uncertainties to the current global atmospheric CH₄ budget, and how these emissions will change under future climate scenarios is also still poorly understood. Bloom et al. (2017b) developed WetCHARTs, a simple, data-driven, ensemble-based model that produces estimates of CH₄ wetland emissions constrained by observations of precipitation and temperature. This study performs the first detailed global and regional evaluation of the WetCHARTs CH₄ emission model ensemble against 9 years of high-quality, validated atmospheric CH₄ observations from GOSAT (the Greenhouse Gases Observing Satellite). A 3-D chemical transport model is used to estimate atmospheric CH₄ mixing ratios based on the WetCHARTs emissions and other sources. Across all years and all ensemble members, the observed global seasonal-cycle amplitude is typically underestimated by WetCHARTs by -7.4 ppb, but the correlation coefficient of 0.83 shows that the seasonality is well-produced at a global scale. The Southern Hemisphere has less of a bias (-1.9 ppb) than the Northern Hemisphere (-9.3 ppb), and our findings show that it is typically the North Tropics where this bias is the worst (-11.9 ppb). We find that WetCHARTs generally performs well in reproducing the observed wetland CH₄ seasonal cycle for the majority of wetland regions although, for some regions, regardless of the ensemble configuration, WetCHARTs does not reproduce the observed seasonal cycle well. In order to investigate this, we performed detailed analysis of some of the more challenging exemplar regions (Paraná River, Congo, Sudd and Yucatán). Our results show that certain ensemble members are more suited to specific regions, due to either deficiencies in the underlying data driving the model or complexities in representing the processes involved. In particular, incorrect definition of the wetland extent is found to be the most common reason for the discrepancy between the modelled and observed CH₄ concentrations. The remaining driving data (i.e. heterotrophic respiration and temperature) are shown to also contribute to the mismatch with observations, with the details differing on a region-by-region basis but generally showing that some degree of temperature dependency is better than none. We conclude that the data-driven approach used by WetCHARTs is

well-suited to producing a benchmark ensemble dataset against which to evaluate more complex process-based land surface models that explicitly model the hydrological behaviour of these complex wetland regions.

107. Contrasting Regional Carbon Cycle Responses to Seasonal Climate Anomalies Across the East-West Divide of Temperate North America

Authors: B. Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. A. Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Z. Butterfield (University of Michigan); J. Joiner (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); T. F. Keenan (Lawrence Berkeley National Laboratory, University of California Berkeley); G. Keppel-Aleks (University of Michigan); N. C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Yin (California Institute of Technology)

Journal: Global Biogeochemical Cycles

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Cited by: 15 publications

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Subjects: N/A

Keywords: terrestrial biosphere model, gross primary production, carbon cycle, net ecosystem exchange, interannual variability (and 1 more)

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Open Access: N/A

Abstract: Across temperate North America, interannual variability (IAV) in gross primary production (GPP) and net ecosystem exchange (NEE) and their relationship with environmental drivers are poorly understood. Here, we examine IAV in GPP and NEE and their relationship to environmental drivers using two state-of-the-science flux products: NEE constrained by surface and space-based atmospheric CO₂ measurements over 2010-2015 and satellite up-scaled GPP from FluxSat over 2001-2017. We show that the arid western half of temperate North America provides a larger contribution to IAV in GPP (104% of east) and NEE (127% of east) than the eastern half, in spite of smaller magnitude of annual mean GPP and NEE. This occurs because anomalies in western ecosystems are temporally coherent across the growing season leading to an amplification of GPP and NEE. In contrast, IAV in GPP and NEE in eastern ecosystems is dominated by seasonal compensation effects, associated with opposite responses to temperature anomalies in spring and summer. Terrestrial biosphere models in the MsTMIP ensemble generally capture these differences between eastern and western temperate North America, although there is considerable spread between models.

108. Definitions and methods to estimate regional land carbon fluxes for the second phase of the REgional Carbon Cycle Assessment and Processes Project (RECCAP-2)

Authors: Philippe Ciais; Ana Bastos; Frédéric Chevallier; Ronny Lauerwald; Ben Poulter; Pep Canadell; Gustaf Hugelius; Robert B. Jackson; Atul Jain; Matthew Jones; Masayuki Kondo; Ingrid Luijkx; Prabir K. Patra; Wouter Peters; Julia Pongratz; A. M. Roxanna Petrescu; Shilong Piao; Chunjing Qiu; Celso Von Randow; Pierre Regnier; Marielle Saunois; Robert Scholes; Anatoli Shvidenko; Hanqin Tian; Hui Yang; Xuhui Wang; Bo Zheng

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Carbon flux|Carbon fibers|Atmospheric carbon cycle

Funding: European Space Agency

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Open Access: N/A

Abstract: . Regional land carbon budgets provide insights on the spatial distribution of the land uptake of atmospheric carbon dioxide, and can be used to evaluate carbon cycle models and to define baselines for land-based additional mitigation efforts. The scientific community has been involved in providing observation-based estimates of regional carbon budgets either by downscaling atmospheric CO₂ observations into surface fluxes with atmospheric inversions, by using inventories of carbon stock changes in terrestrial ecosystems, by upscaling local field observations such as flux towers with gridded climate and remote sensing fields or by integrating data-driven or process-oriented terrestrial carbon cycle models. The first coordinated attempt to collect regional carbon budgets for nine regions covering the entire globe in the RECCAP-1 project has delivered estimates for the decade 2000-2009, but these budgets were not comparable between regions, due to different definitions and component fluxes reported or omitted. The recent recognition of lateral fluxes of carbon by human activities and rivers, that connect CO₂ uptake in one area with its release in another also requires better definition and protocols to reach harmonized regional budgets that can be summed up to the globe and compared with the atmospheric CO₂ growth rate and inversion results. In this study, for the international initiative RECCAP-2 coordinated by the Global Carbon Project, which aims as an update of regional carbon budgets over the last two decades based on observations, for 10 regions covering the globe, with a better harmonization than the precursor project, we provide recommendations for using atmospheric inversions results to match bottom-up carbon accounting and models, and we define the different component fluxes of the net land atmosphere carbon exchange that should be reported by each research group in charge of each region. Special attention is given to lateral fluxes, inland water

fluxes and land use fluxes.

109. Concerns about the completeness of the dataset

Authors: Julia Marshall

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Abstract: N/A

110. The ECCO-Darwin Data-Assimilative Global Ocean Biogeochemistry Model: Estimates of Seasonal to Multidecadal Surface Ocean pCO₂ and Air-Sea CO₂ Flux

Authors: D. Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. F. Adkins (California Institute of Technology); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. Brix (University of California Los Angeles, Institute of Coastal Research Helmholtz-Zentrum Geesthacht Geesthacht Germany); S. Dutkiewicz (Massachusetts Institute of Technology); I. Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. M. Gierach (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. Hill (Massachusetts Institute of Technology); O. Jahn (Massachusetts Institute of Technology); P. Landschützer (Max Planck Institute for Meteorology); J. M. Lauderdale (Massachusetts Institute of Technology); J. Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Manizza (Scripps Institution of Oceanography / University of California San Diego); J. D. Naviaux (California Institute of Technology); C. Rödenbeck (Max Planck Institute for Biogeochemistry); D. S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); T. Van der Stocken (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Advances in Modeling Earth Systems

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Cited by: 63 publications

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Subjects: N/A

Keywords: air-sea CO₂ flux, data assimilation, ocean carbon cycle, ecosystem model, ocean modeling (and 1 more)

Funding: National Science Foundation

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Open Access: N/A

Abstract: Quantifying variability in the ocean carbon sink remains problematic due to sparse observations and spatiotemporal variability in surface ocean pCO₂. To address this challenge, we have updated and improved ECCO-Darwin, a global ocean biogeochemistry model that assimilates both physical and biogeochemical observations. The model consists of an adjoint-based ocean circulation estimate from the Estimating the Circulation and Climate of the Ocean (ECCO) consortium and an

ecosystem model developed by the Massachusetts Institute of Technology Darwin Project. In addition to the data-constrained ECCO physics, a Green's function approach is used to optimize the biogeochemistry by adjusting initial conditions and six biogeochemical parameters. Over seasonal to multidecadal timescales (1995-2017), ECCO-Darwin exhibits broad-scale consistency with observed surface ocean pCO₂ and air-sea CO₂ flux reconstructions in most biomes, particularly in the subtropical and equatorial regions. The largest differences between CO₂ uptake occur in subpolar seasonally stratified biomes, where ECCO-Darwin results in stronger winter uptake. Compared to the Global Carbon Project OBMs, ECCO-Darwin has a time-mean global ocean CO₂ sink (2.47 +/- 0.50 Pg C year⁻¹) and interannual variability that are more consistent with interpolation-based products. Compared to interpolation-based methods, ECCO-Darwin is less sensitive to sparse and irregularly sampled observations. Thus, ECCO-Darwin provides a basis for identifying and predicting the consequences of natural and anthropogenic perturbations to the ocean carbon cycle, as well as the climate-related sensitivity of marine ecosystems. Our study further highlights the importance of physically consistent, property-conserving reconstructions, as are provided by ECCO, for ocean biogeochemistry studies.

111. Atmospheric Simulations of Total Column CO₂ Mole Fractions from Global to Mesoscale within the Carbon Monitoring System Flux Inversion Framework

Authors: Martha P. Butler (Pennsylvania State University); Thomas Lauvaux (Pennsylvania State University); Sha Feng (Pennsylvania State University); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kenneth J. Davis (Pennsylvania State University)

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Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: Model coupling, Mesoscale modeling, Carbon cycle, GOSAT, XCO₂

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Open Access: N/A

Abstract: Quantifying the uncertainty of inversion-derived CO₂ surface fluxes and attributing the uncertainty to errors in either flux or atmospheric transport simulations continue to be challenges in the characterization of surface sources and sinks of carbon dioxide (CO₂). Despite recent studies inferring fluxes while using higher-resolution modeling systems, the utility of regional-scale models remains unclear when compared to existing coarse-resolution global systems. Here, we present an off-line coupling of the mesoscale Weather Research and Forecasting (WRF) model to optimized biogenic CO₂ fluxes and mole fractions from the global Carbon Monitoring System inversion system (CMS-Flux). The coupling framework consists of methods to constrain the mass of CO₂ introduced into WRF, effectively nesting our regional domain covering most of North America (except the northern half of Canada) within the CMS global model. We test the coupling by simulating Greenhouse gases Observing SATellite (GOSAT) column-averaged dry-air mole fractions (XCO₂) over North America for 2010. We find mean model-model differences in summer of ~0.12 ppm, significantly lower than the original coupling scheme (from 0.5 to 1.5 ppm, depending on the boundary). While 85% of the XCO₂ values are due to long-range transport from outside our North American domain, most of the model-model differences appear to be due to transport differences in the fraction of the troposphere below 850 hPa. Satellite data from GOSAT and tower and aircraft data are used to show that vertical transport above the Planetary Boundary Layer is responsible for significant model-model differences in the horizontal distribution of column XCO₂ across North America.

112. Development of a Mesoscale Inversion System for Estimating Continental-Scale CO₂ Fluxes

Authors: Daniel Wesloh (Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement Gif sur Yvette France, Pennsylvania State University); Kenneth J. Davis (Pennsylvania State University)

Journal: Journal of Advances in Modeling Earth Systems

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Subjects: N/A

Keywords: atmospheric trace gas flux inversion, spectral methods, carbon dioxide

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Open Access: N/A

Abstract: Computational requirements often impose limitations on the spatial and temporal resolutions of atmospheric CO₂ inversions, increasing aggregation and representation errors. This study enables higher spatial and temporal resolution inversions with spatial and temporal error structures similar to those used in other published inversions by representing the prior flux error covariances as a Kronecker product of spatial and temporal covariances and by using spectral methods for the spatial correlations. Compared to existing inversion systems that are forced to degrade the resolution of the problem in order to bring the dimensionality down to computationally tractable levels, this inversion framework is able to take advantage of mesoscale transport simulations and more of the complexity of spatial and temporal covariances in the surface CO₂ fluxes. This approach was successfully implemented over one month with an identical-twin observing system simulation experiment (OSSE) using a set of assumptions about the prior flux uncertainties compatible both with continental-scale uncertainty estimates and with comparisons of vegetation models to flux towers. The demonstration illustrates the potential of the newly developed inversion system to use high-temporal-resolution information from the North American tower network, to extract high-resolution information about CO₂ fluxes that is inaccessible to coarser resolution inversion systems, and to simultaneously optimize an ensemble of prior estimates. This demonstration sets the stage for regional flux inversions that can take full advantage of the high-resolution data available in tower CO₂ records and mesoscale atmospheric transport reanalyses, include more realistic prior error structures, and explore specifying prior fluxes with ensembles.

113. NASA's carbon monitoring system (CMS) and arctic-boreal vulnerability experiment (ABoVE) social network and community of practice

Authors: Molly E Brown; Matthew W Cooper; Peter C Griffith

Journal: Environmental Research Letters

Publisher: IOP Publishing

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Open Access: N/A

Abstract: The NASA Carbon Monitoring System (CMS) and Arctic-Boreal Vulnerability Experiment (ABoVE) have been planned and funded by the NASA Earth Science Division. Both programs have a focus on engaging stakeholders and developing science useful for decision making. The resulting programs have funded significant scientific output and advancements in understanding how satellite remote sensing observations can be used to not just study how the Earth is changing, but also create data products that are of high utility to stakeholders and decisions makers. In this paper we focus on documenting thematic diversity of research themes and methods used, and how the CMS and ABoVE themes are related. We do this through developing a Correlated Topic Model on the 521 papers produced by the two programs and plotting the results in a network diagram. Through analysis of the themes in these papers, we document the relationships between researchers and institutions participating in CMS and ABoVE programs and the benefits from sustained engagement with stakeholders due to recurring funding. We note an absence of policy engagement in the papers and conclude that funded researchers need to be more ambitious and explicit in drawing the connection between their research and carbon policy implications in order to meet the stated goals of the CMS and ABoVE programs.

114. Improved Constraints on Northern Extratropical CO₂ Fluxes Obtained by Combining Surface-Based and Space-Based Atmospheric CO₂ Measurements

Authors: B. Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); I. Baker (Colorado State University); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); N. M. Deutscher (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences, University of Wollongong Wollongong New South Wales Australia); D. G. Feist (Institut für Physik der Atmosphäre Deutsches Zentrum für Luft- und Raumfahrt, Oberpfaffenhofen Weßling Germany, Lehrstuhl für Physik der Atmosphäre Ludwig-Maximilians-Universität München Munich Germany, Max Planck Institute for Biogeochemistry); D. W. T. Griffith (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences, University of Wollongong Wollongong New South Wales Australia); L. T. Iraci (National Aeronautics and Space Administration (NASA) Ames Research Center); M. Kiel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. S. Kimball (University of Montana); C. E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); I. Morino (National Institute for Environmental Studies); N. C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. Petri (University of Bremen); C. M. Roehl (California Institute of Technology); M. K. Sha (Royal Belgian Institute for Space Aeronomy); K. Strong (University of Toronto); V. A. Velazco (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences, University of Wollongong Wollongong New South Wales Australia); P. O. Wennberg (California Institute of Technology); D. Wunch (University of Toronto)

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Cited by: 29 publications

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Subjects: N/A

Keywords: carbon cycle, data assimilation, CO₂ flux, TCCON, OCO-2 (and 1 more)

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Open Access: N/A

Abstract: Top-down estimates of CO₂ fluxes are typically constrained by either surface-based or space-based CO₂ observations. Both of these measurement types have spatial and temporal gaps in observational coverage that can lead to differences in inferred fluxes. Assimilating both surface-based and space-based measurements concurrently in a flux inversion framework improves observational

coverage and reduces sampling related artifacts. This study examines the consistency of flux constraints provided by these different observations and the potential to combine them by performing a series of 6-year (2010-2015) CO₂ flux inversions. Flux inversions are performed assimilating surface-based measurements from the in situ and flask network, measurements from the Total Carbon Column Observing Network (TCCON), and space-based measurements from the Greenhouse Gases Observing Satellite (GOSAT), or all three data sets combined. Combining the data sets results in more precise flux estimates for subcontinental regions relative to any of the data sets alone. Combining the data sets also improves the accuracy of the posterior fluxes, based on reduced root-mean-square differences between posterior flux-simulated CO₂ and aircraft-based CO₂ over midlatitude regions (0.33-0.56 ppm) in comparison to GOSAT (0.37-0.61 ppm), TCCON (0.50-0.68 ppm), or in situ and flask measurements (0.46-0.56 ppm) alone. These results suggest that surface-based and GOSAT measurements give complementary constraints on CO₂ fluxes in the northern extratropics and can be combined in flux inversions to improve constraints on regional fluxes. This stands in contrast with many earlier attempts to combine these data sets and suggests that improvements in the NASA Atmospheric CO₂ Observations from Space (ACOS) retrieval algorithm have significantly improved the consistency of space-based and surface-based flux constraints.

115. Amplification of the Ocean Carbon Sink During El Niños: Role of Poleward Ekman Transport and Influence on Atmospheric CO₂

Authors: Enhui Liao (Princeton University); Laure Resplandy (Princeton University); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Keywords: rainfall dilution effect, air-sea CO₂ flux, equatorial upwelling, Ekman transport, tropical Pacific Ocean (and 1 more)

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License: N/A

Open Access: N/A

Abstract: Amplification of the ocean carbon sink during El Niño events partially offset terrestrial biosphere carbon losses to the atmosphere, but uncertainties in the magnitude, timing, and spatial extent of the ocean response confound our understanding of the global carbon budget and its sensitivity to climate. Here, we examine the mechanisms controlling the anomalous tropical Pacific Ocean CO₂ drawdown during El Niño events harnessing multidecadal ocean pCO₂ observations in conjunction with a state-of-the-art ocean biogeochemical model. We show that poleward Ekman transport dramatically amplifies the near-equatorial pCO₂ anomaly identified in prior work and that this amplification varies considerably between El Niño events. During central Pacific events, the CO₂ flux anomaly can vary up to fourfold between events (2002/2003 vs. 2015/2016), but it always recedes during the boreal winter to spring transition and the poleward transport of the anomaly mostly extends into the Northern Hemisphere. During eastern Pacific events characterized by an equatorially centered intertropical convergence zone (e.g., 1997/1998), the intense CO₂ flux anomaly persists into boreal summer and can extend further into the Southern Hemisphere. The hemispheric asymmetry (northward vs. southward extension) and the termination (boreal winter/spring for central Pacific events vs. boreal summer for eastern Pacific events) of the ocean CO₂ response are tied to the El Niño dynamics in the near-equatorial region and also how it is amplified by poleward Ekman transport. Finally, we evaluate how differences in the ocean response influence atmospheric CO₂ and discuss the potential of atmospheric CO₂ satellite data to provide observational constraints.

116. A conjugate BFGS method for accurate estimation of a posterior error covariance matrix in a linear inverse problem

Authors: Yosuke Niwa (Meteorological Research Institute, National Institute for Environmental Studies); Yosuke Fujii (The Institute of Statistical Mathematics, Research Organization of Information and Systems Tokyo Japan, Meteorological Research Institute)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

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Cited by: 6 publications

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Funding: N/A

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Abstract: One effective data assimilation/inversion method is the four-dimensional variational method (4D-Var). However, it is a non-trivial task for a conventional 4D-Var to estimate a posterior error covariance matrix. This study proposes a method to estimate a posterior error covariance matrix applied to the linear inverse problem of an atmospheric constituent. The method was constructed within a 4D-Var framework using a quasi-Newton method with the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm. The proposed method was constructed such that conjugacy among the set of increment vector pairs was ensured. It is theoretically demonstrated that, when this conjugate property is coupled with preconditioning, an analytical solution of a posterior error covariance matrix could be obtained from the same number of vector pairs as observations. Furthermore, to accelerate the speed of convergence, the method can be coupled with an ensemble approach. By performing a simple advection test, it was confirmed that the proposed method could obtain an analytical matrix of the posterior error covariance within the same number of iterations as the observations. Furthermore, the method was also evaluated using an atmospheric CO₂ inverse problem, which demonstrated its practical utility. The evaluation revealed that the proposed method could provide accurate estimates not only of the diagonal but also of the off-diagonal elements of the posterior error covariance matrix. Although far more expensive than optimal state estimation, the computational efficiency was found to be reasonable for practical use, especially in conjunction with an ensemble approach. The accurate estimation of a posterior error covariance matrix resulting from the proposed method could provide valuable quantitative information regarding the uncertainties of estimated variables as well as the observational impacts, which would be beneficial for designing observation networks. Furthermore, error correlations derived from the estimated off-diagonal elements could benefit the interpretation of optimised parameter variations.

117. Geostatistical inverse modeling with very large datasets: an example from the Orbiting Carbon Observatory 2 (OCO-2) satellite

Authors: Scot M. Miller; Arvind K. Saibaba; Michael E. Trudeau; Marikate E. Mountain; Arlyn E. Andrews

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-13-1771-2020

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Abstract: . Geostatistical inverse modeling (GIM) has become a common approach to estimating greenhouse gas fluxes at the Earth's surface using atmospheric observations. GIMs are unique relative to other commonly used approaches because they do not require a single emissions inventory or a bottom-up model to serve as an initial guess of the fluxes. Instead, a modeler can incorporate a wide range of environmental, economic, and/or land use data to estimate the fluxes. Traditionally, GIMs have been paired with in situ observations that number in the thousands or tens of thousands. However, the number of available atmospheric greenhouse gas observations has been increasing enormously as the number of satellites, airborne measurement campaigns, and in situ monitoring stations continues to increase. This era of prolific greenhouse gas observations presents computational and statistical challenges for inverse modeling frameworks that have traditionally been paired with a limited number of in situ monitoring sites. In this article, we discuss the challenges of estimating greenhouse gas fluxes using large atmospheric datasets with a particular focus on GIMs. We subsequently discuss several strategies for estimating the fluxes and quantifying uncertainties, strategies that are adapted from hydrology, applied math, or other academic fields and are compatible with a wide variety of atmospheric models. We further evaluate the accuracy and computational burden of each strategy using a synthetic CO₂ case study based upon NASA's Orbiting Carbon Observatory 2 (OCO-2) satellite. Specifically, we simultaneously estimate a full year of 3-hourly CO₂ fluxes across North America in one case study - a total of 9.4×10^6 unknown fluxes using 9.9×10^4 synthetic observations. The strategies discussed here provide accurate estimates of CO₂ fluxes that are comparable to fluxes calculated directly or analytically. We are also able to approximate posterior uncertainties in the fluxes, but these approximations are, typically, an over- or underestimate depending upon the strategy employed and the degree of approximation required to make the calculations manageable.

118. Cropland Carbon Uptake Delayed and Reduced by 2019 Midwest Floods

Authors: Yi Yin (California Institute of Technology); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Paul O. Wennberg (California Institute of Technology); Kenneth J. Davis (Pennsylvania State University); Troy Magney (California Institute of Technology, Department of Plant Sciences Davis CA USA); Philipp Köhler (California Institute of Technology); Liyin He (California Institute of Technology); Rupesh Jeyaram (California Institute of Technology); Vincent Humphrey (California Institute of Technology); Tobias Gerken (Pennsylvania State University); Sha Feng (Pennsylvania State University); Joshua P. Digangi (National Aeronautics and Space Administration (NASA) Langley Research Center); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: AGU Advances

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: While large-scale floods directly impact human lives and infrastructures, they also profoundly impact agricultural productivity. New satellite observations of vegetation activity and atmospheric CO₂ offer the opportunity to quantify the effects of such extreme events on cropland carbon sequestration. Widespread flooding during spring and early summer 2019 induced conditions that delayed crop planting across the U.S. Midwest. As a result, satellite observations of solar-induced chlorophyll fluorescence from TROPospheric Monitoring Instrument and Orbiting Carbon Observatory reveal a 16-day shift in the seasonal cycle of photosynthesis relative to 2018, along with a 15% lower peak value. We estimate a reduction of 0.21 PgC in cropland gross primary productivity in June and July, partially compensated in August and September (+0.14 PgC). The extension of the 2019 growing season into late September is likely to have benefited from increased water availability and late-season temperature. Ultimately, this change is predicted to reduce the crop productivity in the Midwest Corn/Soy belt by ~15% compared to 2018. Using an atmospheric transport model, we show that a decline of ~0.1 PgC in the net carbon uptake during June and July is consistent with observed CO₂ enhancements of up to 10 ppm in the midday boundary layer from Atmospheric Carbon and Transport-America aircraft and over 3 ppm in column-averaged dry-air mole fractions from Orbiting Carbon Observatory. This study quantifies the impact of floods on cropland productivity and

demonstrates the potential of combining solar-induced chlorophyll fluorescence with atmospheric CO₂ observations to monitor regional carbon flux anomalies.

119. Carbon Flux Variability From a Relatively Simple Ecosystem Model With Assimilated Data Is Consistent With Terrestrial Biosphere Model Estimates

Authors: Gregory R. Quetin (Department of Earth System Science Stanford University Stanford CA USA); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Keywords: data assimilation, carbon cycle, carbon modeling

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Open Access: N/A

Abstract: Modeling the net carbon balance is challenging due to the knowledge gaps in the variability and processes controlling gross carbon fluxes. Terrestrial carbon cycle modeling is susceptible to several sources of bias, including meteorological uncertainty, model structural uncertainty, and model parametric uncertainty. To determine the impact of these uncertainties, we compare three model-derived representations of the global terrestrial carbon balance across 1997-2009: (1) observation-constrained model-data fusion (CARBOn Data Model FraMEwork, CARDAMOM), (2) the reanalysis-driven Trends in Net Land-Atmosphere Carbon Exchange (TRENDY) land biosphere model ensemble, and (3) the Coupled Model Intercomparison Project 5 (CMIP5) Earth System Model ensemble. We consider the spread in carbon cycle simulations attributable primarily to parametric uncertainty (CARDAMOM), structural uncertainty (TRENDY), and combined structural and simulated meteorological uncertainty (CMIP5). We find that the spread across the CARDAMOM ensemble long-term mean--produced by parameter uncertainty--is larger than the spread of TRENDY and CMIP5 for net biosphere exchange (NBE) but similar for gross primary productivity (GPP). The carbon flux dynamics of CARDAMOM compares to models in TRENDY as well as models in TRENDY compare to each other in many regions for NBE seasonal (nine of 12), NBE interannual (11 of 12), and GPP seasonal variability (7 of 12), although not for GPP interannual variability (2 of 12). The simple model structure of CARDAMOM and systematic assimilation of observations is sufficient to produce carbon dynamics within the range of more complex models. These results are encouraging for the use of model-data fusion products with empirically estimated uncertainty for global carbon cycle studies.

120. Enhanced regional terrestrial carbon uptake over Korea revealed by atmospheric CO₂ measurements from 1999 to 2017

Authors: Jeongmin Yun (Seoul National University); Sujong Jeong (Seoul National University); Chang-Hoi Ho (Seoul National University); Hoonyoung Park (Seoul National University); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Haeyoung Lee (Environmental Meteorology Research Division National Institute of Meteorological Sciences Jeju Republic of Korea); Stephen Sitch (University of Exeter); Pierre Friedlingstein (University of Exeter); Sebastian Lienert (University of Bern); Danica Lombardozzi (National Center for Atmospheric Research); Vanessa Haverd (CSIRO Oceans and Atmosphere Canberra ACT Australia); Atul Jain (University of Illinois); Sönke Zaehle (Max Planck Institute for Biogeochemistry); Etsushi Kato (Research & Development Division Institute of Applied Energy (IAE) Tokyo Japan); Hanqin Tian (Auburn University); Nicolas Vuichard (Laboratoire des Sciences du Climat et de l'Environnement Institut Pierre-Simon Laplace CEA-CNRS-UVSQ CE Orme des Merisiers Gif-sur-Yvette CEDEX France); Andy Wiltshire (Met Office Hadley Centre, Met Office); Ning Zeng (University of Maryland)

Journal: Global Change Biology

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Abstract: Understanding changes in terrestrial carbon balance is important to improve our knowledge of the regional carbon cycle and climate change. However, evaluating regional changes in the terrestrial carbon balance is challenging due to the lack of surface flux measurements. This study reveals that the terrestrial carbon uptake over the Republic of Korea has been enhanced from 1999 to 2017 by analyzing long-term atmospheric CO₂ concentration measurements at the Anmyeondo Station (36.53 degreesN, 126.32 degreesE) located in the western coast. The influence of terrestrial carbon flux on atmospheric CO₂ concentrations (Δ CO₂) is estimated from the difference of CO₂ concentrations that were influenced by the land sector (through easterly winds) and the Yellow Sea sector (through westerly winds). We find a significant trend in Δ CO₂ of -4.75 ppm per decade ($p < .05$) during the vegetation growing season (May through October), suggesting that the regional terrestrial carbon uptake has increased relative to the surrounding ocean areas. Combined analysis with satellite measured normalized difference vegetation index and gross primary production shows that the enhanced carbon uptake is associated with significant nationwide increases in vegetation and its production. Process-based terrestrial model and inverse model simulations estimate that regional

terrestrial carbon uptake increases by up to 18.9 and 8.0 Tg C for the study period, accounting for 13.4% and 5.7% of the average annual domestic carbon emissions, respectively. Atmospheric chemical transport model simulations indicate that the enhanced terrestrial carbon sink is the primary reason for the observed ΔCO_2 trend rather than anthropogenic emissions and atmospheric circulation changes. Our results highlight the fact that atmospheric CO_2 measurements could open up the possibility of detecting regional changes in the terrestrial carbon cycle even where anthropogenic emissions are not negligible.

121. Evidence of Carbon Uptake Associated with Vegetation Greening Trends in Eastern China

Authors: Zhonghua He (Purdue University, Chinese Academy of Sciences, Zhejiang Climate Center, Hangzhou 310017, China); Liping Lei (Chinese Academy of Sciences); Zhao-Cheng Zeng (California Institute of Technology); Mengya Sheng (Chinese Academy of Sciences); Lisa R. Welp (Purdue University)

Journal: Remote Sensing

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Keywords: N/A

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Open Access: N/A

Abstract: Persistent and widespread increase of vegetation cover, identified as greening, has been observed in areas of the planet over late 20th century and early 21st century by satellite-derived vegetation indices. It is difficult to verify whether these regions are net carbon sinks or sources by studying vegetation indices alone. In this study, we investigate greening trends in Eastern China (EC) and corresponding trends in atmospheric CO₂ concentrations. We used multiple vegetation indices including NDVI and EVI to characterize changes in vegetation activity over EC from 2003 to 2016. Gap-filled time series of column-averaged CO₂ dry air mole fraction (XCO₂) from January 2003 to May 2016, based on observations from SCIAMACHY, GOSAT, and OCO-2 satellites, were used to calculate XCO₂ changes during growing season for 13 years. We derived a relationship between XCO₂ and surface net CO₂ fluxes from two inversion model simulations, CarbonTracker and Monitoring Atmospheric Composition and Climate (MACC), and used those relationships to estimate the biospheric CO₂ flux enhancement based on satellite observed XCO₂ changes. We observed significant growing period (GP) greening trends in NDVI and EVI related to cropland intensification and forest growth in the region. After removing the influence of large urban center CO₂ emissions, we estimated an enhanced XCO₂ drawdown during the GP of -0.070 to -0.084 ppm yr⁻¹. Increased carbon uptake during the GP was estimated to be 28.41 to 46.04 Tg C, mainly from land management, which could offset about 2-3% of EC's annual fossil fuel emissions. These results show the potential of using multi-satellite observed XCO₂ to estimate carbon fluxes from the regional biosphere, which could be used to verify natural sinks included as national contributions of greenhouse gas emissions reduction in international climate change agreements like the UNFCCC Paris Accord.

122. Lagged effects dominate the inter-annual variability of the 2010-2015 tropical carbon balance

Authors: A. Anthony Bloom; Kevin W. Bowman; Junjie Liu; Alexandra G. Konings; John R. Worden; Nicholas C. Parazoo; Victoria Meyer; John T. Reager; Helen M. Worden; Zhe Jiang; Gregory R. Quetin; T. Luke Smallman; Jean-François Exbrayat; Yi Yin; Sassan S. Saatchi; Mathew Williams; David S. Schimel

Journal: Unknown Journal

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Keywords: Biosphere model|Land Cover|Dominance (genetics)

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Open Access: N/A

Abstract: . Inter-annual variations in the tropical land carbon (C) balance are a dominant component of the global atmospheric CO₂ growth rate. Currently, the lack of quantitative knowledge on processes controlling net tropical ecosystems C balance on inter-annual timescales inhibits accurate understanding and projections of land-atmosphere C exchanges. In particular, uncertainty on the relative contribution of ecosystem C fluxes attributable to concurrent meteorological forcing anomalies (concurrent effects) and those attributable to the continuing influence of past phenomena (lagged effects) stifles efforts to explicitly understand the integrated sensitivity of tropical ecosystem to climatic variability. Here we present a conceptual framework - applicable in principle to any meteorology-forced land biosphere model - to explicitly quantify net biospheric exchange (NBE) as the sum of anomaly-induced concurrent changes and climatology-induced lagged changes to terrestrial ecosystem C states ($NBE = NBECON + NBELAG$). We apply this framework to an observation-constrained analysis of the 2010-2015 tropical C balance: we use a data-model integration approach (CARDAMOM) to merge satellite-retrieved land-surface C observations (leaf area, biomass, solar-induced fluorescence), soil C inventory data and satellite-based atmospheric inversion estimates of CO₂ and CO fluxes to produce a data-constrained analysis of the 2010-2015 tropical C cycle. We find that the inter-annual variability of lagged effects explain the majority of NBE inter-annual variability (IAV) throughout 2010-2015 across the tropics ($NBELAG$ IAV = 112 % of NBE IAV, $r = 0.87$) relative to concurrent effects ($NBECON$ IAV = 54 % of total NBE IAV, $r = 0.03$) and the dominance of $NBELAG$ IAV persists across both wet and dry tropical ecosystems. The magnitude of lagged effect variations on NBE across the tropics is largely attributable to lagged effects on net primary productivity (NPP; $NPPLAG$ IAV 88 % of $NBELAG$ IAV, $r = -0.99$, p-value

123. Seasonal Characteristics of Model Uncertainties From Biogenic Fluxes, Transport, and Large-Scale Boundary Inflow in Atmospheric CO₂ Simulations Over North America

Authors: Sha Feng (Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement, CEA, CNRS, UVSQ/IPSL, Université Paris-Saclay, Orme des Merisiers Gif-sur-Yvette France, Pennsylvania State University); Kenneth J. Davis (Pennsylvania State University); Klaus Keller (Pennsylvania State University); Yu Zhou (Department of Biology, Graduate School of Geography Clark University Worcester MA USA); Christopher Williams (Department of Biology, Graduate School of Geography Clark University Worcester MA USA); Andrew E. Schuh (Colorado State University); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Baker (Colorado State University)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jd031165

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Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: Biogenic fluxes, Regional CO₂ modeling, Transport, Uncertainty, Boundary conditions

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Regional estimates of biogenic carbon fluxes over North America from both atmospheric inversions ("top-down" approach) and terrestrial biosphere models ("bottom-up") remain highly uncertain. We merge these approaches with an ensemble-based, regional modeling system able to diagnose and quantify the causes of uncertainties in top-down atmospheric estimates of the terrestrial sink over North America. Our ensemble approach quantifies and partitions the uncertainty stemming from atmospheric transport, the biosphere, and large-scale CO₂ boundary inflow (boundary conditions). We use meteorological data, CO₂ fluxes, and CO₂ mole fraction measurements to assure the reliability of the ensemble system. Our results show that all uncertainty components have clear seasonal variations. The biogenic flux component dominates modeled boundary layer CO₂ uncertainty, ranging from 2.5 ppm in summer and winter to 1.5 ppm in fall and spring. Spatially, it remains highly uncertain in the U.S. Corn Belt regions. Transport uncertainty reaches a maximum of 2.5 ppm in the summer months and stays at 1.2 ppm for the rest of the year and is highly correlated with the biogenic CO₂ fluxes. Boundary conditions play the smallest role in atmospheric boundary layer CO₂ uncertainty with a magnitude smaller than 1 ppm. However, boundary conditions are the most important uncertainty component in column-averaged CO₂ (XCO₂). The spatiotemporal variations of the uncertainties in modeled XCO₂ are similar to those in atmospheric boundary layer CO₂.

124. A fast and accurate vector radiative transfer model for simulating the near-infrared hyperspectral scattering processes in clear atmospheric conditions

Authors: Wenguang Bai; Peng Zhang; Wenjian Zhang; Gang Ma; Chengli Qi; Hui Liu

Journal: Journal of Quantitative Spectroscopy and Radiative Transfer

Publisher: Elsevier BV

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Keywords: N/A

Funding: National Natural Science Foundation of China, Ministry of Science and Technology of the People's Republic of China, China Meteorological Administration

License: N/A

Open Access: N/A

Abstract: N/A

125. A Road Map for Improving the Treatment of Uncertainties in High-Resolution Regional Carbon Flux Inverse Estimates

Authors: Sha Feng (Pennsylvania State University); Thomas Lauvaux (Laboratoire des Sciences du Climat et de l'Environnement, CEA, CNRS, UVSQ/IPSL, Université Paris-Saclay Gif-sur-Yvette cedex France, Pennsylvania State University); Klaus Keller (Pennsylvania State University); Kenneth J. Davis (Pennsylvania State University); Peter Rayner (University of Melbourne); Tomohiro Oda (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Kevin R. Gurney (Northern Arizona University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Publication Date: 2019-11-11

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Subjects: N/A

Keywords: Fossil fuel emissions, CO₂ modeling, Uncertainty, Regional carbon flux, Boundary inflow (and 1 more)

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Open Access: N/A

Abstract: Atmospheric inversions allow us to estimate the terrestrial carbon sink by combining atmospheric observations with atmospheric transport models. However, these inverse estimates remain highly uncertain. Here we quantify uncertainties in simulations of North American atmospheric CO₂ concentrations using a probabilistic approach. We demonstrate that uncertainty in fossil fuel emissions is a key factor in the uncertainty surrounding biospheric flux estimates. We show that atmospheric transport uncertainties in state-of-the-art numerical weather models diminish when averaged over time, while uncertainties in large-scale CO₂ boundary inflow considerably impair our ability to quantify regional fluxes. Current estimates of the North America land sink that neglect the uncertainties in CO₂ boundary inflow and fossil fuel emissions are likely overconfident. Our findings suggest that targeted use of new atmospheric observations and improved quantification of uncertainty components are a promising avenue to improve atmospheric inversions with the goal to refine estimates of biospheric CO₂ fluxes on regional and continental scales.

126. Prior biosphere model impact on global terrestrial CO₂ fluxes estimated from OCO-2 retrievals

Authors: Sajeev Philip; Matthew S. Johnson; Christopher Potter; Vanessa Genovesse; David F. Baker; Katherine D. Haynes; Daven K. Henze; Junjie Liu; Benjamin Poulter

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

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Keywords: N/A

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Open Access: N/A

Abstract: . This study assesses the impact of different state of the art global biospheric CO₂ flux models, when applied as prior information, on inverse model "top-down" estimates of terrestrial CO₂ fluxes obtained when assimilating Orbiting Carbon Observatory 2 (OCO-2) observations. This is done with a series of observing system simulation experiments (OSSEs) using synthetic CO₂ column-average dry air mole fraction (XCO₂) retrievals sampled at the OCO-2 satellite spatiotemporal frequency. The OSSEs utilized a 4-D variational (4D-Var) assimilation system with the GEOS-Chem global chemical transport model (CTM) to estimate CO₂ net ecosystem exchange (NEE) fluxes using synthetic OCO-2 observations. The impact of biosphere models in inverse model estimates of NEE is quantified by conducting OSSEs using the NASA-CASA, CASA-GFED, SiB-4, and LPJ models as prior estimates and using NEE from the multi-model ensemble mean of the Multiscale Synthesis and Terrestrial Model Intercomparison Project as the "truth". Results show that the assimilation of simulated XCO₂ retrievals at OCO-2 observing modes over land results in posterior NEE estimates which generally reproduce "true" NEE globally and over terrestrial TransCom-3 regions that are well-sampled. However, we find larger spread among posterior NEE estimates, when using different prior NEE fluxes, in regions and seasons that have limited OCO-2 observational coverage and a large range in "bottom-up" NEE fluxes. Seasonally averaged posterior NEE estimates had standard deviations (SD) of ~10 % to ~50 % of the multi-model-mean NEE for different TransCom-3 land regions with significant NEE fluxes (regions/seasons with a NEE flux ≥ 0.5 PgC yr⁻¹). On a global average, the seasonally averaged residual impact of the prior model NEE assumption on the posterior NEE spread is ~10 %-20 % of the posterior NEE mean. Additional OCO-2 OSSE simulations demonstrate that posterior NEE estimates are also sensitive to the assumed prior NEE flux uncertainty statistics, with spread in posterior NEE estimates similar to those when using variable prior model NEE fluxes. In fact, the sensitivity of posterior NEE estimates to prior error statistics was larger than prior flux values in some regions/times in the tropics and Southern Hemisphere where sufficient OCO-2 data were available and large differences between the prior and truth were evident. Overall, even with the availability of

spatiotemporally dense OCO-2 data, noticeable residual differences (up to ~20 %-30 % globally and 50 % regionally) in posterior NEE flux estimates remain that were caused by the choice of prior model flux values and the specification of prior flux uncertainties.

127. On what scales can GOSAT flux inversions constrain anomalies in terrestrial ecosystems?

Authors: Brendan Byrne; Dylan B. A. Jones; Kimberly Strong; Saroja M. Polavarapu; Anna B. Harper; David F. Baker; Shamil Maksyutov

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

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Keywords: N/A

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Open Access: N/A

Abstract: . Interannual variations in temperature and precipitation impact the carbon balance of terrestrial ecosystems, leaving an imprint in atmospheric CO₂. Quantifying the impact of climate anomalies on the net ecosystem exchange (NEE) of terrestrial ecosystems can provide a constraint to evaluate terrestrial biosphere models against and may provide an emergent constraint on the response of terrestrial ecosystems to climate change. We investigate the spatial scales over which interannual variability in NEE can be constrained using atmospheric CO₂ observations from the Greenhouse Gases Observing Satellite (GOSAT). NEE anomalies are calculated by performing a series of inversion analyses using the GEOS-Chem adjoint model to assimilate GOSAT observations. Monthly NEE anomalies are compared to "proxies", variables that are associated with anomalies in the terrestrial carbon cycle, and to upscaled NEE estimates from FLUXCOM. Statistically significant correlations ($P < 0.05$) are obtained between posterior NEE anomalies and anomalies in soil temperature and FLUXCOM NEE on continental and larger scales in the tropics, as well as in the northern extratropics on subcontinental scales during the summer ($R^2 \geq 0.49$), suggesting that GOSAT measurements provide a constraint on NEE interannual variability (IAV) on these spatial scales. Furthermore, we show that GOSAT flux inversions are generally better correlated with the environmental proxies and FLUXCOM NEE than NEE anomalies produced by a set of terrestrial biosphere models (TBMs), suggesting that GOSAT flux inversions could be used to evaluate TBM NEE fluxes.

128. Terrestrial ecosystem carbon flux estimated using GOSAT and OCO-2 XCO₂ retrievals

Authors: Hengmao Wang; Fei Jiang; Jun Wang; Weimin Ju; Jing M. Chen

Journal: Atmospheric Chemistry and Physics

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Open Access: N/A

Abstract: . In this study, both the Greenhouse Gases Observing Satellite (GOSAT) and the Orbiting Carbon Observatory 2 (OCO-2) XCO₂ retrievals produced by the NASA Atmospheric CO₂ Observations from Space (ACOS) project (version b7.3) are assimilated within the GEOS-Chem 4D-Var assimilation framework to constrain the terrestrial ecosystem carbon flux during 1 October 2014 to 31 December 2015. One inversion for the comparison, using in situ CO₂ observations, and another inversion as a benchmark for the simulated atmospheric CO₂ distributions of the real inversions, using global atmospheric CO₂ trends and referred to as the poor-man inversion, are also conducted. The estimated global and regional carbon fluxes for 2015 are shown and discussed. CO₂ observations from surface flask sites and XCO₂ retrievals from Total Carbon Column Observing Network (TCCON) sites are used to evaluate the simulated concentrations with the posterior carbon fluxes. Globally, the terrestrial ecosystem carbon sink (excluding biomass burning emissions) estimated from GOSAT data is stronger than that inferred from OCO-2 data, weaker than the in situ inversion and matches the poor-man inversion the best. Regionally, in most regions, the land sinks inferred from GOSAT data are also stronger than those from OCO-2 data, and in North America, Asia and Europe, the carbon sinks inferred from GOSAT inversion are comparable to those from in situ inversion. For the latitudinal distribution of land sinks, the satellite-based inversions suggest a smaller boreal and tropical sink but larger temperate sinks in both the Northern and Southern Hemisphere than the in situ inversion. However, OCO-2 and GOSAT generally do not agree on which continent contains the smaller or larger sinks. Evaluations using flask and TCCON observations and the comparisons with in situ and poor-man inversions suggest that only GOSAT and the in situ inversions perform better than a poor-man solution. GOSAT data can effectively improve the carbon flux estimates in the Northern Hemisphere, while OCO-2 data, with the specific version used in this study, show only slight improvement. The differences of inferred land fluxes between GOSAT and OCO-2 inversions in different regions are mainly related to the spatial coverage, the data amount and the biases of these two satellite XCO₂ retrievals.

129. The 2015-2016 carbon cycle as seen from OCO-2 and the global in situ network

Authors: Sean Crowell; David Baker; Andrew Schuh; Sourish Basu; Andrew R. Jacobson; Frederic Chevallier; Junjie Liu; Feng Deng; Liang Feng; Kathryn McKain; Abhishek Chatterjee; John B. Miller; Britton B. Stephens; Annmarie Eldering; David Crisp; David Schimel; Ray Nassar; Christopher W. O'Dell; Tomohiro Oda; Colm Sweeney; Paul I. Palmer; Dylan B. A. Jones

Journal: Atmospheric Chemistry and Physics

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Open Access: N/A

Abstract: . The Orbiting Carbon Observatory-2 has been on orbit since 2014, and its global coverage holds the potential to reveal new information about the carbon cycle through the use of top-down atmospheric inversion methods combined with column average CO₂ retrievals. We employ a large ensemble of atmospheric inversions utilizing different transport models, data assimilation techniques, and prior flux distributions in order to quantify the satellite-informed fluxes from OCO-2 Version 7r land observations and their uncertainties at continental scales. Additionally, we use in situ measurements to provide a baseline against which to compare the satellite-constrained results. We find that within the ensemble spread, in situ observations, and satellite retrievals constrain a similar global total carbon sink of 3.7 $\pm$ 0.5 PgC yr⁻¹, and 1.5 $\pm$ 0.6 PgC yr⁻¹ for global land, for the 2015-2016 annual mean. This agreement breaks down in smaller regions, and we discuss the differences between the experiments. Of particular interest is the difference between the different assimilation constraints in the tropics, with the largest differences occurring in tropical Africa, which could be an indication of the global perturbation from the 2015-2016 El Niño. Evaluation of posterior concentrations using TCCON and aircraft observations gives some limited insight into the quality of the different assimilation constraints, but the lack of such data in the tropics inhibits our ability to make strong conclusions there.

130. Detection of fossil fuel emission trends in the presence of natural carbon cycle variability

Authors: Yi Yin; Kevin Bowman; A Anthony Bloom; John Worden

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/ab2dd7

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Abstract: Atmospheric CO₂ observations have the potential to monitor regional fossil fuel emission (FFCO₂) changes to support carbon mitigation efforts such as the Paris Accord, but they must contend with the confounding impacts of the natural carbon cycle. Here, we quantify trend detection time and magnitude in gridded total CO₂ fluxes--the sum of FFCO₂ and natural carbon fluxes--under an idealized assumption that monthly total CO₂ fluxes can be perfectly resolved at a 2 degreesx2 degrees resolution. Using Coupled Model Intercomparison Project 5 (CMIP5) 'business-as-usual' emission scenarios to represent FFCO₂ and simulated net biome exchange (NBE) to represent natural carbon fluxes, we find that trend detection time for the total CO₂ fluxes at such a resolution has a median of 10 years across the globe, with significant spatial variability depending on FFCO₂ magnitude and NBE variability. Differences between trends in the total CO₂ fluxes and the underlying FFCO₂ component highlight the role of natural carbon cycle variability in modulating regional detection of FFCO₂ emission trends using CO₂ observations alone, particularly in the tropics and subtropics where mega-cities with large populations are developing rapidly. Using CO₂ estimates alone at such a spatiotemporal resolution can only quantify fossil fuel trends in a few places--mostly limited to arid regions. For instance, in the Middle East, FFCO₂ can explain more than 75% of the total CO₂ trends in ~70% of the grids, but only ~20% of grids in China can meet such criteria. Only a third of the 25 megacities we analyze here show total CO₂ trends that are primarily explained (>75%) by FFCO₂. Our analysis provides a theoretical baseline at a global scale for the design of regional FFCO₂ monitoring networks and underscores the importance of estimating biospheric interannual variability to improve the accuracy of FFCO₂ trend monitoring. We envision that this can be achieved with a fully integrated carbon cycle assimilation system with explicit constraints on FFCO₂ and NBE, respectively.

131. Global satellite-driven estimates of heterotrophic respiration

Authors: Alexandra G. Konings; A. Anthony Bloom; Junjie Liu; Nicholas C. Parazoo; David S. Schimel; Kevin W. Bowman

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-16-2269-2019

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Abstract: . While heterotrophic respiration (Rh) makes up about a quarter of gross global terrestrial carbon fluxes, it remains among the least-observed carbon fluxes, particularly outside the midlatitudes. In situ measurements collected in the Soil Respiration Database (SRDB) number only a few hundred worldwide. Similarly, only a single data-driven wall-to-wall estimate of annual average heterotrophic respiration exists, based on bottom-up upscaling of SRDB measurements using an assumed functional form to account for climate variability. In this study, we exploit recent advances in remote sensing of terrestrial carbon fluxes to estimate global variations in heterotrophic respiration in a top-down fashion at monthly temporal resolution and 4°x5° spatial resolution. We combine net ecosystem productivity estimates from atmospheric inversions of the NASA Carbon Monitoring System-Flux (CMS-Flux) with an optimally scaled gross primary productivity dataset based on satellite-observed solar-induced fluorescence variations to estimate total ecosystem respiration as a residual of the terrestrial carbon balance. The ecosystem respiration is then separated into autotrophic and heterotrophic components based on a spatially varying carbon use efficiency retrieved in a model-data fusion framework (the CARbon DAta MOdel fraMework, CARDAMOM). The resulting dataset is independent of any assumptions about how heterotrophic respiration responds to climate or substrate variations. It estimates an annual average global average heterotrophic respiration flux of 43.6±19.3 Pg C yr⁻¹. Sensitivity and uncertainty analyses showed that the top-down Rh are more sensitive to the choice of input gross primary productivity (GPP) and net ecosystem productivity (NEP) datasets than to the assumption of a static carbon use efficiency (CUE) value, with the possible exception of the wet tropics. These top-down estimates are compared to bottom-up estimates of annual heterotrophic respiration, using new uncertainty estimates that partially account for sampling and model errors. Top-down heterotrophic respiration estimates are higher than those from bottom-up upscaling everywhere except at high latitudes and are 30 % greater overall (43.6 Pg C yr⁻¹ vs. 33.4 Pg C yr⁻¹). The uncertainty ranges of both methods are comparable, except poleward of 45° N, where bottom-up uncertainties are greater. The ratio of top-down heterotrophic to total ecosystem respiration varies seasonally by as much as 0.6 depending on season and climate, illustrating the importance of

studying the drivers of autotrophic and heterotrophic respiration separately, and thus the importance of data-driven estimates of R_h such as those estimated here.

132. The OCO-3 mission: measurement objectives and expected performance based on 1 year of simulated data

Authors: Annmarie Eldering; Thomas E. Taylor; Christopher W. O'Dell; Ryan Pavlick

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . The Orbiting Carbon Observatory-3 (OCO-3) is NASA's next instrument dedicated to extending the record of the dry-air mole fraction of column carbon dioxide (XCO₂) and solar-induced fluorescence (SIF) measurements from space. The current schedule calls for a launch from the Kennedy Space Center no earlier than April 2019 via a Space-X Falcon 9 and Dragon capsule. The instrument will be installed as an external payload on the Japanese Experimental Module Exposed Facility (JEM-EF) of the International Space Station (ISS) with a nominal mission lifetime of 3 years. The precessing orbit of the ISS will allow for viewing of the Earth at all latitudes less than approximately 52°, with a ground repeat cycle that is much more complicated than the polar-orbiting satellites that so far have carried all of the instruments capable of measuring carbon dioxide from space. The grating spectrometer at the core of OCO-3 is a direct copy of the OCO-2 spectrometer, which was launched into a polar orbit in July 2014. As such, OCO-3 is expected to have similar instrument sensitivity and performance characteristics to OCO-2, which provides measurements of XCO₂ with precision better than 1 ppm at 3 Hz, with each viewing frame containing eight footprints approximately 1.6 km by 2.2 km in size. However, the physical configuration of the instrument aboard the ISS, as well as the use of a new pointing mirror assembly (PMA), will alter some of the characteristics of the OCO-3 data compared to OCO-2. Specifically, there will be significant differences from day to day in the sampling locations and time of day. In addition, the flexible PMA system allows for a much more dynamic observation-mode schedule. This paper outlines the science objectives of the OCO-3 mission and, using a simulation of 1 year of global observations, characterizes the spatial sampling, time-of-day coverage, and anticipated data quality of the simulated L1b. After application of cloud and aerosol prescreening, the L1b radiances are run through the operational L2 full physics retrieval algorithm, as well as post-retrieval filtering and bias correction, to examine the expected coverage and quality of the retrieved XCO₂ and to show how the measurement objectives are met. In addition, results of the SIF from the IMAP-DOAS algorithm are analyzed. This paper focuses only on the nominal nadir-land and glint-water observation modes, although on-orbit measurements will also be made in transition and target modes, similar to OCO-2, as well as the new snapshot area mapping (SAM) mode.

133. Characterization of Regional-Scale CO₂ Transport Uncertainties in an Ensemble with Flow-Dependent Transport Errors

Authors: Hans W. Chen (Lund University, Pennsylvania State University); Fuqing Zhang (Pennsylvania State University); Thomas Lauvaux (Now at Laboratoire des Sciences du Climat et de l'Environnement, CEA, CNRS, UVSQ/IPSL, Université Paris-Saclay, Orme des Merisiers Gif-sur-Yvette France, Pennsylvania State University); Kenneth J. Davis (Pennsylvania State University); Sha Feng (Pennsylvania State University); Martha P. Butler (Pennsylvania State University); Richard B. Alley (Pennsylvania State University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: Inference of CO₂ surface fluxes using atmospheric CO₂ observations in atmospheric inversions depends critically on accurate representation of atmospheric transport. Here we characterize regional-scale CO₂ transport uncertainties due to uncertainties in meteorological fields using a mesoscale atmospheric model and an ensemble of simulations with flow-dependent transport errors. During a 1-month summer period over North America, transport uncertainties yield an ensemble spread in instantaneous CO₂ at 100 m above ground level comparable to the CO₂ uncertainties resulting from 48% relative uncertainty in 3-hourly natural CO₂ fluxes. Temporal averaging reduces transport uncertainties but increases the influence of CO₂ uncertainties from the lateral boundaries. The influence of CO₂ background uncertainties is especially large for column-averaged CO₂. These results suggest that transport errors and CO₂ background errors limit regional atmospheric inversions at two distinct timescales and that the error characteristics of transport and background errors should guide the design of regional inversion systems.

134. Quantifying the Impact of Atmospheric Transport Uncertainty on CO₂ Surface Flux Estimates

Authors: Andrew E. Schuh (Colorado State University); Andrew R. Jacobson (University of Colorado Boulder); Sourish Basu (University of Colorado Boulder); Brad Weir (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); David Baker (Colorado State University); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Frédéric Chevallier (Laboratoire des Sciences du Climat et de l'Environnement, CEA-CNRS-UVSQ, L'Orme des Merisiers, Gif-sur-Yvette Paris France); Sean Crowell (University of Oklahoma); Kenneth J. Davis (Pennsylvania State University); Feng Deng (University of Toronto); Scott Denning (Colorado State University); Liang Feng (University of Edinburgh, National Centre for Earth Observation Edinburgh UK); Dylan Jones (University of Toronto); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Paul I. Palmer (University of Edinburgh, National Centre for Earth Observation Edinburgh UK)

Journal: Global Biogeochemical Cycles

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Abstract: We show that transport differences between two commonly used global chemical transport models, GEOS-Chem and TM5, lead to systematic space-time differences in modeled distributions of carbon dioxide and sulfur hexafluoride. The distribution of differences suggests inconsistencies between the transport simulated by the models, most likely due to the representation of vertical motion. We further demonstrate that these transport differences result in systematic differences in surface CO₂ flux estimated by a collection of global atmospheric inverse models using TM5 and GEOS-Chem and constrained by in situ and satellite observations. While the impact on inferred surface fluxes is most easily illustrated in the magnitude of the seasonal cycle of surface CO₂ exchange, it is the annual carbon budgets that are particularly relevant for carbon cycle science and policy. We show that inverse model flux estimates for large zonal bands can have systematic biases of up to 1.7 PgC/year due to large-scale transport uncertainty. These uncertainties will propagate directly into analysis of the annual meridional CO₂ flux gradient between the tropics and northern midlatitudes, a key metric for understanding the location, and more importantly the processes, responsible for the annual global carbon sink. The research suggests that variability among transport models remains the largest source of uncertainty across global flux inversion systems and highlights the importance both of using model

ensembles and of using independent constraints to evaluate simulated transport.

135. Characterizing uncertainties in atmospheric inversions of fossil fuel CO₂ emissions in California

Authors: Kieran Brophy; Heather Graven; Alistair J. Manning; Emily White; Tim Arnold; Marc L. Fischer; Seongeun Jeong; Xinguang Cui; Matthew Rigby

Journal: Atmospheric Chemistry and Physics

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References: 46

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Funding: N/A

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Open Access: N/A

Abstract: . Atmospheric inverse modelling has become an increasingly useful tool for evaluating emissions of greenhouse gases including methane, nitrous oxide, and synthetic gases such as hydrofluorocarbons (HFCs). Atmospheric inversions for emissions of CO₂ from fossil fuel combustion (ffCO₂) are currently being developed. The aim of this paper is to investigate potential errors and uncertainties related to the spatial and temporal prior representation of emissions and modelled atmospheric transport for the inversion of ffCO₂ emissions in the US state of California. We perform simulation experiments based on a network of ground-based observations of CO₂ concentration and radiocarbon in CO₂ (a tracer of ffCO₂), combining prior (bottom-up) emission models and transport models currently used in many atmospheric studies. The potential effect of errors in the spatial and temporal distribution of prior emission estimates is investigated in experiments by using perturbed versions of the emission estimates used to create the pseudo-data. The potential effect of transport error was investigated by using three different atmospheric transport models for the prior and pseudo-data simulations. We find that the magnitude of biases in posterior total state emissions arising from errors in the spatial and temporal distribution in prior emissions in these experiments are 1 %-15 % of posterior total state emissions and are generally smaller than the 2-sigma uncertainty in posterior emissions. Transport error in these experiments introduces biases of -10 % to +6 % into posterior total state emissions. Our results indicate that uncertainties in posterior total state ffCO₂ estimates arising from the choice of prior emissions or atmospheric transport model are on the order of 15 % or less for the ground-based network in California we consider. We highlight the need for temporal variations to be included in prior emissions and for continuing efforts to evaluate and improve the representation of atmospheric transport for regional ffCO₂ inversions.

136. Mass-conserving coupling of total column CO₂ (XCO₂) from global to mesoscale models: Case study with CMS-Flux inversion system and WRF-Chem (v3.6.1)

Authors: Martha P. Butler; Thomas Lauvaux; Sha Feng; Junjie Liu; Kevin W. Bowman; Kenneth J. Davis

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Representative Concentration Pathways|Mole fraction

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Open Access: N/A

Abstract: . Quantifying the uncertainty of inversion-derived fluxes and attributing the uncertainty to errors in either flux or transport continue to be challenges in the characterization of surface sources and sinks of carbon dioxide (CO₂). It is also not clear if fluxes inferred in a coarse-resolution global system will remain optimal in a higher-resolution modeling environment. Here we present an off-line coupling of the mesoscale Weather Research and Forecasting (WRF) model to optimized biogenic CO₂ fluxes and mole fractions from the global Carbon Monitoring System inversion system (CMS-Flux). The coupling framework consists of methods to constrain the mass of CO₂ introduced into WRF, effectively nesting our North American domain within the global model. We test the coupling by simulating Greenhouse gases Observing SATellite (GOSAT) column-averaged dry-air mole fractions (XCO₂) over North American for 2010. We find mean model-model differences in summer of ~ 0.12 ppm. While 85 % of the XCO₂ values are due to long-range transport from outside our North American domain, most of the model-model differences appear to be due to transport differences in the fraction of the troposphere below 850 hPa. The framework methods can be used to couple other global model inversion results to WRF for further study using different boundary layer and transport parameterizations.

137. Modeling the Recent Changes in the Arctic Ocean CO₂Sink (2006-2013)

Authors: Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Global Biogeochemical Cycles

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References: 68

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Keywords: sea-ice, carbon cycle, Arctic Ocean, polar oceans, climate change (and 1 more)

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Abstract: The Arctic Ocean (AO) and its associated marginal seas have recently experienced rapid climate and environmental changes, most notably sea-ice area (SIA) loss and warming potentially impacting its uptake of carbon dioxide (CO₂). We used the state-of-the-art ECCO2-Darwin coupled ocean-biogeochemistry model to simulate the 2006-2013 period and investigate the impact of changing SIA on the CO₂uptake of the AO. We find that the mean annual CO₂sink of the AO is 153 +/- 14 TgC a⁻¹and the CO₂sink decreased at a rate of 3.6 TgC a⁻¹even though SIA decreased by 8 x 10⁴ km² a⁻¹over the same period. Extreme SIA loss in 2007 resulted in a 185.4 TgC CO₂sink, an increase ~20% over the 2006-2013 mean. In contrast, extreme SIA loss of 2012 resulted in a CO₂sink of the AO of only 146.3 TgC due to two main factors: (1) increased both wind speed and stratification in the Eastern Siberian Sea absorbing less CO₂and (2) decreased primary production and area of air-sea gas exchange in the Chukchi and Nordic Seas. Our model captures a trend of decreasing CO₂sink in most of the Chukchi Sea during fall but does not show the changes in winter CO₂sink in the Nordic and Barents Seas as previous independent studies have suggested. Our results indicate that future AO-atmosphere CO₂exchange will be determined by complex interplay of SIA and other environmental drivers.

138. Identifying local anthropogenic CO2 emissions with satellite retrievals: a case study in South Korea

Authors: Changsub Shim (Korea Environment Institute, Sejong, South Korea); Jihyun Han (Korea Environment Institute, Sejong, South Korea); Daven K. Henze (University of Colorado Boulder); Taeyeon Yoon (Sunmoon University, Chungcheongnam-do, South Korea)

Journal: International Journal of Remote Sensing

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Abstract: N/A

139. A comparison of posterior atmospheric CO₂ adjustments obtained from in situ and GOSAT constrained flux inversions

Authors: Saroja M. Polavarapu; Feng Deng; Brendan Byrne; Dylan B. A. Jones; Michael Neish

Journal: Atmospheric Chemistry and Physics

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Abstract: . Posterior fluxes obtained from inverse modelling are difficult to verify because there is no dense network of flux measurements available to evaluate estimates against. Here we present a new diagnostic to evaluate structures in posterior fluxes. First, we simulate the change in atmospheric CO₂ fields between posterior and prior fluxes, referred to as the posterior atmospheric adjustments due to updated fluxes (PAAFs). Second, we calculate the uncertainty in atmospheric CO₂ fields due solely to uncertainty in the meteorological fields, referred to as the posterior atmospheric adjustments due to imperfect meteorology (PAAMs). We argue that PAAF can only be considered robust if it exceeds PAAM, that is, the changes in atmospheric CO₂ between the posterior and prior fluxes should at least exceed atmospheric CO₂ changes arising from imperfect meteorology. This diagnostic is applied to two CO₂ flux inversions: one which assimilates observations from the in situ CO₂ network and the other which assimilates observations from the Greenhouse Gases Observing SATellite (GOSAT). On the global scale, PAAF in the troposphere reflects northern extratropical fluxes, whereas stratospheric adjustments primarily reflect tropical fluxes. In general, larger spatiotemporal variations in PAAF are obtained for the GOSAT inversion than for the in situ inversion. Zonal standard deviations of the PAAF exceed the PAAM through most of the year when GOSAT observations are used, but the minimum value is exceeded only in boreal summer when in situ observations are used. Zonal spatial structures in GOSAT-based PAAF exceed PAAM throughout the year in the tropics and through most of the year in the northern extratropics, suggesting GOSAT flux inversions can constrain zonal asymmetries in fluxes. However, we cannot discount the possibility that these structures are influenced by biases in GOSAT retrievals. Verification of such spatial structures will require a dense network of independent observations. Because PAAF depends on the choice of prior fluxes, the comparison with PAAM is system dependent and thus can be used to monitor a given assimilation system's behaviour.

140. The TanSat mission: preliminary global observations

Authors: Yi Liu; Jing Wang; Lu Yao; Xi Chen; Zhaonan Cai; Dongxu Yang; Zengshan Yin; Songyan Gu; Longfei Tian; Naimeng Lu; Daren Lyu

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Open Access: N/A

Abstract: N/A

141. A global synthesis inversion analysis of recent variability in CO₂ fluxes using GOSAT and in situ observations

Authors: James S. Wang; S. Randolph Kawa; G. James Collatz; Motoki Sasakawa; Luciana V. Gatti; Toshinobu Machida; Yuping Liu; Michael E. Manyin

Journal: Atmospheric Chemistry and Physics

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Open Access: N/A

Abstract: . The precise contribution of the two major sinks for anthropogenic CO₂ emissions, terrestrial vegetation and the ocean, and their location and year-to-year variability are not well understood. Top-down estimates of the spatiotemporal variations in emissions and uptake of CO₂ are expected to benefit from the increasing measurement density brought by recent in situ and remote CO₂ observations. We uniquely apply a batch Bayesian synthesis inversion at relatively high resolution to in situ surface observations and bias-corrected GOSAT satellite column CO₂ retrievals to deduce the global distributions of natural CO₂ fluxes during 2009–2010. The GOSAT inversion is generally better constrained than the in situ inversion, with smaller posterior regional flux uncertainties and correlations, because of greater spatial coverage, except over North America and northern and southern high-latitude oceans. Complementarity of the in situ and GOSAT data enhances uncertainty reductions in a joint inversion; however, remaining coverage gaps, including those associated with spatial and temporal sampling biases in the passive satellite measurements, still limit the ability to accurately resolve fluxes down to the sub-continental or sub-ocean basin scale. The GOSAT inversion produces a shift in the global CO₂ sink from the tropics to the north and south relative to the prior, and an increased source in the tropics of $\sim 2 \text{ Pg C yr}^{-1}$ relative to the in situ inversion, similar to what is seen in studies using other inversion approaches. This result may be driven by sampling and residual retrieval biases in the GOSAT data, as suggested by significant discrepancies between posterior CO₂ distributions and surface in situ and HIPPO mission aircraft data. While the shift in the global sink appears to be a robust feature of the inversions, the partitioning of the sink between land and ocean in the inversions using either in situ or GOSAT data is found to be sensitive to prior uncertainties because of negative correlations in the flux errors. The GOSAT inversion indicates significantly less CO₂ uptake in the summer of 2010 than in 2009 across northern regions, consistent with the impact of observed severe heat waves and drought. However, observations from an in situ network in Siberia imply that the GOSAT inversion exaggerates the 2010–2009 difference in uptake in that region, while the prior CASA-GFED model of net ecosystem production and fire emissions reasonably estimates that quantity.

The prior, in situ posterior, and GOSAT posterior all indicate greater uptake over North America in spring to early summer of 2010 than in 2009, consistent with wetter conditions. The GOSAT inversion does not show the expected impact on fluxes of a 2010 drought in the Amazon; evaluation of posterior mole fractions against local aircraft profiles suggests that time-varying GOSAT coverage can bias the estimation of interannual flux variability in this region.

142. Detecting drought impact on terrestrial biosphere carbon fluxes over contiguous US with satellite observations

Authors: Junjie Liu; Kevin Bowman; Nicholas C Parazoo; A Anthony Bloom; Debra Wunch; Zhe Jiang; Kevin R Gurney; Dave Schimel

Journal: Environmental Research Letters

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143. Observing carbon cycle-climate feedbacks from space

Authors: Piers J. Sellers (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Berrien Moore (University of Oklahoma); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Annmarie Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: The impact of human emissions of carbon dioxide and methane on climate is an accepted central concern for current society. It is increasingly evident that atmospheric concentrations of carbon dioxide and methane are not simply a function of emissions but that there are myriad feedbacks forced by changes in climate that affect atmospheric concentrations. If these feedbacks change with changing climate, which is likely, then the effect of the human enterprise on climate will change. Quantifying, understanding, and articulating the feedbacks within the carbon-climate system at the process level are crucial if we are to employ Earth system models to inform effective mitigation regimes that would lead to a stable climate. Recent advances using space-based, more highly resolved measurements of carbon exchange and its component processes--photosynthesis, respiration, and biomass burning--suggest that remote sensing can add key spatial and process resolution to the existing in situ systems needed to provide enhanced understanding and advancements in Earth system models. Information about emissions and feedbacks from a long-term carbon-climate observing system is essential to better stewardship of the planet.

144. Characterizing biospheric carbon balance using CO₂ observations from the OCO-2 satellite

Authors: Scot M. Miller; Anna M. Michalak; Vineet Yadav; Jovan M. Tadi■

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

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Abstract: . NASA's Orbiting Carbon Observatory 2 (OCO-2) satellite launched in summer of 2014. Its observations could allow scientists to constrain CO₂ fluxes across regions or continents that were previously difficult to monitor. This study explores an initial step toward that goal; we evaluate the extent to which current OCO-2 observations can detect patterns in biospheric CO₂ fluxes and constrain monthly CO₂ budgets. Our goal is to guide top-down, inverse modeling studies and identify areas for future improvement. We find that uncertainties and biases in the individual OCO-2 observations are comparable to the atmospheric signal from biospheric fluxes, particularly during Northern Hemisphere winter when biospheric fluxes are small. A series of top-down experiments indicate how these errors affect our ability to constrain monthly biospheric CO₂ budgets. We are able to constrain budgets for between two and four global regions using OCO-2 observations, depending on the month, and we can constrain CO₂ budgets at the regional level (i.e., smaller than seven global biomes) in only a handful of cases (16 % of all regions and months). The potential of the OCO-2 observations, however, is greater than these results might imply. A set of synthetic data experiments suggests that retrieval errors have a salient effect. Advances in retrieval algorithms and to a lesser extent atmospheric transport modeling will improve the results. In the interim, top-down studies that use current satellite observations are best-equipped to constrain the biospheric carbon balance across only continental or hemispheric regions.

145. A Bayesian ensemble data assimilation to constrain model parameters and land-use carbon emissions

Authors: Sebastian Lienert; Fortunat Joos

Journal: Biogeosciences

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/bg-15-2909-2018>

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Abstract: . A dynamic global vegetation model (DGVM) is applied in a probabilistic framework and benchmarking system to constrain uncertain model parameters by observations and to quantify carbon emissions from land-use and land-cover change (LULCC). Processes featured in DGVMs include parameters which are prone to substantial uncertainty. To cope with these uncertainties Latin hypercube sampling (LHS) is used to create a 1000-member perturbed parameter ensemble, which is then evaluated with a diverse set of global and spatiotemporally resolved observational constraints. We discuss the performance of the constrained ensemble and use it to formulate a new best-guess version of the model (LPX-Bern v1.4). The observationally constrained ensemble is used to investigate historical emissions due to LULCC (ELUC) and their sensitivity to model parametrization. We find a global ELUC estimate of 158 (108, 211) PgC (median and 90 % confidence interval) between 1800 and 2016. We compare ELUC to other estimates both globally and regionally. Spatial patterns are investigated and estimates of ELUC of the 10 countries with the largest contribution to the flux over the historical period are reported. We consider model versions with and without additional land-use processes (shifting cultivation and wood harvest) and find that the difference in global ELUC is on the same order of magnitude as parameter-induced uncertainty and in some cases could potentially even be offset with appropriate parameter choice.

146. Development of the WRF-CO2 4D-Var assimilation system v1.0

Authors: Tao Zheng; Nancy H. F. French; Martin Baxter

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Abstract: . Regional atmospheric CO₂ inversions commonly use Lagrangian particle trajectory model simulations to calculate the required influence function, which quantifies the sensitivity of a receptor to flux sources. In this paper, an adjoint-based four-dimensional variational (4D-Var) assimilation system, WRF-CO₂ 4D-Var, is developed to provide an alternative approach. This system is developed based on the Weather Research and Forecasting (WRF) modeling system, including the system coupled to chemistry (WRF-Chem), with tangent linear and adjoint codes (WRFPLUS), and with data assimilation (WRFDA), all in version 3.6. In WRF-CO₂ 4D-Var, CO₂ is modeled as a tracer and its feedback to meteorology is ignored. This configuration allows most WRF physical parameterizations to be used in the assimilation system without incurring a large amount of code development. WRF-CO₂ 4D-Var solves for the optimized CO₂ flux scaling factors in a Bayesian framework. Two variational optimization schemes are implemented for the system: the first uses the limited memory Broyden-Fletcher-Goldfarb-Shanno (BFGS) minimization algorithm (L-BFGS-B) and the second uses the Lanczos conjugate gradient (CG) in an incremental approach. WRFPLUS forward, tangent linear, and adjoint models are modified to include the physical and dynamical processes involved in the atmospheric transport of CO₂. The system is tested by simulations over a domain covering the continental United States at 48 km x 48 km grid spacing. The accuracy of the tangent linear and adjoint models is assessed by comparing against finite difference sensitivity. The system's effectiveness for CO₂ inverse modeling is tested using pseudo-observation data. The results of the sensitivity and inverse modeling tests demonstrate the potential usefulness of WRF-CO₂ 4D-Var for regional CO₂ inversions.

147. Spring photosynthetic onset and net CO₂ uptake in Alaska triggered by landscape thawing

Authors: Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Almut Arneth (Karlsruhe Institute of Technology); Thomas A. M. Pugh (Birmingham Institute of Forest Research University of Birmingham Birmingham UK, School of Geography, Earth and Environmental Sciences University of Birmingham Birmingham UK); Ben Smith (Lund University); Nicholas Steiner (Department of Earth and Atmospheric Sciences The City College of New York New York New York); Kristina Luus (Centre for Applied Data Analytics Dublin Ireland); Roisin Commann (Harvard University); Josh Benmergui (Harvard University); Eric Stofferahn (Conservation Science Partners Truckee California); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christian Rödenbeck (Max Planck Institute for Biogeochemistry Jena Germany); Randy Kawa (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Eugenie Euskirchen (University of Alaska Fairbanks); Donatella Zona (Biology Department San Diego State University San Diego California); Kyle Arndt (Biology Department San Diego State University San Diego California); Walt Oechel (Biology Department San Diego State University San Diego California); Charles Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: The springtime transition to regional-scale onset of photosynthesis and net ecosystem carbon uptake in boreal and tundra ecosystems are linked to the soil freeze-thaw state. We present evidence from diagnostic and inversion models constrained by satellite fluorescence and airborne CO₂ from 2012 to 2014 indicating the timing and magnitude of spring carbon uptake in Alaska correlates with landscape thaw and ecoregion. Landscape thaw in boreal forests typically occurs in late April (DOY 111 $\pm$ 7) with a 29 $\pm$ 6 day lag until photosynthetic onset. North Slope tundra thaws 3 weeks later (DOY 133 $\pm$ 5) but experiences only a 20 $\pm$ 5 day lag until photosynthetic onset. These time lag differences reflect efficient cold season adaptation in tundra shrub and the longer dehardening period for boreal evergreens. Despite the short transition from thaw to photosynthetic onset in tundra, synchrony of tundra respiration with snow melt and landscape thaw delays the transition from net carbon loss (at photosynthetic onset) to net uptake by 13 $\pm$ 7 days, thus reducing the tundra net

carbon uptake period. Two global CO₂ inversions using a CASA-GFED model prior estimate earlier northern high latitude net carbon uptake compared to our regional inversion, which we attribute to (i) early photosynthetic-onset model prior bias, (ii) inverse method (scaling factor + optimization window), and (iii) sparsity of available Alaskan CO₂ observations. Another global inversion with zero prior estimates the same timing for net carbon uptake as the regional model but smaller seasonal amplitude. The analysis of Alaskan eddy covariance observations confirms regional scale findings for tundra, but indicates that photosynthesis and net carbon uptake occur up to 1 month earlier in evergreens than captured by models or CO₂ inversions, with better correlation to above-freezing air temperature than date of primary thaw. Further collection and analysis of boreal evergreen species over multiple years and at additional subarctic flux towers are critically needed.

148. Top-down constraints on global N₂O emissions at optimal resolution: application of a new dimension reduction technique

Authors: Kelley C. Wells; Dylan B. Millet; Nicolas Bousserez; Daven K. Henze; Timothy J. Griffis; Sreelekha Chaliyakunnel; Edward J. Dlugokencky; Eri Saikawa; Gao Xiang; Ronald G. Prinn; Simon O'Doherty; Dickon Young; Ray F. Weiss; Geoff S. Dutton; James W. Elkins; Paul B. Krummel; Ray Langenfelds; L. Paul Steele

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Abstract: . We present top-down constraints on global monthly N₂O emissions for 2011 from a multi-inversion approach and an ensemble of surface observations. The inversions employ the GEOS-Chem adjoint and an array of aggregation strategies to test how well current observations can constrain the spatial distribution of global N₂O emissions. The strategies include (1) a standard 4D-Var inversion at native model resolution (4 degrees x 5 degrees), (2) an inversion for six continental and three ocean regions, and (3) a fast 4D-Var inversion based on a novel dimension reduction technique employing randomized singular value decomposition (SVD). The optimized global flux ranges from 15.9 Tg N yr⁻¹ (SVD-based inversion) to 17.5-17.7 Tg N yr⁻¹ (continental-scale, standard 4D-Var inversions), with the former better capturing the extratropical N₂O background measured during the HIPER Pole-to-Pole Observations (HIPPO) airborne campaigns. We find that the tropics provide a greater contribution to the global N₂O flux than is predicted by the prior bottom-up inventories, likely due to underestimated agricultural and oceanic emissions. We infer an overestimate of natural soil emissions in the extratropics and find that predicted emissions are seasonally biased in northern midlatitudes. Here, optimized fluxes exhibit a springtime peak consistent with the timing of spring fertilizer and manure application, soil thawing, and elevated soil moisture. Finally, the inversions reveal a major emission underestimate in the US Corn Belt in the bottom-up inventory used here. We extensively test the impact of initial conditions on the analysis and recommend formally optimizing the initial N₂O distribution to avoid biasing the inferred fluxes. We find that the SVD-based approach provides a powerful framework for deriving emission information from N₂O observations: by defining the optimal resolution of the solution based on the information content of the inversion, it provides spatial information that is lost when aggregating to political or geographic regions, while also providing more temporal information than a standard 4D-Var inversion.

149. Thriving on Our Changing Planet

Authors: Unknown Author; Unknown Author; Unknown Author; Unknown Author

Journal: Unknown Journal

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Abstract: N/A

150. Contrasting carbon cycle responses of the tropical continents to the 2015-2016 El Niño

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicolas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhe Jiang (National Center for Atmospheric Research); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Debra Wunch (University of Toronto); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ying Sun (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christopher W. O'Dell (Colorado State University); Kevin R. Gurney (Arizona State University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michelle Gierach (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David Crisp (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Annmarie Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: INTRODUCTION The influence of El Niño on climate is accompanied by large changes to the carbon cycle, and El Niño-induced variability in the carbon cycle has been attributed mainly to the tropical continents. However, owing to a dearth of observations in the tropics, tropical carbon fluxes are poorly quantified, and considerable debate exists over the dominant mechanisms (e.g., plant growth, respiration, fire) and regions (e.g., humid versus semiarid tropics) on the net carbon balance.

RATIONALE The launch of the Orbiting Carbon Observatory-2 (OCO-2) shortly before the 2015-2016 El Niño, the second strongest since the 1950s, has provided an opportunity to understand how tropical

land carbon fluxes respond to the warm and dry climate characteristics of El Niño conditions. The El Niño events may also provide a natural experiment to study the response of tropical land carbon fluxes to future climate changes, because anomalously warm and dry tropical environments typical of El Niño are expected to be more frequent under most emission scenarios.

RESULTS The tropical regions of three continents (South America, Asia, and Africa) had heterogeneous responses to the 2015-2016 El Niño, in terms of both climate drivers and the carbon cycle. The annual mean precipitation over tropical South America and tropical Asia was lower by 3.0σ and 2.8σ , respectively, in 2015 relative to the 2011 La Niña year. Tropical Africa, on the other hand, had near equal precipitation and the same number of dry months between 2015 and 2011; however, surface temperatures were higher by 1.6σ , dominated by the positive anomaly over its eastern and southern regions. In response to the warmer and drier climate anomaly in 2015, the pantropical biosphere released 2.5 ± 0.34 gigatons more carbon into the atmosphere than in 2011, which accounts for 83.3% of the global total 3.0 -gigatons of carbon (gigatons C) net biosphere flux differences and 92.6% of the atmospheric CO₂ growth-rate differences between 2015 and 2011. It indicates that the tropical land biosphere flux anomaly was the driver of the highest atmospheric CO₂ growth rate in 2015. The three tropical continents had an approximately even contribution to the pantropical net carbon flux anomaly in 2015, but had diverse dominant processes: gross primary production (GPP) reduced carbon uptake (0.9 ± 0.96 gigatons C) in tropical South America, fire increased carbon release (0.4 ± 0.08 gigatons C) in tropical Asia, and respiration increased carbon release (0.6 ± 1.01 gigatons C) in Africa. We found that most of the excess carbon release in 2015 was associated with either extremely low precipitation or high temperatures, or both.

CONCLUSION Our results indicate that the global El Niño effect is a superposition of regionally specific effects. The heterogeneous climate forcing and carbon response over the three tropical continents to the 2015-2016 El Niño challenges previous studies that suggested that a single dominant process determines carbon cycle interannual variability, which could also be due to previous disturbance and soil and vegetation structure. The similarity between the 2015 tropical climate anomaly and the projected climate changes imply that the role of the tropical land as a buffer for fossil fuel emissions may be reduced in the future. The heterogeneous response may reflect differences in temperature and rainfall anomalies, but intrinsic differences in vegetation species, soils, and prior disturbance may contribute as well. A synergistic use of multiple satellite observations and a long time series of spatially resolved fluxes derived from sustained satellite observations will enable tests of these hypotheses, allow for a more process-based understanding, and, ultimately, aid improved carbon-climate model projections. Diverse climate driver anomalies and carbon cycle responses to the 2015-2016 El Niño over the three tropical continents. Schematic of climate anomaly patterns over the three tropical continents and the anomalies of the net carbon flux and its dominant constituent flux (i.e., GPP, respiration, and fire) relative to the 2011 La Niña during the 2015-2016 El Niño. GtC, gigatons C.

151. The Orbiting Carbon Observatory-2 early science investigations of regional carbon dioxide fluxes

Authors: A. Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); P. O. Wennberg (California Institute of Technology); D. Crisp (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. R. Gunson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Chatterjee (NASA Global Modeling and Assimilation Office, Greenbelt, MD, USA., Universities Space Research Association); J. Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); F. M. Schwandner (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Sun (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. W. O'Dell (Colorado State University); C. Frankenberg (California Institute of Technology); T. Taylor (Colorado State University); B. Fisher (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); G. B. Osterman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Wunch (California Institute of Technology); J. Hakkarainen (Finnish Meteorological Institute); J. Tamminen (Finnish Meteorological Institute); B. Weir (NASA Global Modeling and Assimilation Office, Greenbelt, MD, USA., Universities Space Research Association)

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Abstract: INTRODUCTION Earth's carbon cycle involves large fluxes of carbon dioxide (CO₂) between the atmosphere, land biosphere, and oceans. Over the past several decades, net loss of CO₂ from the atmosphere to the land and oceans has varied considerably from year to year, equaling 20 to 80% of CO₂ emissions from fossil fuel combustion and land use change. On average, the uptake is about 50%. The imbalance between CO₂ emissions and removal is seen in increasing atmospheric CO₂ concentrations. In recent years, an increase of 2 to 3 parts per million (ppm) per year in the atmospheric mole fraction, which is currently about 400 ppm, has been observed. Almost a quarter of the CO₂ emitted by human activities is being absorbed by the ocean, and another quarter is absorbed

by processes on land. The identity and location of the terrestrial sinks are poorly understood. This absorption has been attributed by some to tropical or Eurasian temperate forests, whereas others argue that these regions may be net sources of CO₂. The efficiency of these land sinks appears to vary dramatically from year to year. Because the identity, location, and processes controlling these natural sinks are not well constrained, substantial additional uncertainty is added to projections of future CO₂ levels.

RATIONALE The NASA satellite, the Orbiting Carbon Observatory-2 (OCO-2), which was launched on 2 July 2014, is designed to collect global measurements with sufficient precision, coverage, and resolution to aid in resolving sources and sinks of CO₂ on regional scales. Since 6 September 2014, the OCO-2 mission has been producing about 2 million estimates of the column-averaged CO₂ dry-air mole fraction (X_{CO_2}) each month after quality screening, with spatial resolution of <3 km² per sounding. Solar-induced chlorophyll fluorescence (SIF), a small amount of light emitted during photosynthesis, is detected in remote sensing measurements of radiance within solar Fraunhofer lines and is another data product from OCO-2.

RESULTS The measurements from OCO-2 provide a global view of the seasonal cycles and spatial patterns of atmospheric CO₂, with the anticipated year-over-year growth rate. The buildup of CO₂ in the Northern Hemisphere during winter and its rapid decrease in concentration as spring arrives (and the SIF increases) is seen in unprecedented detail. The enhanced CO₂ in urban areas relative to nearby background areas is observed with a single overpass of OCO-2. Increases in CO₂ due to the biomass burning in Africa are also clearly observed. The dense, global, X_{CO_2} and SIF data sets from OCO-2 are combined with other remote sensing data sets and used to disentangle the processes driving the carbon cycle on regional scales during the recent 2015-2016 El Niño event. This analysis shows more carbon release in 2015 relative to 2011 over Africa, South America, and Southeast Asia. Now, the fundamental driver for the change in carbon release can be assessed continent by continent, rather than treating the tropics as a single, integrated region. Small changes in X_{CO_2} were also observed early in the El Niño over the equatorial eastern Pacific, due to less upwelling of cold, carbon-rich water than is typical.

CONCLUSION NASA's OCO-2 mission is collecting a dense, global set of high-spectral resolution measurements that are used to estimate X_{CO_2} and SIF. The OCO-2 mission data set can now be used to assess regional-scale sources and sinks of CO₂ around the globe. The papers in this collection present early scientific findings from this new data set. El Niño impact on carbon flux in 2015 relative to 2011, detected from Greenhouse Gases Observing Satellite (GOSAT) and OCO-2 data. OCO-2 uses reflected sunlight to derive X_{CO_2} and SIF. This shows OCO-2 X_{CO_2} data over North America from 12 August 2015 to 26 August 2015.

152. Sensitivity of CO₂ surface flux constraints to observational coverage

Authors: B. Byrne (University of Toronto); D. B. A. Jones (University of California Los Angeles, University of Toronto); K. Strong (University of Toronto); Z.-C. Zeng (California Institute of Technology, Chinese University of Hong Kong, Chinese Academy of Sciences); F. Deng (University of Toronto); J. Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: Inverse modeling of regional CO₂ fluxes using atmospheric CO₂ data is sensitive to the observational coverage of the observing network. Here we use the GEOS-Chem adjoint model to examine the sensitivity to CO₂ fluxes of observations from the in situ surface network, the Total Carbon Column Observing Network (TCCON), the Greenhouse Gases Observing Satellite (GOSAT), and the Orbiting Carbon Observatory (OCO-2). We find that OCO-2 has high sensitivity to fluxes throughout the tropics and Southern Hemisphere, while surface observations have high sensitivity to fluxes in the northern extratropics throughout the year. For GOSAT viewing modes, ocean glint data provide the strongest constraints on fluxes in the tropics and Southern Hemisphere during Northern Hemisphere fall and winter relative to other viewing modes. In contrast, GOSAT nadir land data offer the greater sensitivity to fluxes in these regions during Northern Hemisphere spring and summer. For OCO-2 viewing modes, ocean glint data provided the dominant sensitivity to the surface fluxes in the northern subtropics, tropics, and Southern Hemisphere. We performed a series of inversion analyses using pseudodata and found that the varying sensitivities can result in large differences in regional flux estimates. However, combining measurements from different observing systems to exploit their complementarity may lead to a posteriori flux estimates with improved accuracy.

153. A global wetland methane emissions and uncertainty dataset for atmospheric chemical transport models (WetCHARTs version 1.0)

Authors: A. Anthony Bloom; Kevin W. Bowman; Meemong Lee; Alexander J. Turner; Ronny Schroeder; John R. Worden; Richard Weidner; Kyle C. McDonald; Daniel J. Jacob

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Abstract: . Wetland emissions remain one of the principal sources of uncertainty in the global atmospheric methane (CH₄) budget, largely due to poorly constrained process controls on CH₄ production in waterlogged soils. Process-based estimates of global wetland CH₄ emissions and their associated uncertainties can provide crucial prior information for model-based top-down CH₄ emission estimates. Here we construct a global wetland CH₄ emission model ensemble for use in atmospheric chemical transport models (WetCHARTs version 1.0). Our 0.5 degrees x 0.5 degrees resolution model ensemble is based on satellite-derived surface water extent and precipitation reanalyses, nine heterotrophic respiration simulations (eight carbon cycle models and a data-constrained terrestrial carbon cycle analysis) and three temperature dependence parameterizations for the period 2009-2010; an extended ensemble subset based solely on precipitation and the data-constrained terrestrial carbon cycle analysis is derived for the period 2001-2015. We incorporate the mean of the full and extended model ensembles into GEOS-Chem and compare the model against surface measurements of atmospheric CH₄; the model performance (site-level and zonal mean anomaly residuals) compares favourably against published wetland CH₄ emissions scenarios. We find that uncertainties in carbon decomposition rates and the wetland extent together account for more than 80 % of the dominant uncertainty in the timing, magnitude and seasonal variability in wetland CH₄ emissions, although uncertainty in the temperature CH₄ : C dependence is a significant contributor to seasonal variations in mid-latitude wetland CH₄ emissions. The combination of satellite, carbon cycle models and temperature dependence parameterizations provides a physically informed structural a priori uncertainty that is critical for top-down estimates of wetland CH₄ fluxes. Specifically, our ensemble can provide enhanced information on the prior CH₄ emission uncertainty and the error covariance structure, as well as a means for using posterior flux estimates and their uncertainties to quantitatively constrain the biogeochemical process controls of global wetland CH₄ emissions.

154. Consistent regional fluxes of CH₄ and CO₂ inferred from GOSAT proxy XCH₄ : XCO₂ retrievals, 2010-2014

Authors: Liang Feng; Paul I. Palmer; Hartmut Bösch; Robert J. Parker; Alex J. Webb; Caio S. C. Correia; Nicholas M. Deutscher; Lucas G. Domingues; Dietrich G. Feist; Luciana V. Gatti; Emanuel Gloor; Frank Hase; Rigel Kivi; Yi Liu; John B. Miller; Isamu Morino; Ralf Sussmann; Kimberly Strong; Osamu Uchino; Jing Wang; Andreas Zahn

Journal: Atmospheric Chemistry and Physics

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References: 76

Cited by: 57 publications

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Abstract: . We use the GEOS-Chem global 3-D model of atmospheric chemistry and transport and an ensemble Kalman filter to simultaneously infer regional fluxes of methane (CH₄) and carbon dioxide (CO₂) directly from GOSAT retrievals of XCH₄ : XCO₂, using sparse ground-based CH₄ and CO₂ mole fraction data to anchor the ratio. This work builds on the previously reported theory that takes into account that (1) these ratios are less prone to systematic error than either the full-physics data products or the proxy CH₄ data products; and (2) the resulting CH₄ and CO₂ fluxes are self-consistent. We show that a posteriori fluxes inferred from the GOSAT data generally outperform the fluxes inferred only from in situ data, as expected. GOSAT CH₄ and CO₂ fluxes are consistent with global growth rates for CO₂ and CH₄ reported by NOAA and have a range of independent data including new profile measurements (0-7 km) over the Amazon Basin that were collected specifically to help validate GOSAT over this geographical region. We find that large-scale multi-year annual a posteriori CO₂ fluxes inferred from GOSAT data are similar to those inferred from the in situ surface data but with smaller uncertainties, particularly over the tropics. GOSAT data are consistent with smaller peak-to-peak seasonal amplitudes of CO₂ than either the a priori or in situ inversion, particularly over the tropics and the southern extratropics. Over the northern extratropics, GOSAT data show larger uptake than the a priori but less than the in situ inversion, resulting in small net emissions over the year. We also find evidence that the carbon balance of tropical South America was perturbed following the droughts of 2010 and 2012 with net annual fluxes not returning to an approximate annual balance until 2013. In contrast, GOSAT data significantly changed the a priori spatial distribution of CH₄ emission with a 40 % increase over tropical South America and tropical Asia and a smaller decrease over Eurasia and temperate South America. We find no evidence from GOSAT that tropical South American CH₄ fluxes were dramatically affected by the two large-scale Amazon droughts. However, we find that GOSAT data are consistent with double seasonal peaks in Amazonian fluxes that are reproduced over the 5

years we studied: a small peak from January to April and a larger peak from June to October, which are likely due to superimposed emissions from different geographical regions.

155. A 4D-Var inversion system based on the icosahedral grid model (NICAM-TM 4D-Var v1.0) - Part 1: Offline forward and adjoint transport models

Authors: Yosuke Niwa; Hirofumi Tomita; Masaki Satoh; Ryoichi Imasu; Yousuke Sawa; Kazuhiro Tsuboi; Hidekazu Matsueda; Toshinobu Machida; Motoki Sasakawa; Boris Belan; Nobuko Saigusa

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-10-1157-2017

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Open Access: N/A

Abstract: . A four-dimensional variational (4D-Var) method is a popular algorithm for inverting atmospheric greenhouse gas (GHG) measurements. In order to meet the computationally intense 4D-Var iterative calculation, offline forward and adjoint transport models are developed based on the Nonhydrostatic ICosahedral Atmospheric Model (NICAM). By introducing flexibility into the temporal resolution of the input meteorological data, the forward model developed in this study is not only computationally efficient, it is also found to nearly match the transport performance of the online model. In a transport simulation of atmospheric carbon dioxide (CO₂), the data-thinning error (error resulting from reduction in the time resolution of the meteorological data used to drive the offline transport model) is minimized by employing high temporal resolution data of the vertical diffusion coefficient; with a low 6-hourly temporal resolution, significant concentration biases near the surface are introduced. The new adjoint model can be run in discrete or continuous adjoint mode for the advection process. The discrete adjoint is characterized by perfect adjoint relationship with the forward model that switches off the flux limiter, while the continuous adjoint is characterized by an imperfect but reasonable adjoint relationship with its corresponding forward model. In the latter case, both the forward and adjoint models use the flux limiter to ensure the monotonicity of tracer concentrations and sensitivities. Trajectory analysis for high CO₂ concentration events are performed to test adjoint sensitivities. We also demonstrate the potential usefulness of our adjoint model for diagnosing tracer transport. Both the offline forward and adjoint models have computational efficiency about 10 times higher than the online model. A description of our new 4D-Var system that includes an optimization method, along with its application in an atmospheric CO₂ inversion and the effects of using either the discrete or continuous adjoint method, is presented in an accompanying paper Niwa et al.(2016).

156. Simulating estimation of California fossil fuel and biosphere carbon dioxide exchanges combining in situ tower and satellite column observations

Authors: Marc L. Fischer (Lawrence Berkeley National Laboratory); Nicholas Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kieran Brophy (Imperial College London); Xinguang Cui (Lawrence Berkeley National Laboratory); Seongeun Jeong (Lawrence Berkeley National Laboratory); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ralph Keeling (Scripps Institution of Oceanography / University of California San Diego); Thomas E. Taylor (Colorado State University); Kevin Gurney (Arizona State University); Tomohiro Oda (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Heather Graven (Imperial College London)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jd025617

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Keywords: N/A

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Abstract: We report simulation experiments estimating the uncertainties in California regional fossil fuel and biosphere CO₂ exchanges that might be obtained by using an atmospheric inverse modeling system driven by the combination of ground-based observations of radiocarbon and total CO₂, together with column-mean CO₂ observations from NASA's Orbiting Carbon Observatory (OCO-2). The work includes an initial examination of statistical uncertainties in prior models for CO₂ exchange, in radiocarbon-based fossil fuel CO₂ measurements, in OCO-2 measurements, and in a regional atmospheric transport modeling system. Using these nominal assumptions for measurement and model uncertainties, we find that flask measurements of radiocarbon and total CO₂ at 10 towers can be used to distinguish between different fossil fuel emission data products for major urban regions of California. We then show that the combination of flask and OCO-2 observations yields posterior uncertainties in monthly-mean fossil fuel emissions of ~5-10%, levels likely useful for policy relevant evaluation of bottom-up fossil fuel emission estimates. Similarly, we find that inversions yield uncertainties in monthly biosphere CO₂ exchange of ~6%-12%, depending on season, providing useful information on net carbon uptake in California's forests and agricultural lands. Finally, initial sensitivity analysis suggests that obtaining the above results requires control of systematic biases below approximately 0.5 ppm, placing requirements on accuracy of the atmospheric measurements, background subtraction, and atmospheric transport modeling.

157. Development of the WRF-CO2 4DVar assimilation system

Authors: Tao Zheng; Nancy French; Martin Baxter

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Regional atmospheric CO₂ inversions commonly use Lagrangian particle trajectory model simulations to calculate the required influence function. To provide an alternative, we developed an adjoint based four-dimensional variational (4DVar) assimilation system, WRF-CO₂ 4DVar. This system is developed based on the Weather Research and Forecasting (WRF) model system, including WRF-Chem, WRFPLUS, and WRFDA. In WRF-CO₂ 4DVR, CO₂ is modeled as a tracer and its feedback to meteorology is ignored. This configuration allows most WRF physical parameterizations to be used in the assimilation system without incurring a large amount of code development. WRF-CO₂ 4DVar solves for the optimized CO₂ emission scaling factors in a Bayesian framework. Two variational optimization schemes are implemented for the system: the first uses the L-BFGS-B and the second uses the Lanczos conjugate gradient (CG) in an incremental approach. We modified WRFPLUS forward, tangent linear, and adjoint models to include CO₂ related processes. The system is tested by simulations over a domain covering the continental United States at 48 km x 48 km grid spacing. The accuracy of the tangent linear and adjoint models are assessed by comparing against finite difference sensitivity. The system's effectiveness for CO₂ inverse modeling is tested using pseudo-observation data. The results of the sensitivity and inverse modeling tests demonstrate the potential usefulness of WRF-CO₂ 4DVar for regional CO₂ inversions.

158. Enhanced Methane Emissions during Amazonian Drought by Biomass Burning

Authors: Makoto Saito; Heon-Sook Kim; Akihiko Ito; Tatsuya Yokota; Shamil Maksyutov

Journal: PLOS ONE

Publisher: Public Library of Science (PLoS)

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Subjects: N/A

Keywords: N/A

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Abstract: N/A

159. Gridded National Inventory of U.S. Methane Emissions

Authors: Joannes D. Maasakkers (Harvard University); Daniel J. Jacob (Harvard University); Melissa P. Sulprizio (Harvard University); Alexander J. Turner (Harvard University); Melissa Weitz (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Tom Wirth (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Cate Hight (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Mark DeFigueiredo (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Mausami Desai (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Rachel Schmeltz (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Leif Hockstad (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Anthony A. Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Seongeun Jeong (Lawrence Berkeley National Laboratory); Marc L. Fischer (Lawrence Berkeley National Laboratory)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

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Language: en

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Keywords: N/A

Funding: National Aeronautics and Space Administration, Office of Science, U.S. Department of Energy

License: N/A

Open Access: N/A

Abstract: N/A

160. Comparison between the Local Ensemble Transform Kalman Filter (LETKF) and 4D-Var in atmospheric CO₂ flux inversion with the Goddard Earth Observing System-Chem model and the observation impact diagnostics from the LETKF

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jd025100

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Cited by: 19 publications

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Keywords: N/A

Funding: NASA OCO-2 science team grant, NASA Carbon Cycle Science program, NASA Carbon Monitoring System program

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Open Access: N/A

Abstract: Ensemble Kalman filter (EnKF) and 4D-Variational (4D-Var) are two advanced data assimilation methods that are the basis of numerical weather prediction and have been extensively used in trace gas assimilation and inverse modeling. In this study, we compare 4D-Var and the Local Ensemble Transform Kalman Filter (LETKF), one type of EnKF, in estimating CO₂ fluxes with both simulated and real satellite data from Greenhouse gases Observing Satellite (GOSAT) and propose a method to calculate flux changes and flux error reductions from assimilating each observation within the LETKF. The results show that the mean posterior flux accuracy across 11 land regions defined by the Atmospheric Tracer Transport Model Intercomparison Project is comparable between 4D-Var and the LETKF, as shown in the Observing System Simulation Experiment, but the differences between the LETKF and 4D-Var are relatively larger over data sparse regions. We show that this is most likely due to the fact that the observations from a much broader region have impact on flux estimation in 4D-Var than in the LETKF. As a result, the posterior fluxes from 4D-Var are more consistent with the atmospheric CO₂ growth rate. We find that the inversion results are less dependent on inversion methods with the increase of observations. With real GOSAT observations, we show that the posterior flux changes in 2011 relative to 2010 are more consistent between these two methods than the absolute estimates.

161. Comparison of national level biomass maps for conterminous US: understanding pattern and causes of differences

Authors: N. Neeti; R. Kennedy

Journal: Carbon Balance and Management

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s13021-016-0060-y

URL: <https://doi.org/10.1186/s13021-016-0060-y>

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Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: N/A

162. Technical note: 3-hourly temporal downscaling of monthly global terrestrial biosphere model net ecosystem exchange

Authors: Joshua B. Fisher; Munish Sikka; Deborah N. Huntzinger; Christopher Schwalm; Junjie Liu

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-13-4271-2016

URL: <https://doi.org/10.5194/bg-13-4271-2016>

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References: 33

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The land surface provides a boundary condition to atmospheric forward and flux inversion models. These models require prior estimates of CO₂ fluxes at relatively high temporal resolutions (e.g., 3-hourly) because of the high frequency of atmospheric mixing and wind heterogeneity. However, land surface model CO₂ fluxes are often provided at monthly time steps, typically because the land surface modeling community focuses more on time steps associated with plant phenology (e.g., seasonal) than on sub-daily phenomena. Here, we describe a new dataset created from 15 global land surface models and 4 ensemble products in the Multi-scale Synthesis and Terrestrial Model Intercomparison Project (MsTMIP), temporally downscaled from monthly to 3-hourly output. We provide 3-hourly output for each individual model over 7 years (2004-2010), as well as an ensemble mean, a weighted ensemble mean, and the multi-model standard deviation. Output is provided in three different spatial resolutions for user preferences: 0.5 degrees x 0.5 degrees, 2.0 degrees x 2.5 degrees, and 4.0 degrees x 5.0 degrees (latitude x longitude). These data are publicly available from [doi:10.3334/ORNLDAAAC/1315](https://doi.org/10.3334/ORNLDAAAC/1315).

163. Constraints on methane emissions in North America from future geostationary remote-sensing measurements

Authors: Nicolas Bousserez; Daven K. Henze; Brigitte Rooney; Andre Perkins; Kevin J. Wecht; Alexander J. Turner; Vijay Natraj; John R. Worden

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-16-6175-2016

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Language: en

Subjects: N/A

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Open Access: N/A

Abstract: . The success of future geostationary (GEO) satellite observation missions depends on our ability to design instruments that address their key scientific objectives. In this study, an Observation System Simulation Experiment (OSSE) is performed to quantify the constraints on methane (CH₄) emissions in North America obtained from shortwave infrared (SWIR), thermal infrared (TIR), and multi-spectral (SWIR+TIR) measurements in geostationary orbit and from future SWIR low-Earth orbit (LEO) measurements. An efficient stochastic algorithm is used to compute the information content of the inverted emissions at high spatial resolution (0.5 degrees x 0.7 degrees) in a variational framework using the GEOS-Chem chemistry-transport model and its adjoint. Our results show that at sub-weekly timescales, SWIR measurements in GEO orbit can constrain about twice as many independent flux patterns than in LEO orbit, with a degree of freedom for signal (DOF) for the inversion of 266 and 115, respectively. Comparisons between TIR GEO and SWIR LEO configurations reveal that poor boundary layer sensitivities for the TIR measurements cannot be compensated for by the high spatiotemporal sampling of a GEO orbit. The benefit of a multi-spectral instrument compared to current SWIR products in a GEO context is shown for sub-weekly timescale constraints, with an increase in the DOF of about 50 % for a 3-day inversion. Our results further suggest that both the SWIR and multi-spectral measurements on GEO orbits could almost fully resolve CH₄ fluxes at a spatial resolution of at least 100 km x 100 km over source hotspots (emissions > 4 x 10⁵ kg day⁻¹). The sensitivity of the optimized emission scaling factors to typical errors in boundary and initial conditions can reach 30 and 50 % for the SWIR GEO or SWIR LEO configurations, respectively, while it is smaller than 5 % in the case of a multi-spectral GEO system. Overall, our results demonstrate that multi-spectral measurements from a geostationary satellite platform would address the need for higher spatiotemporal constraints on CH₄ emissions while greatly mitigating the impact of inherent uncertainties in source inversion methods on the inferred fluxes.

164. Separation of biospheric and fossil fuel fluxes of CO₂ by atmospheric inversion of CO₂ and 14CO₂ measurements: Observation System Simulations

Authors: Sourish Basu; John Bharat Miller; Scott Lehman

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-16-5665-2016

URL: <https://doi.org/10.5194/acp-16-5665-2016>

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References: 58

Cited by: 47 publications

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Abstract: . National annual total CO₂ emissions from combustion of fossil fuels are likely known to within 5-10 % for most developed countries. However, uncertainties are inevitably larger (by unknown amounts) for emission estimates at regional and monthly scales, or for developing countries. Given recent international efforts to establish emission reduction targets, independent determination and verification of regional and national scale fossil fuel CO₂ emissions are likely to become increasingly important. Here, we take advantage of the fact that precise measurements of 14C in CO₂ provide a largely unbiased tracer for recently added fossil-fuel-derived CO₂ in the atmosphere and present an atmospheric inversion technique to jointly assimilate observations of CO₂ and 14CO₂ in order to simultaneously estimate fossil fuel emissions and biospheric exchange fluxes of CO₂. Using this method in a set of Observation System Simulation Experiments (OSSEs), we show that given the coverage of 14CO₂ measurements available in 2010 (969 over North America, 1063 globally), we can recover the US national total fossil fuel emission to better than 1 % for the year and to within 5 % for most months. Increasing the number of 14CO₂ observations to ~ 5000 per year over North America, as recently recommended by the National Academy of Science (NAS) (Pacala et al., 2010), we recover monthly emissions to within 5 % for all months for the US as a whole and also for smaller, highly emissive regions over which the specified data coverage is relatively dense, such as for the New England states or the NY-NJ-PA tri-state area. This result suggests that, given continued improvement in state-of-the art transport models, a measurement program similar in scale to that recommended by the NAS can provide for independent verification of bottom-up inventories of fossil fuel CO₂ at the regional and national scale. In addition, we show that the dual tracer inversion framework can detect and minimize biases in estimates of the biospheric flux that would otherwise arise in a traditional CO₂-only inversion when prescribing fixed but inaccurate fossil fuel fluxes.

165. A global perspective of atmospheric carbon dioxide concentrations

Authors: William M Putman; Lesley Ott; Anton Darmanov; Arlindo daSilva

Journal: Parallel Computing

Publisher: Elsevier BV

DOI: 10.1016/j.parco.2016.03.001

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Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Modeling, Analysis, and Prediction (MAP) Program, NASA High End Computing program

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Abstract: N/A

166. Orbiting Carbon Observatory-2 (OCO-2) cloud screening algorithms: validation against collocated MODIS and CALIOP data

Authors: Thomas E. Taylor; Christopher W. O'Dell; Christian Frankenberg; Philip T. Partain; Heather Q. Cronk; Andrey Savtchenko; Robert R. Nelson; Emily J. Rosenthal; Albert Y. Chang; Brenden Fisher; Gregory B. Osterman; Randy H. Pollock; David Crisp; Annmarie Eldering; Michael R. Gunson

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

DOI: 10.5194/amt-9-973-2016

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References: 42

Cited by: 76 publications

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Open Access: N/A

Abstract: . The objective of the National Aeronautics and Space Administration's (NASA) Orbiting Carbon Observatory-2 (OCO-2) mission is to retrieve the column-averaged carbon dioxide (CO₂) dry air mole fraction (XCO₂) from satellite measurements of reflected sunlight in the near-infrared. These estimates can be biased by clouds and aerosols, i.e., contamination, within the instrument's field of view. Screening of the most contaminated soundings minimizes unnecessary calls to the computationally expensive Level 2 (L2) XCO₂ retrieval algorithm. Hence, robust cloud screening methods have been an important focus of the OCO-2 algorithm development team. Two distinct, computationally inexpensive cloud screening algorithms have been developed for this application. The A-Band Preprocessor (ABP) retrieves the surface pressure using measurements in the 0.76 μm O₂ A band, neglecting scattering by clouds and aerosols, which introduce photon path-length differences that can cause large deviations between the expected and retrieved surface pressure. The Iterative Maximum A Posteriori (IMAP) Differential Optical Absorption Spectroscopy (DOAS) Preprocessor (IDP) retrieves independent estimates of the CO₂ and H₂O column abundances using observations taken at 1.61 μm (weak CO₂ band) and 2.06 μm (strong CO₂ band), while neglecting atmospheric scattering. The CO₂ and H₂O column abundances retrieved in these two spectral regions differ significantly in the presence of cloud and scattering aerosols. The combination of these two algorithms, which are sensitive to different features in the spectra, provides the basis for cloud screening of the OCO-2 data set. To validate the OCO-2 cloud screening approach, collocated measurements from NASA's Moderate Resolution Imaging Spectrometer (MODIS), aboard the Aqua platform, were compared to results from the two OCO-2 cloud screening algorithms. With tuning of algorithmic threshold parameters that allows for processing of ■ 20-25 % of all OCO-2 soundings, agreement between the OCO-2 and MODIS cloud screening methods is found to be ■ 85 % over four 16-day orbit repeat cycles in both the winter (December) and spring (April-May) for OCO-2 nadir-land, glint-land and glint-water observations. No major, systematic, spatial or temporal dependencies were found, although

slight differences in the seasonal data sets do exist and validation is more problematic with increasing solar zenith angle and when surfaces are covered in snow and ice and have complex topography. To further analyze the performance of the cloud screening algorithms, an initial comparison of OCO-2 observations was made to collocated measurements from the Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP) aboard the Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO). These comparisons highlight the strength of the OCO-2 cloud screening algorithms in identifying high, thin clouds but suggest some difficulty in identifying some clouds near the surface, even when the optical thicknesses are greater than 1.

167. A method for independent validation of surface fluxes from atmospheric inversion: Application to CO₂

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl067828

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Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: CO₂ flux inversion, atmospheric inversion, independent validation

Funding: OCO-2 science team, NASA Carbon Monitoring System

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Abstract: Validating fluxes from an atmospheric inversion is a challenging problem because there are often no direct flux measurements at comparable spatiotemporal scales whereas there are often relevant independent observables, e.g., trace gas concentrations. In this paper, we propose a method that validates posterior fluxes by projecting the errors between posterior and prior observable model states and independent data to the spatiotemporal differences between posterior and prior fluxes with an atmospheric transport adjoint model. We prove theoretically the conditions for which observed error reductions lead to error reductions in fluxes. We apply this approach to the atmospheric CO₂ inversion problem using the NASA Carbon Monitoring System Flux project with an Observing System Simulation Experiment. We show that the posterior fluxes are more accurate than the prior over the region that significantly contributes to the reduction of CO₂ errors, which is consistent with the theory.

168. Does GOSAT capture the true seasonal cycle of carbon dioxide?

Authors: H. Lindqvist; C. W. O'Dell; S. Basu; H. Boesch; F. Chevallier; N. Deutscher; L. Feng; B. Fisher; F. Hase; M. Inoue; R. Kivi; I. Morino; P. I. Palmer; R. Parker; M. Schneider; R. Sussmann; Y. Yoshida

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License: Creative Commons

Open Access: N/A

Abstract: . The seasonal cycle accounts for a dominant mode of total column CO₂ (XCO₂) annual variability and is connected to CO₂ uptake and release; it thus represents an important quantity to test the accuracy of the measurements from space. We quantitatively evaluate the XCO₂ seasonal cycle of the Greenhouse Gases Observing Satellite (GOSAT) observations from the Atmospheric CO₂ Observations from Space (ACOS) retrieval system and compare average regional seasonal cycle features to those directly measured by the Total Carbon Column Observing Network (TCCON). We analyse the mean seasonal cycle amplitude, dates of maximum and minimum XCO₂, as well as the regional growth rates in XCO₂ through the fitted trend over several years. We find that GOSAT/ACOS captures the seasonal cycle amplitude within 1.0 ppm accuracy compared to TCCON, except in Europe, where the difference exceeds 1.0 ppm at two sites, and the amplitude captured by GOSAT/ACOS is generally shallower compared to TCCON. This bias over Europe is not as large for the other GOSAT retrieval algorithms (NIES v02.21, RemoTeC v2.35, UoL v5.1, and NIES PPDF-S v.02.11), although they have significant biases at other sites. We find that the ACOS bias correction partially explains the shallow amplitude over Europe. The impact of the co-location method and aerosol changes in the ACOS algorithm were also tested and found to be few tenths of a ppm and mostly non-systematic. We find generally good agreement in the date of minimum XCO₂ between ACOS and TCCON, but ACOS generally infers a date of maximum XCO₂ 2-3 weeks later than TCCON. We further analyse the latitudinal dependence of the seasonal cycle amplitude throughout the Northern Hemisphere and compare the dependence to that predicted by current optimized models that assimilate in situ measurements of CO₂. In the zonal averages, models are consistent with the GOSAT amplitude to within 1.4 ppm, depending on the model and latitude. We also show that the seasonal cycle of XCO₂ depends on longitude especially at the mid-latitudes: the amplitude of GOSAT XCO₂ doubles from western USA to East Asia at 45-50 degrees N, which is only partially shown by the models. In general, we find that model-to-model differences can be larger than GOSAT-to-model differences. These results suggest that GOSAT/ACOS retrievals of the XCO₂ seasonal cycle may be

sufficiently accurate to evaluate land surface models in regions with significant discrepancies between the models.

169. Continuity of NASA Earth Observations from Space

Authors: No authors listed

Journal: Unknown Journal

Publisher: National Academies Press

DOI: 10.17226/21789

URL: <https://doi.org/10.17226/21789>

Publication Date: 2015-10-24

Volume/Issue: N/A

References: N/A

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

170. Biogenic carbon fluxes from global agricultural production and consumption

Authors: Julie Wolf (Pacific Northwest National Laboratory); Tristram O. West (Pacific Northwest National Laboratory); Yannick Le Page (Pacific Northwest National Laboratory); G. Page Kyle (Pacific Northwest National Laboratory); Xuesong Zhang (Pacific Northwest National Laboratory); G. James Collatz (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Marc L. Imhoff (Pacific Northwest National Laboratory)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015gb005119

URL: <https://doi.org/10.1002/2015gb005119>

Publication Date: 2015-09-08

Volume/Issue: Volume 29, Issue 10

References: 81

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: methane, remote Sensing, land cover, agricultural sustainability, atmospheric inverse modeling (and 1 more)

Funding: National Aeronautics and Space Administration Carbon Monitoring System

License: N/A

Open Access: N/A

Abstract: Quantification of biogenic carbon fluxes from agricultural lands is needed to generate comprehensive bottom-up estimates of net carbon exchange for global and regional carbon monitoring. We estimated global agricultural carbon fluxes associated with annual crop net primary production (NPP), harvested biomass, and consumption of biomass by humans and livestock. These estimates were combined for a single estimate of net carbon exchange and spatially distributed to 0.05 degrees resolution using Moderate Resolution Imaging Spectroradiometer satellite land cover data. Global crop NPP in 2011 was estimated at 5.25 +/- 0.46 Pg C yr⁻¹, of which 2.05 +/- 0.05 Pg C yr⁻¹ was harvested and 0.54 Pg C yr⁻¹ was collected from crop residues for livestock fodder. Total livestock feed intake in 2011 was 2.42 +/- 0.21 Pg C yr⁻¹, of which 2.31 +/- 0.21 Pg C yr⁻¹ was emitted as CO₂, 0.07 +/- 0.01 Pg C yr⁻¹ was emitted as CH₄, and 0.04 Pg C yr⁻¹ was contained within milk and egg production. Livestock grazed an estimated 1.27 Pg C yr⁻¹ in 2011, which constituted 52.4% of total feed intake. Global human food intake was 0.57 +/- 0.03 Pg C yr⁻¹ in 2011, the majority of which was respired as CO₂. Completed global cropland carbon budgets accounted for the ultimate use of approximately 80% of harvested biomass. The spatial distribution of these fluxes may be used for global carbon monitoring, estimation of regional uncertainty, and for use as input to Earth system models.

171. Quantifying lower tropospheric methane concentrations using GOSAT near-IR and TES thermal IR measurements

Authors: J. R. Worden; A. J. Turner; A. Bloom; S. S. Kulawik; J. Liu; M. Lee; R. Weidner; K. Bowman; C. Frankenberg; R. Parker; V. H. Payne

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

DOI: 10.5194/amt-8-3433-2015

URL: <https://doi.org/10.5194/amt-8-3433-2015>

Publication Date: 2015-08-25

Volume/Issue: Volume 8, Issue 8

References: 43

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Evaluating surface fluxes of CH₄ using total column data requires models to accurately account for the transport and chemistry of methane in the free troposphere and stratosphere, thus reducing sensitivity to the underlying fluxes. Vertical profiles of methane have increased sensitivity to surface fluxes because lower tropospheric methane is more sensitive to surface fluxes than a total column, and quantifying free-tropospheric CH₄ concentrations helps to evaluate the impact of transport and chemistry uncertainties on estimated surface fluxes. Here we demonstrate the potential for estimating lower tropospheric CH₄ concentrations through the combination of free-tropospheric methane measurements from the Aura Tropospheric Emission Spectrometer (TES) and XCH₄ (dry-mole air fraction of methane) from the Greenhouse gases Observing SATellite - Thermal And Near-infrared for carbon Observation (GOSAT TANSO, herein GOSAT for brevity). The calculated precision of these estimates ranges from 10 to 30 ppb for a monthly average on a 4 degrees x 5 degrees latitude/longitude grid making these data suitable for evaluating lower-tropospheric methane concentrations. Smoothing error is approximately 10 ppb or less. Comparisons between these data and the GEOS-Chem model demonstrate that these lower-tropospheric CH₄ estimates can resolve enhanced concentrations over flux regions that are challenging to resolve with total column measurements. We also use the GEOS-Chem model and surface measurements in background regions across a range of latitudes to determine that these lower-tropospheric estimates are biased low by approximately 65 ppb, with an accuracy of approximately 6 ppb (after removal of the bias) and an actual precision of approximately 30 ppb. This 6 ppb accuracy is consistent with the accuracy of TES and GOSAT methane retrievals.

172. Using Green's Functions to initialize and adjust a global, eddying ocean biogeochemistry general circulation model

Authors: H. Brix; D. Menemenlis; C. Hill; S. Dutkiewicz; O. Jahn; D. Wang; K. Bowman; H. Zhang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2015.07.008

URL: <https://doi.org/10.1016/j.ocemod.2015.07.008>

Publication Date: 2015-07-26

Volume/Issue: Volume 95

References: 86

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

173. Observing System Simulation Experiments for air quality

Authors: R.M.A. Timmermans; W.A. Lahoz; J.-L. Attié; V.-H. Peuch; R.L. Curier; D.P. Edwards; H.J. Eskes; P.J.H. Builtjes

Journal: Atmospheric Environment

Publisher: Elsevier BV

DOI: 10.1016/j.atmosenv.2015.05.032

URL: <https://doi.org/10.1016/j.atmosenv.2015.05.032>

Publication Date: 2015-05-18

Volume/Issue: Volume 115

References: 64

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: TNO, NILU

License: N/A

Open Access: N/A

Abstract: N/A

174. An intercomparison of inverse models for estimating sources and sinks of CO₂ using GOSAT measurements

Authors: S. Houweling (Institute for Marine and Atmospheric Research Utrecht Utrecht Netherlands, SRON Netherlands Institute for Space Research Utrecht Netherlands); D. Baker (Colorado State University); S. Basu (NOAA ESRL Boulder Colorado USA); H. Boesch (University of Leicester); A. Butz (Karlsruhe Institute of Technology); F. Chevallier (Le Laboratoire des Sciences du Climat et l'Environnement Gif-Sur-Yvette France); F. Deng (University of Toronto); E. J. Dlugokencky (NOAA ESRL Boulder Colorado USA); L. Feng (University of Edinburgh); A. Ganshin (Central Aerological Observatory); O. Hasekamp (SRON Netherlands Institute for Space Research Utrecht Netherlands); D. Jones (University of Toronto); S. Maksyutov (National Institute for Environmental Studies); J. Marshall (Max Planck Institute for Biogeochemistry); T. Oda (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, NOAA ESRL Boulder Colorado USA, Goddard Earth Sciences Technology and Research (GESTAR) Columbia USA); C. W. O'Dell (Colorado State University); S. Oshchepkov (National Institute for Environmental Studies); P. I. Palmer (University of Edinburgh); P. Peylin (Le Laboratoire des Sciences du Climat et l'Environnement Gif-Sur-Yvette France); Z. Poussi (Climmod Orsay France); F. Reum (Max Planck Institute for Biogeochemistry); H. Takagi (National Institute for Environmental Studies); Y. Yoshida (National Institute for Environmental Studies); R. Zhuravlev (Central Aerological Observatory)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

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Volume/Issue: Volume 120, Issue 10

References: 42

Cited by: 112 publications

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Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: This study presents the outcome of an inverse modeling intercomparison experiment on the use of total column CO₂ retrievals from Greenhouse Gas Observing Satellite (GOSAT) for quantifying global sources and sinks of CO₂. Eight research groups submitted inverse modeling results for the first year of GOSAT measurements. Inversions were carried out using only GOSAT data, a combination of GOSAT and surface measurements, and using only surface measurements. As expected, the most robust flux estimates are obtained at large scales (e.g., within 20% of the annual flux at the global scale), and they quickly diverge toward the scale of the subcontinental TRANSCOM regions and beyond (to >100% of the annual flux). We focus our analysis on a shift in the CO₂ uptake over land from the Tropics toward the Northern Hemisphere Extra tropics of ~1 PgC/yr when GOSAT data are used in the inversions. This shift is largely driven by TRANSCOM regions Europe and Northern Africa, showing, respectively, an increased uptake and release of 0.7 and 0.9 PgC/yr. Inversions using

GOSAT data show a reduced gradient between midlatitudes of the Northern Hemisphere and the Tropics, consistent with the latitudinal shift in carbon uptake. However, the reduced gradients degrade the agreement with background aircraft and surface measurements. To narrow the range of inversion-derived flux, estimates will require further efforts to understand the differences not only between the retrieval schemes but also between inverse models, as their contributions to the overall uncertainty are estimated to be of similar magnitude.

175. Source-receptor relationships of column-average CO₂ and implications for the impact of observations on flux inversions

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Daven K. Henze (University of Colorado Boulder)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jd022914

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References: 47

Cited by: 22 publications

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Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Source-receptor relationships are the fundamental quantities used in atmosphere CO₂ flux inversions. In this study, we systematically investigate the global source-receptor relationships of column CO₂ (XCO₂) in 12 continental-scale receptors in terms of transport and local versus nonlocal flux contributions using the GEOS-Chem adjoint model. Using simulated Greenhouse gases Observing Satellite (GOSAT) XCO₂, we quantify the impact of inclusion (add-on) or exclusion of observations (data-denial) within a receptor region on flux inversion. We discuss the connections between the observation impact and the underlying source-receptor relationships. The strong sensitivity of XCO₂ to nonlocal fluxes makes the XCO₂ observations have strong impact on nonlocal flux estimation. On an annual mean, the impact of GOSAT XCO₂ over Europe on North America flux estimation is 10%-39% of the full observation impact. Because the Southern Hemisphere midlatitude XCO₂ are most sensitive to local and tropical fluxes, the mean impact of GOSAT XCO₂ over Southern South America on Northern South America (N-S-America) flux estimation is 30%-59% of the full observation impact. Because of the strong sensitivity of the XCO₂ over N-S-America to Central Africa (C-Africa) fluxes, the mean impact on C-Africa flux estimation is between 14% and 31% of the full observation impact in spite of the sparse observation coverage. The results also show that XCO₂ have similar sensitivity to local and nonlocal fluxes occurring 3 months before, which indicates that XCO₂ observations cannot differentiate any fluxes occurring beyond 3 months.

176. Assessing the magnitude of CO₂ flux uncertainty in atmospheric CO₂ records using products from NASA's Carbon Monitoring Flux Pilot Project

Authors: Lesley E. Ott (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Steven Pawson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); George J. Collatz (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Watson W. Gregg (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Holger Brix (Now at Helmholtz-Zentrum Geesthacht Institute of Coastal Research Geesthacht Germany, University of California Los Angeles); Cecile S. Rousseaux (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Annmarie Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael R. Gunson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Stephan R. Kawa (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jd022411

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Publication Date: 2014-12-03

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References: 110

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA's CMS FPP and Phase One activities

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Open Access: N/A

Abstract: NASA's Carbon Monitoring System Flux Pilot Project (FPP) was designed to better understand contemporary carbon fluxes by bringing together state-of-the art models with remote sensing data sets. Here we report on simulations using NASA's Goddard Earth Observing System Model, version 5 (GEOS-5) which was used to evaluate the consistency of two different sets of observationally informed land and ocean fluxes with atmospheric CO₂ records. Despite the observation inputs, the average difference in annual terrestrial biosphere flux between the two land (NASA Ames Carnegie-Ames-Stanford-Approach (CASA) and CASA-Global Fire Emissions Database version 3 (GFED)) models is 1.7 Pg C for 2009-2010. Ocean models (NASA's Ocean Biogeochemical Model (NOBM) and Estimating the Circulation and Climate of the Ocean Phase II (ECCO₂)-Darwin) differ by 35% in their global estimates of carbon flux with particularly strong disagreement in high latitudes. Based upon combinations of terrestrial and ocean fluxes, GEOS-5 reasonably simulated the seasonal

cycle observed at Northern Hemisphere surface sites and by the Greenhouse gases Observing SATellite (GOSAT) while the model struggled to simulate the seasonal cycle at Southern Hemisphere surface locations. Though GEOS-5 was able to reasonably reproduce the patterns of XCO₂ observed by GOSAT, it struggled to reproduce these aspects of Atmospheric Infrared Sounder observations. Despite large differences between land and ocean flux estimates, resulting differences in atmospheric mixing ratio were small, typically less than 5 ppm at the surface and 3 ppm in the XCO₂ column. A statistical analysis based on the variability of observations shows that flux differences of these magnitudes are difficult to distinguish from inherent measurement variability, regardless of the measurement platform.

177. Improved analysis-error covariance matrix for high-dimensional variational inversions: application to source estimation using a 3D atmospheric transport model

Authors: N. Bousserez (University of Colorado Boulder); D. K. Henze (University of Colorado Boulder); A. Perkins (University of Colorado Boulder); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); F. Deng (University of Toronto); D. B. A. Jones (University of California Los Angeles, University of Toronto)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2495

URL: <https://doi.org/10.1002/qj.2495>

Publication Date: 2014-11-14

Volume/Issue: Volume 141, Issue 690

References: 42

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, National Aeronautics and Space Administration (and 1 more)

License: N/A

Open Access: N/A

Abstract: Variational methods are widely used to solve geophysical inverse problems. Although gradient-based minimization algorithms are available for high-dimensional problems (dimension $>10^6$), they do not provide an estimate of the errors in the optimal solution. In this study, we assess the performance of several numerical methods to approximate the analysis-error covariance matrix, assuming reasonably linear models. The evaluation is performed for a CO₂ flux estimation problem using synthetic remote-sensing observations of CO₂ columns. A low-dimensional experiment is considered in order to compare the analysis error approximations to a full-rank finite-difference inverse Hessian estimate, followed by a realistic high-dimensional application. Two stochastic approaches, a Monte-Carlo simulation and a method based on random gradients of the cost function, produced analysis error variances with a relative error $<10\%$. The long-distance error correlations due to sampling noise are significantly less pronounced for the gradient-based randomization, which is also particularly attractive when implemented in parallel. Deterministic evaluations of the inverse Hessian using the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm are also tested. While existing BFGS preconditioning techniques yield poor approximations of the error variances (relative error $>120\%$), a new preconditioner that efficiently accumulates information on the diagonal of the inverse Hessian dramatically improves the results (relative error $<50\%$). Furthermore, performing several cycles of the BFGS algorithm using the same gradient and vector pairs enhances its performance (relative error

<30%) and is necessary to obtain convergence. Leveraging those findings, we proposed a BFGS hybrid approach which combines the new preconditioner with several BFGS cycles using information from a few (3-5) Monte-Carlo simulations. Its performance is comparable to the stochastic approximations for the low-dimensional case, while good scalability is obtained for the high-dimensional experiment. Potential applications of these new BFGS methods range from characterizing the information content of high-dimensional inverse problems to improving the convergence rate of current minimization algorithms.

178. Carbon monitoring system flux estimation and attribution: impact of ACOS-GOSAT XCO₂ sampling on the inference of terrestrial biospheric sources and sinks

Authors: Junjie Liu; Kevin W. Bowman; Meemong Lee; Daven K. Henze; Nicolas Bousserez; Holger Brix; G. James Collatz; Dimitris Menemenlis; Lesley Ott; Steven Pawson; Dylan Jones; Ray Nassar

Journal: Tellus B: Chemical and Physical Meteorology

Publisher: Stockholm University Press

DOI: 10.3402/tellusb.v66.22486

URL: <https://doi.org/10.3402/tellusb.v66.22486>

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References: 66

Cited by: 94 publications

Language: N/A

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

179. Inferring regional sources and sinks of atmospheric CO₂ from GOSAT XCO₂ data

Authors: F. Deng; D. B. A. Jones; D. K. Henze; N. Bousserez; K. W. Bowman; J. B. Fisher; R. Nassar; C. O'Dell; D. Wunch; P. O. Wennberg; E. A. Kort; S. C. Wofsy; T. Blumenstock; N. M. Deutscher; D. W. T. Griffith; F. Hase; P. Heikkinen; V. Sherlock; K. Strong; R. Sussmann; T. Warneke

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-14-3703-2014

URL: <https://doi.org/10.5194/acp-14-3703-2014>

Publication Date: 2014-04-11

Volume/Issue: Volume 14, Issue 7

References: 79

Cited by: 108 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We have examined the utility of retrieved column-averaged, dry-air mole fractions of CO₂ (XCO₂) from the Greenhouse Gases Observing Satellite (GOSAT) for quantifying monthly, regional flux estimates of CO₂, using the GEOS-Chem four-dimensional variational (4D-Var) data assimilation system. We focused on assessing the potential impact of biases in the GOSAT CO₂ data on the regional flux estimates. Using different screening and bias correction approaches, we selected three different subsets of the GOSAT XCO₂ data for the 4D-Var inversion analyses, and found that the inferred global fluxes were consistent across the three XCO₂ inversions. However, the GOSAT observational coverage was a challenge for the regional flux estimates. In the northern extratropics, the inversions were more sensitive to North American fluxes than to European and Asian fluxes due to the lack of observations over Eurasia in winter and over eastern and southern Asia in summer. The regional flux estimates were also sensitive to the treatment of the residual bias in the GOSAT XCO₂ data. The largest differences obtained were for temperate North America and temperate South America, for which the largest spread between the inversions was 1.02 and 0.96 Pg C, respectively. In the case of temperate North America, one inversion suggested a strong source, whereas the second and third XCO₂ inversions produced a weak and strong sink, respectively. Despite the discrepancies in the regional flux estimates between the three XCO₂ inversions, the a posteriori CO₂ distributions were in good agreement (with a mean difference between the three inversions of typically less than 0.5 ppm) with independent data from the Total Carbon Column Observing Network (TCCON), the surface flask network, and from the HIAPER Pole-to-Pole Observations (HIPPO) aircraft campaign. The discrepancy in the regional flux estimates from the different inversions, despite the agreement of the global flux estimates suggests the need for additional work to determine the minimum spatial scales at which we can reliably quantify the fluxes using GOSAT XCO₂. The fact that the a posteriori CO₂ from the different inversions were in good agreement with the independent data although the regional flux estimates differed significantly, suggests that innovative ways of exploiting existing data sets, and

possibly additional observations, are needed to better evaluate the inferred regional flux estimates.

180. Definition, capabilities and components of a terrestrial carbon monitoring system

Authors: Tristram O West; Molly E Brown; Riley M Duren; Stephen M Ogle; Richard H Moss

Journal: Carbon Management

Publisher: Informa UK Limited

DOI: 10.4155/cmt.13.36

URL: <https://doi.org/10.4155/cmt.13.36>

Publication Date: 2013-08-01

Volume/Issue: Volume 4, Issue 4

References: 53

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: Earth system science|Carbon fibers|Carbon accounting

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

181. Profiling tropospheric CO₂ using Aura TES and TCCON instruments

Authors: L. Kuai; J. Worden; S. Kulawik; K. Bowman; M. Lee; S. C. Biraud; J. B. Abshire; S. C. Wofsy; V. Natraj; C. Frankenberg; D. Wunch; B. Connor; C. Miller; C. Roehl; R.-L. Shia; Y. Yung

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

DOI: 10.5194/amt-6-63-2013

URL: <https://doi.org/10.5194/amt-6-63-2013>

Publication Date: 2013-01-10

Volume/Issue: Volume 6, Issue 1

References: 63

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Monitoring the global distribution and long-term variations of CO₂ sources and sinks is required for characterizing the global carbon budget. Total column measurements are useful for estimating regional-scale fluxes; however, model transport remains a significant error source, particularly for quantifying local sources and sinks. To improve the capability of estimating regional fluxes, we estimate lower tropospheric CO₂ concentrations from ground-based near-infrared (NIR) measurements with space-based thermal infrared (TIR) measurements. The NIR measurements are obtained from the Total Carbon Column Observing Network (TCCON) of solar measurements, which provide an estimate of the total CO₂ column amount. Estimates of tropospheric CO₂ that are co-located with TCCON are obtained by assimilating Tropospheric Emission Spectrometer (TES) free tropospheric CO₂ estimates into the GEOS-Chem model. We find that quantifying lower tropospheric CO₂ by subtracting free tropospheric CO₂ estimates from total column estimates is a linear problem, because the calculated random uncertainties in total column and lower tropospheric estimates are consistent with actual uncertainties as compared to aircraft data. For the total column estimates, the random uncertainty is about 0.55 ppm with a bias of -5.66 ppm, consistent with previously published results. After accounting for the total column bias, the bias in the lower tropospheric CO₂ estimates is 0.26 ppm with a precision (one standard deviation) of 1.02 ppm. This precision is sufficient for capturing the winter to summer variability of approximately 12 ppm in the lower troposphere; double the variability of the total column. This work shows that a combination of NIR and TIR measurements can profile CO₂ with the precision and accuracy needed to quantify lower tropospheric CO₂ variability.

182. Communicating the Needs of Climate Change Policy Makers to Scientists

Authors: Molly E.; Vanessa M.; Heather Lovell

Journal: Human and Social Dimensions of Climate Change

Publisher: InTech

DOI: 10.5772/50607

URL: <https://doi.org/10.5772/50607>

Publication Date: 2012-11-16

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

183. Bringing together users and developers of forest biomass maps

Authors: Molly E. Brown; Molly Macauley

Journal: Eos, Transactions American Geophysical Union

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012eo030011

URL: <https://doi.org/10.1029/2012eo030011>

Publication Date: 2012-01-19

Volume/Issue: Volume 93, Issue 3

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: NASA Carbon Monitoring System Briefing: Steps Towards Improved Measurements of Biomass and Resources for the Future; Washington, D. C., 9 September 2011 Forests store carbon and thus represent important sinks for atmospheric carbon dioxide. Reducing uncertainty in current estimates of the amount of carbon in standing forests will improve precision of estimates of anthropogenic contributions to carbon dioxide in the atmosphere due to deforestation. Although satellite remote sensing has long been an important tool for mapping land cover, until recently aboveground forest biomass estimates have relied mostly on systematic ground sampling of forests.